Statistics

Lecture notes, Chapters 1–12

Based on *General Statistics*, 3rd ed. (Hanbit Academy)

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Chapter 1. Generating Data

1.1 What is statistics?

Motivation. Statistics draws conclusions about a large group from a small part of it, with assessable reliability.

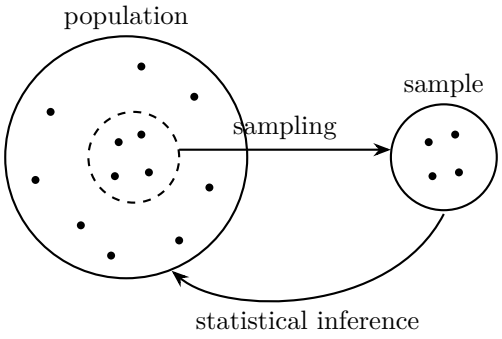
Table 1-1 shows a poll taken before a mayoral election.

Candidate	Support
John Smith	42%
Emily Johnson	37%
Michael Lee	15%
Sarah Davis	6%

Three tasks arise: *whom* to survey (how many, chosen how, so as to represent all voters); how to *summarize* the data and conclude about all voters (does Smith lead Johnson overall?); and how to *assess the reliability* of the conclusion. **Statistics** is the science that collects, organizes and interprets numerical information (**data**) to reach rational conclusions with a measurable degree of reliability. Since surveying every unit is usually too costly or impossible, we observe a part and reason about the whole.

Definition (Population and sample)

Each individual unit making up the whole is a **sampling unit**; the value describing the feature of interest of a unit (support for a candidate, sleep quality, drug effect) is a **characteristic**. The collection of the characteristic values of *all* sampling units is the **population**; the collection of values actually observed is the **sample**.
A population made of finitely many units is a **finite population** (all voters of a city). A population with infinitely many units, e.g. all present and future smartphone users, is an **infinite population**; it is a conceptual, not a physically existing, object.



A brief history. "Statistics" comes from *State Arithmetics* (censuses, land valuation for taxes). In the 17th–18th centuries astronomers and surveyors met **variability** of repeated measurements; in the 19th century statistics spread to agriculture and biology, and in the 20th it became a unified science, now used from epidemic forecasting and polling to AI evaluation and climate research.

1.2 Simple random sampling

Motivation. Inference from a part is trustworthy only if chance, not people, picks the part.

A satisfaction survey of all Korean students that covers only students near the surveyor's home is not representative.

Illustration 1-1: a biased product sample

To test the strength of steel plates, an inspector cuts ten $5\,\mathrm{cm}^2$ pieces but avoids visible defects. The test says the plates are excellent, though the sample does not reflect actual quality.

Illustration 1-2: voluntary response

Of 5 million viewers of a singing contest, 10% (500,000) voted by web, text or phone; contestant A won with 50%. A random survey of 500 viewers found only 10% preferred A: the voters were mostly A's fans. A sample of self-selected respondents is a **voluntary response sample**; it can badly distort opinion.

Giving every unit an equal chance of selection removes the surveyor's intervention (1-1) and self-selection (1-2), the sources of structural error; this requires careful *sample design*.

Definition 1-1 (Simple random sampling)

A sampling method designed so that, from a finite population, every subset consisting of n sampling units has the same probability of being selected as the sample is called **simple random sampling**.

With population size N, each of the $\binom{N}{n}$ subsets has probability $1/\binom{N}{n}$. Sequentially: the first unit is picked with probability $1/N$, and after k units one of the remaining $N-k$ with probability $1/(N-k)$ (**sampling without replacement**, probabilities $\frac{1}{N}, \frac{1}{N-1}, \ldots, \frac{1}{N-n+1}$). In **sampling with replacement** the chosen unit is returned, so each draw picks one of N units with probability $1/N$.

Proposition (Properties of simple random sampling)

Simple random sampling (without replacement) of size n from a population of size N satisfies
(1) every possible sample of size n has the same chance of being selected;
(2) every unit of the population has the same chance of being selected, namely n/N.

Proof. [Supplement]

Step 1. An ordered list of n distinct units is drawn with probability $\frac{1}{N}\cdots\frac{1}{N-n+1}$; a subset arises from $n!$ orderings, so

$$\frac{n!}{N(N-1)\cdots(N-n+1)} = \frac{n!\,(N-n)!}{N!} = \frac{1}{\binom{N}{n}}$$

for every subset; this is (1).

Step 2. A fixed unit lies in $\binom{N-1}{n-1}$ of the $\binom{N}{n}$ subsets: probability $\binom{N-1}{n-1}/\binom{N}{n} = n/N$. ■

Systematic sampling (random start, then every k-th unit) satisfies (2) but not (1). Statistical designs use **probability sampling**, with explicitly defined selection probabilities (Chapter 12).

1.3 Cautions in sample surveys

Motivation. Random selection removes the surveyor's bias, but practical errors remain.

(1) Incomplete frame. Units missing from the list (homeless people, households without phones) cause **bias** if they differ from the rest.

(2) Non-response. Non-response often reaches 30%; replacing non-respondents does not help if their views differ.

(3) Response and interviewer effects. Sensitive questions ("Have you ever downloaded illegally?") are answered untruthfully; tone or leading phrasing ("Most people exercise three times a week; how often do you?") biases answers. Interviewers must be trained.

(4) Question wording. "Environmental regulation places an excessive burden on small firms, costing jobs and hindering growth. Do you favour relaxing it?" induces support for deregulation.

1.4 Statistical experiments

Motivation. Surveys observe units as they are; experiments act on them and watch the response.

Definition 1-2 (Experimental unit and treatment)

The individual on which an experiment is performed is an **experimental unit**; applying a specific experimental environment or condition to an experimental unit is a **treatment**.

Definition 1-3 (Response variable, factor, factor level)

In a statistical experiment, a variable expressing the experimental environment or condition is a **factor**, and the variable expressing the reaction to it is the **response variable**. The values a factor takes are the **levels** of the factor.

Illustration 1-3: vaccine dose

One vaccine dose is given to 25 people; the response is blood concentration after 30 minutes. One factor (vaccine) at one level; three doses would give three levels.

Example 1-1 (Diabetes treatment)

Half of 18,000 diabetic patients take a new treatment daily, the other half the existing one; after six months the new-treatment group has lower mean blood sugar. Identify the experimental units, factor, levels and response variable.
Solution. Units: the 18,000 patients. Factor: type of treatment, with two levels (new, existing). Response: blood sugar after six months.

Illustration 1-4: two factors

Do practice problems work best at the start or end of a chapter, depending on problem type? Six textbook versions are used, with equal study time, homework and final test.

Position \ Type	Simple fact	Computation	Concept
End of chapter	Group 1	Group 2	Group 3
Start of chapter	Group 4	Group 5	Group 6

Units: students; response: test score; two factors, position (2 levels) and type (3 levels), whose $2 \times 3 = 6$ combinations are the treatments.

1.5 Comparative experiments and randomization

Motivation. Control groups and random assignment separate a treatment's effect from everything else.

Illustration 1-5: gastric freezing

Gastric freezing (a swallowed balloon with circulating coolant) was claimed to relieve ulcer pain, based on

$$\text{observe pain} \;\rightarrow\; \text{gastric freezing} \;\rightarrow\; \text{observe pain}.$$

Patients improved, but through the **placebo effect**, a positive response to any treatment, which affects even physiological responses such as acid secretion. A controlled study of 160 patients found 34% of 82 frozen and 38% of 78 given saline improved.

Both groups of the second study received the placebo influence, so their difference is the effect of freezing itself, which was nil. The group receiving the treatment is the **treatment group**; the group receiving the dummy treatment is the **control group**.

First principle of experimental design

Minimize the effect of extraneous factors other than the factor of interest.

Neither patient nor experimenter should know the assignment (**double-blind experiment**), and the groups must be alike in extraneous factors; otherwise the group difference mixes the treatment effect with that factor: **confounding**.

Illustration 1-6: confounded study

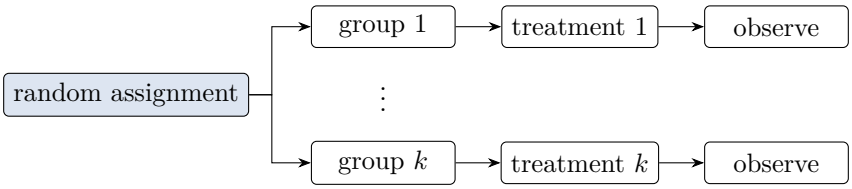
After a chemical leak, an agency reported genetic damage in residents, but age and living habits were not controlled, so the damage could not be attributed to the chemicals. A similar control group (age, sex, region), identical measurement and blind assessment were needed.

Extraneous factors are too many to equalize by hand, and assignment by judgement (e.g. the new drug to milder cases) can bias the result; units should be assigned by chance.

Illustration 1-7: random assignment

Of 30 newly weaned rats, 10 chosen by simple random sampling get a new food and 20 (control) the existing food; weight gain after 4 weeks is measured.

Each rat is treated with probability 1/3. Assigning units by chance so that each has the same chance of each treatment is **randomization** (R. A. Fisher, Rothamsted).



Completely randomized design (Figures 1-2 and 1-3; $k = 2$ is the randomized comparative experiment)

Second principle of experimental design (randomization)

When deciding which treatment each experimental unit receives, give every unit the same chance of receiving any particular treatment.

Subjective assignment produces unexpected errors, while random results obey the laws of probability. "Random" is not "haphazard": taking the first ten rats caught may select smaller or slower ones. For k treatments, draw groups successively by simple random sampling (e.g. 30 units into three groups of 10); this is a **completely randomized design**.

A response difference is still not automatically a treatment effect (fast growers may cluster by chance); randomization lets probability judge whether it is too large for chance (Chapter 6), and more units make groups more alike.

Third principle of experimental design (replication)

Apply each treatment repeatedly to a sufficiently large number of experimental units.

1.6 Blocking

Motivation. Grouping similar units and randomizing within groups gives more precise comparisons.

Illustration 1-8: paired design

Does room temperature (15 °C vs. 20 °C) affect manual dexterity (pegs inserted in 30 minutes)? Individual dexterity varies so much that two separate groups may hide a small effect. Instead every subject works at both temperatures on different days, in random order (heads ⇒ 15 °C first).

This **randomized paired comparison design** keeps comparison, randomization and replication; each unit is its own control, removing between-person variation. Pairing two similar subjects and randomizing who works at 15 °C is also a paired design.

Definition (Blocking)

Blocking means forming, before the experiment, groups of experimental units expected to respond similarly to the same treatment; each such group is a **block**. A design that uses blocking and randomization (within blocks) together is a **randomized block design**; the randomized paired comparison design is a special case (block size 2).

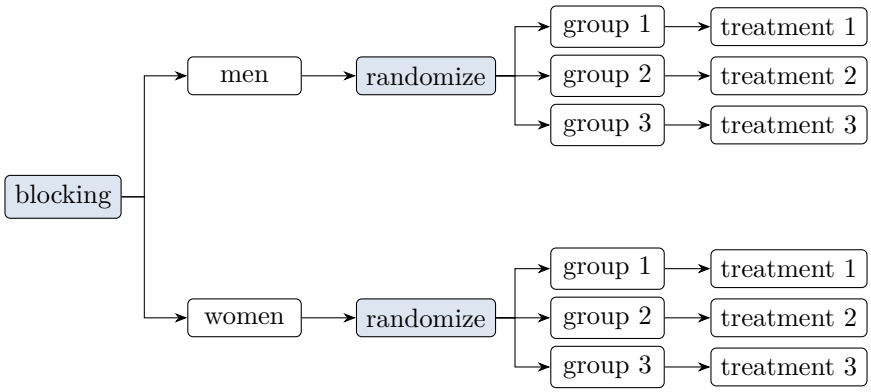
Blocks are built from factors expected to affect the response. Usually block size equals the number of treatments (1-9); larger blocks are split by randomization into treatment groups (1-10).

Illustration 1-9: field blocks

For 2 soybean varieties × 3 pesticides, each field (soil, fertility) is a block of 6 plots, with the 6 treatments randomized to plots.

Illustration 1-10: blocking by sex

A cancer progresses differently in men and women. Three treatments are tested on 30 male and 30 female patients as two blocks; within each block randomization forms three groups of 10, and survival is compared within blocks.



Randomized block design of Illustration 1-10 (Figure 1-4)

Survey or experimental data are analysed as a *sample from a population*, often a conceptual infinite one: in 1-9, each variety–pesticide combination defines an infinite population of yields.

In an *experiment* the experimenter decides who is treated, so extraneous factors can be controlled; in a *sample survey* units decide themselves. When experiments are unethical (forcing people to smoke), an **observational study** is needed, and hidden extraneous factors require special care.

1.7 Selected exercises

Motivation. Systematic sampling, confounding and blocking, and how randomization balances extraneous factors.

Exercise 1.5 (Systematic sampling)

In systematic sampling the units are listed and selected at a fixed interval. For instance, to choose 4 addresses from a list of 100, the interval is $100/4 = 25$; a number r is drawn at random from $1, \dots, 25$, and the sample is the addresses numbered r, $r+25$, $r+50$, $r+75$ (for $r = 13$: $13, 38, 63, 88$).
(a) Describe how to select a systematic sample of size 5 from a list of 200 addresses.
(b) Show that every address has the same chance of being selected, but that the method is not simple random sampling.

Solution.

(a) $k = 200/5 = 40$: draw r uniformly from $\{1, \dots, 40\}$ and take $r,\ r+40,\ r+80,\ r+120,\ r+160$.

(b)

Step 1. The 40 samples $S_r = \{r, r+40, \dots, r+160\}$ partition the 200 addresses, so $P(j \text{ selected}) = 1/40 = n/N$, as in property (2) (Section 1.2).

Step 2. Only the 40 sets S_r have positive probability, not all $\binom{200}{5} \approx 2.5 \times 10^9$ subsets (addresses $1, 2$ never occur together), so property (1) fails.

If the list has a period matching the interval (every 40th address a corner flat), the sample is all one type and biased. ■

Exercise 1.9 (Fungus and wind)

An experiment compares whether fungal disease spreads differently among three crops A, B and C. Fungal spread is known to be affected by the wind direction. The field is divided into three adjacent strips in a row along the wind direction, and A, B and C are planted in the three strips, C being nearest to the side from which the wind blows. Point out the problem with this experiment and propose a suitable design.

Solution.

Step 1. (The problem.) Crop and position along the wind are completely *confounded* (C always nearest the wind), and with one plot per crop there is no replication.

Step 2. (A design.) Form *blocks* as bands across the wind, so plots in a block are equally exposed; randomize A, B, C to the three plots of each block independently.

Step 3. (Why it works.) In this randomized block design (Section 1.6) every crop meets every exposure level, within-block comparisons remove exposure differences, randomization guards against unknown plot differences, and the blocks give replication. ■

Exercise 1.11 (A confounded comparison)

To test whether a new learning programme is more effective than the current one, a school randomly splits the 40 students of one class into two groups of 20. One group studies with the new programme, taught by one teacher; the other group studies with the current programme, taught by another teacher. At the end the new-programme group has improved more, and the school concludes that the new programme is effective. Although the students were randomized, the experiment has a serious defect. Explain why the conclusion cannot be trusted.

Solution.

Step 1. The treatment is "programme with teacher": the programme effect is confounded with the teacher effect. Randomizing students balances students, not teachers.

Step 2. With one teacher per programme there is no replication at the programme level, so no measure of teacher-to-teacher variation.

Step 3. Let each teacher teach both programmes to randomly assigned halves (teachers as blocks), or randomize several classes and teachers to programmes. ■

Exercise 1.13 (Randomization balances a covariate)

There are $2N$ experimental units with values $u_1, \dots, u_{2N}$ of some extraneous variable (age, fertility, ...), with mean $\bar{u}$ and $\sigma^2 = \frac{1}{2N}\sum_k (u_k - \bar{u})^2$. A completely randomized design sends a simple random sample of N units to treatment and the rest to control. Let $\bar{u}_T$ and $\bar{u}_C$ be the mean values of u in the two groups. Show that

$$E(\bar{u}_T - \bar{u}_C) = 0, \qquad \operatorname{Var}(\bar{u}_T - \bar{u}_C) = \frac{4\sigma^2}{2N - 1}.$$

Solution [Supplement].

Step 1. Let $\delta_k = 1$ if unit k is treated, else 0, and $d_k = u_k - \bar{u}$, so $\sum_k d_k = 0$, $\sum_k d_k^2 = 2N\sigma^2$. As $\sum_k \delta_k = N$ and $\bar{u}_C = 2\bar{u} - \bar{u}_T$,

$$\bar{u}_T - \bar{u} = \frac{1}{N}\sum_k \delta_k u_k - \bar{u} = \frac{1}{N}\sum_k \delta_k d_k, \qquad \bar{u}_T - \bar{u}_C = 2(\bar{u}_T - \bar{u}).$$

Step 2. $P(\delta_k = 1) = \frac{1}{2}$ (property (2), Section 1.2); a pair lies in $\binom{2N-2}{N-2}$ of the $\binom{2N}{N}$ groups, so for $k \neq l$

$$P(\delta_k = \delta_l = 1) = \frac{N(N-1)}{2N(2N-1)}, \qquad \mathrm{Cov}(\delta_k, \delta_l) = \frac{N-1}{2(2N-1)} - \frac{1}{4} = -\frac{1}{4(2N-1)},$$

and $\mathrm{Var}\,\delta_k = \frac{1}{4}$.

Step 3. (Mean.) $E\sum_k \delta_k d_k = \frac{1}{2}\sum_k d_k = 0$.

Step 4. (Variance.) With $\sum_{k\neq l} d_k d_l = -\sum_k d_k^2$,

$$\mathrm{Var}\Big(\sum_k \delta_k d_k\Big) = \frac{1}{4}\sum_k d_k^2 + \frac{1}{4(2N-1)}\sum_k d_k^2 = \frac{2N}{4(2N-1)} \cdot 2N\sigma^2 = \frac{N^2\sigma^2}{2N-1}.$$

So $\mathrm{Var}(\bar{u}_T - \bar{u}) = \sigma^2/(2N-1)$ and $\mathrm{Var}(\bar{u}_T - \bar{u}_C) = 4\sigma^2/(2N-1)$.

Randomization balances u *on average*, even unmeasured, with chance imbalance shrinking like $1/N$; blocking on u removes it altogether. ■

Chapter 2. Populations and Samples

2.1 Distribution of a population

Motivation. A distribution records how a population's values are spread, as nonnegative weights summing to one.

Table 1-1 summarized a *sample*; the election result summarizes the *population*:

Candidate	J. Smith	E. Johnson	M. Lee	S. Davis	Total
Vote share	0.367	0.281	0.271	0.081	1.000

The shares are nonnegative and sum to 1; drawn as bars they form a bar graph (Figure 2-1).

Definition 2-1 (Distribution of a population)

The way the characteristic values of a population are spread, expressed by nonnegative numbers summing to 1, is called the **distribution** of the population.

A pre-election poll aims to estimate this distribution.

Illustration 2-1: Mendel's peas

Crossing RRYY with rryy (R round, r wrinkled, Y yellow, y green), the second generation shows round–yellow (C1), round–green (C2), wrinkled–yellow (C3), wrinkled–green (C4) in ratio $9:3:3:1$; endless repetition gives an infinite population with distribution

Value	C1	C2	C3	C4
Proportion	9/16	3/16	3/16	1/16

Qualitative or discrete values have distributions given by relative frequencies (or their limits).

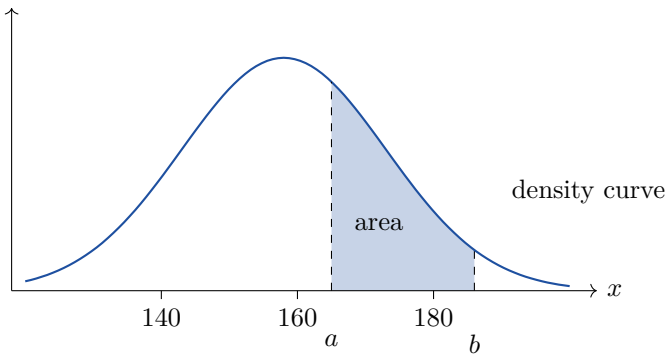
Continuous characteristics. Height, weight etc. are grouped into classes; with width 10 cm, say

Class (cm)	130–140	140–150	150–160	160–170	170–180	180–190	Total
Rel. freq.	0.05	0.1	0.3	0.3	0.2	0.05	1.00

(class $[a, b)$). Over each class draw a rectangle of *area* equal to its relative frequency (height = rel. freq./width); total area is 1. Width 5 cm gives a finer picture (Table 2-4: $0.01, 0.04, 0.04, 0.06, 0.2, 0.1, 0.2, 0.1, 0.05, 0.15, 0.03, 0.02$ for $130, 135, \ldots, 185$); a smooth curve removes the dependence on the classes.

Definition (Density curve)

A curve describing the distribution of an infinite population with a continuous characteristic is a **density curve**. The area under it over $[a, b]$ equals the limit of the relative frequency of values in $[a, b]$, so the total area between curve and horizontal axis is always 1. Even for a finite population, a density curve is a convenient approximation when the population is large.



Area over $[a, b]$ = limit of the relative frequency of values in $[a, b]$ (Figure 2-5)

For several characteristics, a **joint distribution** spreads mass 1 over multivariate values (bars summing to 1, or volume 1 under a surface).

2.2 Summary measures of a population

Motivation. A few numbers summarize location, spread and association of a numerical distribution.

2.2.1 Location: population mean, percentiles

For a finite population $\{c_1, \ldots, c_N\}$ (c_i the value of unit i), the **population mean** is

$$\mu = \frac{1}{N}(c_1 + c_2 + \cdots + c_N) = \frac{1}{N}\sum_{i=1}^{N} c_i.$$

If the distinct values are $c_1^*, \ldots, c_k^*$ with frequencies $f_1, \ldots, f_k$, then μ is the sum of (value)×(relative frequency):

Definition 2-2 (Population mean of a finite population)

$$\mu = \sum_{i=1}^{k} c_i^* \frac{f_i}{N}.$$

Example 2-1 (Number of children)

Number of children of the 16,549 thousand married Korean women aged 15+ (2020 census; 5* means "5 or more", counted as 5):

Children	0	1	2	3	4	5*
Rel. freq.	0.083	0.189	0.477	0.150	0.054	0.047

Find the population mean.
Solution.

$$\mu = 0(0.083) + 1(0.189) + 2(0.477) + 3(0.150) + 4(0.054) + 5(0.047) = 2.044.$$

μ is the **balance point** of the bar graph (Figure 2-7), or of a plate shaped like the density curve.

Definition 2-3 (Population mean of an infinite population)

$$\mu = \begin{cases} \sum_{\text{all } x} x\,p(x) & \text{(discrete case)} \\ \int_{-\infty}^{\infty} x\,p(x)\,dx & \text{(continuous case)} \end{cases}$$

Here $p(x)$ is the limiting relative frequency of the value x (discrete case) or the density curve (continuous case); p is called the **density function**.

Example 2-2 (Mean of an infinite population)

Find the population mean of the infinite population

x	0	1	2	3	4
$p(x)$	1/10	3/10	3/10	2/10	1/10

Solution.

$$\mu = 0 \cdot \tfrac{1}{10} + 1 \cdot \tfrac{3}{10} + 2 \cdot \tfrac{3}{10} + 3 \cdot \tfrac{2}{10} + 4 \cdot \tfrac{1}{10} = 1.9.$$

The p**th percentile** has $p\%$ of the ordered values below it; the 25th, 50th, 75th are the **first quartile** Q_1, **median** Q_2, **third quartile** Q_3. The median halves the area under the density; $\mu = Q_2$ for symmetric densities, and a right tail pulls μ above Q_2 (Figure 2-9). μ (with σ) is the key inference target.

2.2.2 Spread: MAD, IQR, variance

For a finite population, the **mean absolute deviation** and **interquartile range** are

$$\text{MAD} = \frac{1}{N}\sum_{i=1}^{N} |c_i - Q_2|, \qquad \text{IQR} = Q_3 - Q_1.$$

The standard measures are the **standard deviation** σ and **variance** σ^2:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(c_i - \mu)^2}, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(c_i - \mu)^2.$$

In terms of distinct values c_i^* with frequencies f_i:

Definition 2-4 (Variance and s.d. of a finite population)

(1) Population variance: $\sigma^2 = \sum_{i=1}^{k}(c_i^* - \mu)^2 \frac{f_i}{N}$.

(2) Population standard deviation: $\sigma = \sqrt{\sigma^2}$.

Definition 2-5 (Variance and s.d. of an infinite population)

For an infinite population with density function $p(x)$:

(1) Population variance:

$$\sigma^2 = \begin{cases} \displaystyle\sum_{\text{all } x} (x-\mu)^2 p(x) & \text{(discrete case)} \\ \displaystyle\int_{-\infty}^{\infty} (x-\mu)^2 p(x)\,dx & \text{(continuous case)} \end{cases}$$

(2) Population standard deviation: $\sigma = \sqrt{\sigma^2}$.

Example 2-3 (Variance)

Find the population variance and standard deviation for Example 2-2.
Solution. With $\mu = 1.9$,

$$\begin{aligned}\sigma^2 &= (0-1.9)^2\tfrac{1}{10} + (1-1.9)^2\tfrac{3}{10} + (2-1.9)^2\tfrac{3}{10} + (3-1.9)^2\tfrac{2}{10} + (4-1.9)^2\tfrac{1}{10} \\ &= 1.29,\end{aligned}$$

and $\sigma = \sqrt{1.29} = 1.1358$.

2.2.3 Association: correlation

Table 2-5 gives two bivariate populations; each entry is the relative frequency of the pair (characteristic 1, characteristic 2).

Population A

char. 2 \ char. 1	0	1	2	Total
0	0.1	0.2	0.1	0.4
1	0.0	0.2	0.1	0.3
2	0.0	0.0	0.3	0.3
Total	0.1	0.4	0.5	1.0

Population B

char. 2 \ char. 1	0	1	2	Total
0	0.0	0.0	0.4	0.4
1	0.0	0.2	0.1	0.3
2	0.1	0.2	0.0	0.3
Total	0.1	0.4	0.5	1.0

The bottom margin gives the distribution of characteristic 1, the right margin that of characteristic 2. In both populations

$$\mu_1 = 1.4,\ \sigma_1 = 0.6633, \qquad \mu_2 = 0.9,\ \sigma_2 = 0.8307,$$

yet A's mass rises and B's falls, so a measure of association is needed. For a bivariate finite population $(c_{11}, c_{21}), \ldots, (c_{1N}, c_{2N})$ the **correlation coefficient** (Galton, Pearson) is

$$\rho = \frac{1}{N}\sum_{i=1}^{N}\Big(\frac{c_{1i}-\mu_1}{\sigma_1}\Big)\Big(\frac{c_{2i}-\mu_2}{\sigma_2}\Big).$$

Proposition (Range of the correlation coefficient)

$-1 \le \rho \le 1$, because

$$0 \le \frac{1}{N}\sum_{i=1}^{N}\Big(\frac{c_{1i}-\mu_1}{\sigma_1} - \rho\,\frac{c_{2i}-\mu_2}{\sigma_2}\Big)^2 = 1-\rho^2.$$

Proof. With $u_i = (c_{1i}-\mu_1)/\sigma_1$, $v_i = (c_{2i}-\mu_2)/\sigma_2$,

$$\frac{1}{N}\sum u_i^2 = 1, \qquad \frac{1}{N}\sum v_i^2 = 1, \qquad \frac{1}{N}\sum u_i v_i = \rho,$$

so

$$\frac{1}{N}\sum(u_i-\rho v_i)^2=\frac{1}{N}\sum u_i^2-2\rho\,\frac{1}{N}\sum u_iv_i+\rho^2\,\frac{1}{N}\sum v_i^2=1-2\rho^2+\rho^2=1-\rho^2.$$

and $\rho^2\le 1$. ■

Grouping distinct values c_{1i}^* $(i\le k)$ of characteristic 1 and c_{2j}^* $(j\le l)$ of characteristic 2, with f_{ij} the frequency of (c_{1i}^*,c_{2j}^*):

Definition 2-6 (Correlation of a finite population)

$$\rho=\sum_{i=1}^{k}\sum_{j=1}^{l}\left(\frac{c_{1i}^*-\mu_1}{\sigma_1}\right)\left(\frac{c_{2j}^*-\mu_2}{\sigma_2}\right)\frac{f_{ij}}{N}=\frac{\sum_{i=1}^{k}\sum_{j=1}^{l}(c_{1i}^*-\mu_1)(c_{2j}^*-\mu_2)\,f_{ij}/N}{\sigma_1\sigma_2}.$$

Example 2-4 (Correlations of Table 2-5)

Find the population correlation coefficients of populations A and B.
Solution. For A,

$$\begin{aligned}&\sum_i\sum_j(c_{1i}^*-\mu_1)(c_{2j}^*-\mu_2)\frac{f_{ij}}{N}\\&\quad=(0-1.4)(0-0.9)(0.1)+(1-1.4)(0-0.9)(0.2)+\cdots+(2-1.4)(2-0.9)(0.3)\\&\quad=0.34,\end{aligned}$$

so $\rho=0.34/(0.6633\times 0.8307)=0.6171$. For B the numerator is -0.46 and $\rho=-0.8348$.

For an infinite bivariate population let $p(x,y)$ be the **joint density function** (limiting relative frequency, or density surface):

Definition 2-7 (Correlation of an infinite population)

$$\rho=\begin{cases}\dfrac{\sum_{\text{all }x}\sum_{\text{all }y}(x-\mu_1)(y-\mu_2)\,p(x,y)}{\sigma_1\sigma_2} & \text{(discrete case)}\\[2ex]\dfrac{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x-\mu_1)(y-\mu_2)\,p(x,y)\,dx\,dy}{\sigma_1\sigma_2} & \text{(continuous case)}\end{cases}$$

The numerator is the **covariance**, $\rho\sigma_1\sigma_2$. $\rho>0$ for rising mass, $\rho<0$ for falling, and $\rho\to\pm1$ as mass concentrates on a line. Numbers describing a population (mean, variance, covariance, correlation) are **parameters**.

2.3 Graphical display of samples

Motivation. Tables and graphs of a sample give a first picture of the population's distribution.

2.3.1 Qualitative or few-valued data

A **frequency table** lists each value with its **frequency** (count) and **relative frequency** (count / sample size). It is drawn as a **bar graph** (bar heights proportional to relative frequencies) or a **pie graph** (sector areas, i.e. central angles, proportional to relative frequencies).

Example 2-5 (Bar and pie graphs)

Draw a bar graph and a pie graph of the sample in Table 1-1.
Solution. The bar graph has heights $0.42, 0.37, 0.15, 0.06$. The central angles of the pie are

$$360^\circ \times 0.42 = 151.2^\circ, \quad 360^\circ \times 0.37 = 133.2^\circ, \quad 360^\circ \times 0.15 = 54^\circ, \quad 360^\circ \times 0.06 = 21.6^\circ.$$

2.3.2 Frequency tables, histograms, polygons

For continuous or many-valued data, divide the range into classes: not too sparse, not so wide as to hide the shape, with no observation on a boundary.

Constructing a frequency table

(1) Range = (largest value) − (smallest value).
(2) Choose about 5–20 classes; class width = range ÷ (number of classes), rounded *up* to the smallest unit of the data.
(3) Start the first class at (smallest value) $-\frac{1}{2}\times$(smallest unit), and build the classes with the width from (2).
(4) Count the (relative) frequency of each class.

Example 2-6 (Heights of 40 people)

Heights (cm) of 40 randomly selected people:

183	168	165	158	180	176	168	160	160	150
180	183	170	170	185	170	165	170	168	154
152	173	173	165	163	163	160	178	138	171
163	168	165	165	178	170	168	183	145	174

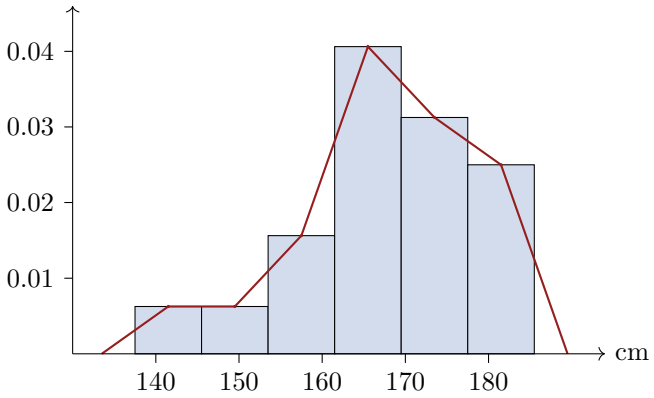
Make a frequency table.
Solution. Max 185, min 138, range 47. With 6 classes, $47/6 = 7.83\ldots$, rounded up to width 8. The first class starts at $138 - \frac{1}{2} \times 1 = 137.5$.

Class (cm)	Freq.	Rel. freq.
$[137.5, 145.5)$	2	0.050
$[145.5, 153.5)$	2	0.050
$[153.5, 161.5)$	5	0.125
$[161.5, 169.5)$	13	0.325
$[169.5, 177.5)$	10	0.250
$[177.5, 185.5)$	8	0.200
Total	40	1

A **histogram** marks the class intervals on the horizontal axis and draws over each class a rectangle with

$$\text{area} = \text{relative frequency}, \qquad \text{i.e.} \qquad \text{height} = \frac{\text{relative frequency}}{\text{class width}}.$$

Total area is 1, so the histogram estimates the density curve. Widths may vary: narrow where data are dense, wide where sparse (Figure 2-13 splits the 4th class in two). A **frequency polygon** joins the midpoints of the rectangle tops, ending on the axis half a width beyond the end classes; its area is also 1.



Histogram (height = rel. freq./8) and frequency polygon for Example 2-6 (Figures 2-12, 2-14)

2.3.3 Stem-and-leaf display

Leading digits form the **stems**; the last digit of each observation is a sorted **leaf** beside its stem. For Example 2-6:

18	0 0 3 3 3 5
17	0 0 0 0 0 1 3 3 4 6 8 8
16	0 0 0 3 3 3 5 5 5 5 5 8 8 8 8 8
15	0 2 4 8
14	5
13	8

This **stem-and-leaf display** is a histogram rotated by 90° with constant leaf width (here 10; split stems 14^-, 14^+ give 5). It keeps the data values but is impractical for large samples.

2.3.4 Bivariate samples

A **contingency table** lists the (relative) frequency of each pair of values; e.g. favourite subject of 400 students:

	Korean	English	Math	Total
Male	0.18	0.24	0.18	0.60
Female	0.14	0.16	0.10	0.40
Total	0.32	0.40	0.28	1.00

For continuous characteristics, class both ranges; the **bivariate histogram** erects over each cell a box of volume equal to its relative frequency.

Example 2-7 (Height and weight)

Adding weight to the heights of Example 2-6 (height cm, weight kg):

(183, 82)	(168, 52)	(165, 52)	(158, 55)	(180, 93)
(176, 57)	(168, 60)	(160, 48)	(160, 50)	(150, 49)
(180, 79)	(183, 73)	(170, 60)	(170, 66)	(185, 74)
(170, 82)	(165, 57)	(170, 54)	(168, 57)	(154, 55)
(152, 50)	(173, 63)	(173, 60)	(165, 49)	(163, 63)
(163, 57)	(160, 57)	(178, 77)	(138, 54)	(171, 69)
(163, 50)	(168, 57)	(165, 54)	(165, 59)	(178, 79)
(170, 54)	(168, 79)	(183, 70)	(145, 37)	(174, 65)

Make a contingency table and a bivariate histogram.
Solution. Using the 6 height classes of Example 2-6 and 6 weight classes of width 10 starting at 36.5:

Height \ Weight	36.5–46.5	46.5–56.5	56.5–66.5	66.5–76.5	76.5–86.5	86.5–96.5	Total
137.5–145.5	0.025	0.025	0	0	0	0	0.050
145.5–153.5	0	0.050	0	0	0	0	0.050
153.5–161.5	0	0.100	0.025	0	0	0	0.125
161.5–169.5	0	0.125	0.175	0	0.025	0	0.325
169.5–177.5	0	0.050	0.150	0.025	0.025	0	0.250
177.5–185.5	0	0	0	0.075	0.100	0.025	0.200
Total	0.025	0.350	0.350	0.100	0.150	0.025	1.000

The bivariate histogram erects over each cell a box of volume equal to the entry (Figure 2-17).

2.3.5 Scatter plots and regressograms

A **scatter plot** plots each pair as a point. A **regressogram** classes one characteristic and plots the class means of the other; for Example 2-7 by weight class:

Weight class	36.5–46.5	46.5–56.5	56.5–66.5	66.5–76.5	76.5–86.5	86.5–96.5
Mean height	145.0	159.9	168.3	180.5	176.2	180.0

As a step function this is the regressogram (Figure 2-19), usually more effective than the scatter plot for analysing the relationship.

2.4 Summary measures of a sample

Motivation. Unknown parameters are estimated by the corresponding sample quantities.

2.4.1 Location

Let the sample values be $x_1, \dots, x_n$.

Definition 2-8 (Sample mean)

$$\bar{x} = \frac{1}{n}(x_1 + x_2 + \cdots + x_n) = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

Sample percentiles, quartiles and median $\hat{Q}_1, \hat{Q}_2, \hat{Q}_3$ are defined as for the population. Since a few outlying values strongly affect $\bar{x}$, the $100p\%$ **trimmed mean** $\bar{x}_p$ drops the smallest and largest $100p\%$ and averages the remaining $100(1-2p)\%$.

Example 2-8 (Sample mean, quartiles, trimmed mean)

For the data of Example 2-6, find the sample mean, the sample quartiles and the 10% trimmed mean.

Solution. $\bar{x} = \frac{1}{40}(183 + 168 + \cdots + 174) = \frac{6698}{40} = 167.45$. From the stem-and-leaf display (midpoints of the 10th/11th, 20th/21st, 30th/31st values):

$$\hat{Q}_1 = 163, \qquad \hat{Q}_2 = 168, \qquad \hat{Q}_3 = 173.5.$$

The 10% trimmed mean drops 4 values at each end (138, 145, 150, 152 and 183, 183, 183, 185) and averages the middle 32:

$$\bar{x}_{0.1} = \frac{1}{32}(180 + 180 + \cdots + 154) = \frac{5379}{32} = 168.0938.$$

2.4.2 Spread

The **sample interquartile range** is $\widehat{\mathrm{IQR}} = \hat{Q}_3 - \hat{Q}_1$. Replacing μ by $\bar{x}$ in $\frac{1}{n}\sum(x_i - \mu)^2$ *underestimates* σ^2:

Proposition (Centring at the sample mean)

For any number μ,

$$\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2 - (\bar{x} - \mu)^2.$$

Proof. [Supplement] Square $x_i - \mu = (x_i - \bar{x}) + (\bar{x} - \mu)$:

$$\sum_{i=1}^{n}(x_i - \mu)^2 = \sum_{i=1}^{n}(x_i - \bar{x})^2 + 2(\bar{x} - \mu)\sum_{i=1}^{n}(x_i - \bar{x}) + n(\bar{x} - \mu)^2.$$

The middle sum is $\sum x_i - n\bar{x} = 0$; divide by n. ∎

Dividing by $n - 1$ instead corrects this, giving the **sample variance**

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Definition 2-9 (Sample standard deviation)

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

Example 2-9 (Sample IQR and s.d.)

For the data of Example 2-6, find the sample interquartile range and the sample standard deviation.

Solution. From Example 2-8, $\widehat{\mathrm{IQR}} = 173.5 - 163 = 10.5$. Next

$$s^2 = \frac{1}{40-1}\{(183 - 167.45)^2 + (168 - 167.45)^2 + \cdots + (174 - 167.45)^2\} = 108.6128,$$

so $s = \sqrt{108.6128} = 10.4218$.

2.4.3 Bivariate samples

For a bivariate sample $(x_1, y_1), \ldots, (x_n, y_n)$:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad s_1 = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}; \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \quad s_2 = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2}.$$

Definition 2-10 (Sample correlation coefficient)

$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s_1}\right)\left(\frac{y_i - \bar{y}}{s_2}\right) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\,\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

(The two expressions agree since $(n-1)s_1 s_2 = \sqrt{\sum (x_i - \bar{x})^2}\sqrt{\sum (y_i - \bar{y})^2}$.) The **sample covariance**, estimating the population covariance, is

$$r s_1 s_2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}).$$

Example 2-10 (Sample correlation)

Find the sample correlation coefficient for the data of Example 2-7.

Solution. $\bar{x} = 167.45$, $s_1 = 10.42$ (Examples 2-8, 2-9); $\bar{y} = 61.45$, $s_2 = 11.94$. The sample covariance is

$$r s_1 s_2 = \frac{1}{40-1}\{(183 - 167.45)(82 - 61.45) + \cdots + (174 - 167.45)(65 - 61.45)\} = 90.59,$$

so $r = 90.59/(10.42 \times 11.94) = 0.73$.

Sample quantities are **statistics**; one used to estimate a parameter is an **estimator**.

Chapter 3. Probability and Probability Distributions

3.1 Definition of probability

Motivation. Probability makes inductive conclusions from samples quantifiable; it is assigned to events.

Definition (Sample space and events)

The set of all possible outcomes of a statistical survey (experiment) is the **sample space** S. A subset of S—the set of outcomes satisfying some condition of interest—is an **event**; the event A *occurs* if the observed outcome is an element of A. An event consisting of a single element is an **elementary** (**simple**) **event**.
$A \cup B$, $A \cap B$, A^c are the **union**, **intersection** and **complement**. If $A \cap B = \varnothing$, A and B are **mutually disjoint** (exclusive): they cannot occur together.

Illustration 3-1: sample spaces

(a) Number shown by one die: $S = \{1, 2, 3, 4, 5, 6\}$.
(b) Number of items produced until the 10th defective: $S = \{10, 11, 12, \dots\}$.
(c) Lifetime of a light bulb drawn from a factory: $S = \{t \mid t \geq 0\}$.

For finitely many equally likely elementary events (Laplace):

Definition 3-1 (Classical definition of probability)

If $S = \{e_1, \dots, e_N\}$ has N elements whose elementary events are equally likely, the **probability** of an event A with m elements is

$$P(A) = \frac{m}{N}.$$

Example 3-1 (Table-tennis balls)

A box contains 3 white balls and 1 yellow ball. Two balls are drawn by simple random sampling. Find the probability of getting one white and one yellow ball.
Solution. Label the white balls a, b, c and the yellow one d:

$$S = \{\{a,b\},\{a,c\},\{a,d\},\{b,c\},\{b,d\},\{c,d\}\}, \qquad A = \{\{a,d\},\{b,d\},\{c,d\}\}.$$

All 6 pairs are equally likely, so $P(A) = 3/6 = 1/2$.

This fails for infinitely many or unequally likely outcomes. Under repetition the *relative frequency* of an event settles at a constant (Figure 3-1: proportion of defectives); Kolmogorov's axioms abstract its properties.

Definition 3-2 (Axiomatic definition of probability)

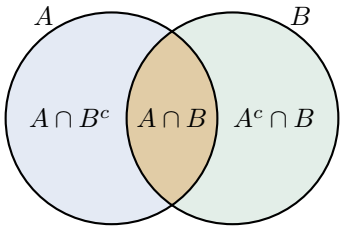
A rule P assigning numbers to events of S is a probability if
(1) $0 \leq P(A) \leq 1$ for every event A;
(2) $P(S) = 1$;

(3) for mutually disjoint events $A_1, A_2, \ldots$,

$$P(A_1 \cup A_2 \cup \cdots) = P(A_1) + P(A_2) + \cdots.$$

$P(A)$ is called the **probability** of A.

This covers empirical (limiting frequency) and subjective probability.



Partition of $A \cup B$ into three disjoint pieces (Figure 3-2)

Theorem 3-1 (Properties of probability)

(1) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
(2) $P(A^c) = 1 - P(A)$.
(3) If $A \subset B$, then $P(A) \leq P(B)$.

Proof. (1) $A \cap B^c$, $A \cap B$, $A^c \cap B$ are disjoint with union $A \cup B$; by axiom (3),

$$P(A \cup B) = P(A \cap B^c) + P(A \cap B) + P(A^c \cap B),$$
$$P(A) = P(A \cap B^c) + P(A \cap B),$$
$$P(B) = P(A \cap B) + P(A^c \cap B).$$

Subtracting, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

(2) $P(A) + P(A^c) = P(S) = 1$ (axioms (3), (2)).

(3) $B = A \sqcup (A^c \cap B)$, so $P(B) = P(A) + P(A^c \cap B) \geq P(A)$ (axioms (3), (1)). ■

Example 3-2 (Three bags)

Each of three bags contains 5 marbles numbered 1 to 5. One marble is drawn from each bag. Find the probability that the sum of the numbers is at least 5.
Solution. $A =$ "sum ≥ 5" has $A^c = B \sqcup C$, $B =$ "sum $= 3$", $C =$ "sum $= 4$"; of $5^3 = 125$ outcomes,

$$B = \{(1,1,1)\}, \qquad C = \{(1,1,2), (1,2,1), (2,1,1)\},$$

so $P(B) = 1/125$, $P(C) = 3/125$ and

$$P(A) = 1 - \{P(B) + P(C)\} = 1 - \frac{4}{125} = \frac{121}{125}.$$

3.2 Conditional probability and independence

Motivation. Conditional probability updates for information; independence means no update is needed.

Given A, A acts as the sample space, and B gets the share $P(A \cap B)/P(A)$.

Definition 3-3 (Conditional probability)

The **conditional probability** of B given A is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} \qquad (P(A) > 0).$$

Example 3-3 (Two dice)

Two dice are thrown. Given that the first die shows more than the second, find the probability that the sum is 7.

Solution. Let $A =$ "first $>$ second", $B =$ "sum $= 7$". Of 36 outcomes, A has 15, so $P(A) = 15/36$. Also $B = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$ and $A \cap B = \{(4,3),(5,2),(6,1)\}$, so

$$P(B) = \frac{6}{36} = \frac{1}{6}, \qquad P(A \cap B) = \frac{3}{36} = \frac{1}{12}, \qquad P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{3}{15} = \frac{1}{5}.$$

Theorem 3-2 (Multiplication rule)

If $P(A) > 0$ and $P(B) > 0$, then

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A).$$

Proof. Multiply $P(B \mid A) = P(A \cap B)/P(A)$ (Definition 3-3) by $P(A)$; exchange A and B. ∎

Example 3-4 (Prize draw)

A couple joins a draw at a supermarket: a box holds 10 winning and 50 losing balls; each person draws once and a drawn ball is not returned. Find the probability that both win.

Solution. Let $A =$ "husband wins", $B =$ "wife wins". Then

$$P(A \cap B) = P(B \mid A)P(A) = \frac{9}{59} \times \frac{10}{60} = \frac{3}{118}.$$

If $A_1, \dots, A_n$ *partition* S, the $B \cap A_i$ partition B.

Theorem 3-3 (Law of total probability)

If $A_i \cap A_j = \varnothing$ $(i \neq j)$, $A_1 \cup A_2 \cup \cdots \cup A_n = S$ and $P(A_i) > 0$, then

$$P(B) = P(B \mid A_1)P(A_1) + \cdots + P(B \mid A_n)P(A_n).$$

Proof.

Step 1. For $i \neq j$, $(B \cap A_i) \cap (B \cap A_j) \subset A_i \cap A_j = \varnothing$, and

$$(B \cap A_1) \cup \cdots \cup (B \cap A_n) = B \cap (A_1 \cup \cdots \cup A_n) = B \cap S = B.$$

Step 2. By axiom (3), $P(B) = \sum_{i=1}^{n} P(B \cap A_i)$, and by Theorem 3-2 each term equals $P(B \mid A_i)P(A_i)$. ∎

Example 3-5 (Joint training)

Three football teams train together: 50% of the players are from team 1, 30% from team 2, 20% from team 3; 20%, 15%, 30% of the respective teams are forwards. A player is chosen at random. Find the probability that he is a forward.
Solution. With A_i = "from team i" (a partition) and B = "forward",

$$P(B) = 0.2 \times 0.5 + 0.15 \times 0.3 + 0.3 \times 0.2 = 0.205.$$

If $P(B \mid A) = P(B)$, then $P(A \cap B) = P(A)P(B)$ and, by the multiplication rule, $P(A \mid B) = P(A)$.

Definition 3-4 (Independent events)

Events A and B are **independent** if one of the following holds (when the conditional probabilities are defined, they are equivalent):

$$P(A \cap B) = P(A)P(B), \qquad P(A \mid B) = P(A), \qquad P(B \mid A) = P(B).$$

Example 3-6 (Prize draw revisited)

In Example 3-4:
(a) Are "husband wins" and "wife wins" independent?
(b) Are they independent if the drawn ball is returned to the box?
Solution.
(a) By the law of total probability,

$$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c) = \frac{9}{59} \cdot \frac{10}{60} + \frac{10}{59} \cdot \frac{50}{60} = \frac{1}{6}.$$

Since $P(B) \neq P(B \mid A) = 9/59$, A and B are not independent.
(b) With replacement, $P(B \mid A) = 10/60 = 1/6 = P(B)$, so A and B are independent.

Draws with replacement are independent, without replacement not. Non-independent events are **dependent**. In general, $A_1, \ldots, A_n$ are **independent** if

$$\begin{aligned} P(A_i \cap A_j) &= P(A_i)P(A_j) && (1 \le i < j \le n), \\ P(A_i \cap A_j \cap A_k) &= P(A_i)P(A_j)P(A_k) && (1 \le i < j < k \le n), \\ &\vdots \\ P(A_1 \cap A_2 \cap \cdots \cap A_n) &= P(A_1)P(A_2)\cdots P(A_n). \end{aligned}$$

3.3 Random variables and probability distributions

Motivation. Mapping outcomes to numbers moves probability from the sample space to sets of numbers.

Illustration 3-2: coin toss

For $S = \{\mathrm{H}, \mathrm{T}\}$, $X(\mathrm{H}) = 1$, $X(\mathrm{T}) = 0$ transfers the probability on S to $\{0, 1\}$: $P(X = 0) = P(X = 1) = 1/2$.

Definition (Random variable, distribution)

A real-valued function defined on a sample space is a **random variable**. If an outcome is mapped to the value a, we say "X takes the value a". The description of how probability is spread over the values of X, by nonnegative numbers summing to 1, is the **probability distribution** (or simply **distribution**) of X.

Illustration 3-3

In Illustration 3-2 the distribution of X is $P(X=0)=P(X=1)=1/2$.

If the possible values $x_1, x_2, \dots$ of X can be listed (countable), X is a **discrete random variable**, and

$$p(x)=\begin{cases} P(X=x_i), & x=x_i \ (i=1,2,\dots) \\ 0, & \text{otherwise} \end{cases}$$

is its **probability mass function**.

Definition 3-5 (Probability mass function of a discrete r.v.)

A function $p(x)$ is the **probability mass function** of the discrete random variable X if
(1) $0 \le p(x) \le 1$ and $\sum_{\text{all } x} p(x)=1$;
(2) $P(a<X\le b)=\sum_{a<x\le b} p(x)$.

The p.m.f. determines the distribution; with few values, list it as a table.

Example 3-7 (Popular posts)

Of an influencer's 12 posts, 4 have more than 10,000 likes. Three posts are chosen by simple random sampling; X is the number among them with more than 10,000 likes. Find the distribution of X.
Solution. There are $\binom{12}{3}=220$ equally likely choices, and $p(x)=\binom{8}{3-x}\binom{4}{x}/\binom{12}{3}$:

$$p(0)=\frac{56\times 1}{220}=\frac{14}{55}, \qquad p(1)=\frac{28\times 4}{220}=\frac{28}{55},$$
$$p(2)=\frac{8\times 6}{220}=\frac{12}{55}, \qquad p(3)=\frac{1\times 4}{220}=\frac{1}{55}.$$

If X can take every value in an interval, it is a **continuous random variable**.

Illustration 3-4: spinner

A pointer on a dial marked 0 to 1 spins freely; $X\in[0,1]$ is where it stops.

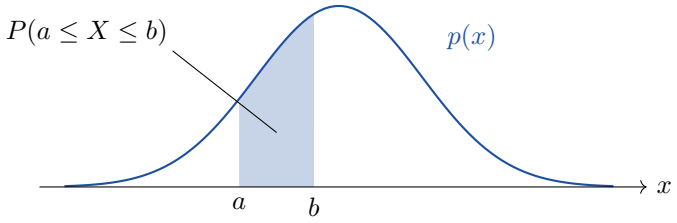
By the ratio of lengths,

$$P(a\le X\le b)=\frac{b-a}{1-0}=b-a \qquad (0\le a<b\le 1).$$

With $p=1$ on $[0,1]$ and 0 otherwise, $P(a\le X\le b)=\int_a^b p(x)\,dx$: p is the **probability density function**.

Definition 3-6 (Probability density function of a continuous r.v.)

A function $p(x)$ is the **probability density function** of the continuous random variable X if
(1) $p(x) \geq 0$ and $\int_{-\infty}^{\infty} p(x)\,dx = 1$;
(2) $P(a \leq X \leq b) = \int_a^b p(x)\,dx$ $(-\infty \leq a < b \leq \infty)$.



$P(a \leq X \leq b)$ is the area under the density over $[a, b]$ (Figure 3-5)

Theorem 3-4 (Properties of continuous random variables)

For a continuous random variable X,
(1) $P(X = c) = 0$ for every c;
(2) $P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b) = P(a < X < b)$.

Proof. (1) $\{X = c\} \subset \{c - h \leq X \leq c + h\}$, so by Theorem 3-1(3)

$$0 \leq P(X = c) \leq \int_{c-h}^{c+h} p(x)\,dx \longrightarrow 0 \qquad (h \to 0).$$

(2) The sets differ by $\{X = a\}$, $\{X = b\}$, of probability 0 by (1); apply axiom (3). ■

Example 3-8 (Finding the constant)

A continuous random variable X has density $p(x) = bx(1 - x)$ for $0 \leq x \leq 1$, and 0 for $x < 0$ or $x > 1$. Find b and $P(0 \leq X \leq \frac{2}{3})$.
Solution. $1 = \int_0^1 bx(1 - x)\,dx = b/6$, so $b = 6$. Then

$$P\Big(0 \leq X \leq \frac{2}{3}\Big) = \int_0^{2/3} 6x(1 - x)\,dx = \Big[3x^2 - 2x^3\Big]_0^{2/3} = \frac{4}{3} - \frac{16}{27} = \frac{20}{27}.$$

An observed value of a population characteristic is a random variable with the population distribution; populations are modelled by such random variables.

3.4 Expected value and its properties

Motivation. The mean and variance of Chapter 2, for random variables.

Definition 3-7 (Expected value of a random variable)

If the p.m.f. (p.d.f.) of the discrete (continuous) random variable X is $p(x)$,

$$E(X) = \begin{cases} \sum_{\text{all } x} x\,p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} x\,p(x)\,dx & \text{(continuous)} \end{cases}$$

Example 3-9 (A mean)

X has distribution $p(1) = \frac{1}{8}$, $p(2) = \frac{3}{8}$, $p(3) = \frac{3}{8}$, $p(4) = \frac{1}{8}$. Find $E(X)$.
Solution. $E(X) = 1 \cdot \frac{1}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{3}{8} + 4 \cdot \frac{1}{8} = 2.5$, the centre of the distribution graph (Figure 3-6).

$E(X)$ is the centre of mass of masses $p(x_i)$ at x_i (or density p); for a population model it is the population mean μ.

Example 3-10 (A function of X)

A coin is tossed 3 times; X is the number of heads.
(a) Find the distribution of $Y = X(3 - X)$.
(b) Find $E(Y)$.
Solution.
(a) $p_X(0) = \frac{1}{8}$, $p_X(1) = \frac{3}{8}$, $p_X(2) = \frac{3}{8}$, $p_X(3) = \frac{1}{8}$. $Y = 0$ when $X \in \{0, 3\}$ and $Y = 2$ when $X \in \{1, 2\}$, so $p_Y(0) = \frac{1}{4}$, $p_Y(2) = \frac{3}{4}$.
(b) $E[X(3 - X)] = E(Y) = 0 \cdot \frac{1}{4} + 2 \cdot \frac{3}{4} = \frac{3}{2}$.

$E(Y)$ can be computed without the distribution of Y:

$$\begin{aligned} E[X(3-X)] &= 0(3-0)p_X(0) + 1(3-1)p_X(1) + 2(3-2)p_X(2) + 3(3-3)p_X(3) \\ &= 0 \cdot \tfrac{1}{8} + 2 \cdot \tfrac{3}{8} + 2 \cdot \tfrac{3}{8} + 0 \cdot \tfrac{1}{8} = \tfrac{3}{2}. \end{aligned}$$

Theorem 3-5 (Expectation of a function of a random variable)

If the p.m.f. (p.d.f.) of the discrete (continuous) random variable X is $p(x)$, then for any function g

$$E[g(X)] = \begin{cases} \sum_{\text{all } x} g(x)\,p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} g(x)\,p(x)\,dx & \text{(continuous)} \end{cases}$$

Proof. [Supplement]

Step 1 (discrete case). $\{Y = y\} = \bigsqcup_{x:\,g(x)=y}\{X = x\}$, so $p_Y(y) = \sum_{x:\,g(x)=y} p(x)$ and

$$E(Y) = \sum_y y\,p_Y(y) = \sum_y \sum_{x:\,g(x)=y} g(x)\,p(x) = \sum_{\text{all } x} g(x)\,p(x),$$

each x in exactly one group (regrouping valid under absolute convergence, the condition for $E(Y)$ to exist).

Step 2 (continuous case, monotone g). For differentiable strictly increasing g with inverse h, $P(Y \le y) = \int_{-\infty}^{h(y)} p(x)\,dx$, so $p_Y(y) = p(h(y))h'(y)$ and, with $x = h(y)$,

$$E(Y) = \int y\,p(h(y))h'(y)\,dy = \int g(x)\,p(x)\,dx.$$

For decreasing g, $p_Y(y) = -p(h(y))h'(y)$ and the limits reverse; same result.

Step 3 (general g). Split into intervals of monotonicity and apply Step 2; for arbitrary g this is the measure-theoretic change of variables, which we accept. ■

Example 3-11 (Waiting for a bus)

Buses leave a stop at 0, 15, 30 and 45 minutes past each hour. A person arrives at a random time. Find the expected waiting time.

Solution. The arrival time X (minutes past the hour) has density $\frac{1}{60}$ on $(0, 60]$; the wait is

$$g(x) = 15 - x, \quad 30 - x, \quad 45 - x, \quad 60 - x \qquad \text{on } (0, 15],\ (15, 30],\ (30, 45],\ (45, 60].$$

By Theorem 3-5,

$$\begin{aligned} E[g(X)] &= \frac{1}{60}\Big\{ \int_0^{15} (15 - x)\,dx + \int_{15}^{30} (30 - x)\,dx + \int_{30}^{45} (45 - x)\,dx + \int_{45}^{60} (60 - x)\,dx \Big\} \\ &= \frac{1}{60} \times 4 \times 112.5 = 7.5, \end{aligned}$$

i.e. 7 minutes 30 seconds.

Theorem 3-6 (Properties of expected value)

(1) $E(aX + b) = aE(X) + b$ (a, b constants).
(2) $E[c_1 g_1(X) + c_2 g_2(X)] = c_1 E[g_1(X)] + c_2 E[g_2(X)]$ (c_1, c_2 constants, g_1, g_2 any functions).

Proof. [Supplement] (2) By Theorem 3-5, in the discrete case

$$E[c_1 g_1(X) + c_2 g_2(X)] = \sum_x \{c_1 g_1(x) + c_2 g_2(x)\} p(x) = c_1 \sum_x g_1(x) p(x) + c_2 \sum_x g_2(x) p(x),$$

likewise with integrals.

(1) Take $g_1(x) = x$, $g_2 = 1$, $c_1 = a$, $c_2 = b$; $E(1) = 1$. ■

Definition 3-8 (Variance and standard deviation of a random variable)

If X has mean μ and p.m.f. (or p.d.f.) $p(x)$,

(1) $\mathrm{Var}(X) = E[(X - \mu)^2] = \begin{cases} \sum_{\text{all } x} (x - \mu)^2 p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} (x - \mu)^2 p(x)\,dx & \text{(continuous)} \end{cases}$

(2) $\mathrm{sd}(X) = \sqrt{\mathrm{Var}(X)} = \sqrt{E[(X - \mu)^2]}$.

For a population model these are σ^2 and σ.

Theorem 3-7 (Shortcut formula for the variance)

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2.$$

Proof. By Theorem 3-6(2) with $\mu = E(X)$,

$$\mathrm{Var}(X) = E[(X - \mu)^2] = E(X^2 - 2\mu X + \mu^2) = E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - \mu^2.$$

■

Example 3-12 (Two coin tosses)

X is the number of heads in 2 tosses of a coin. Find the mean and standard deviation of X.
Solution.

x	0	1	2	Total
$p(x)$	1/4	1/2	1/4	1
$xp(x)$	0	1/2	1/2	1
$x^2p(x)$	0	1/2	1	3/2

So $E(X) = 1$, $E(X^2) = \frac{3}{2}$, $\mathrm{Var}(X) = \frac{3}{2} - 1^2 = \frac{1}{2}$ and $\mathrm{sd}(X) = 1/\sqrt{2}$.

Example 3-13 (Spinner)

For the spinner of Illustration 3-4, find the mean and standard deviation of X.
Solution. $p(x) = 1$ on $[0, 1]$, so

$$E(X) = \int_0^1 x\,dx = \frac{1}{2}, \qquad E(X^2) = \int_0^1 x^2\,dx = \frac{1}{3},$$

$$\mathrm{Var}(X) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}, \qquad \mathrm{sd}(X) = \frac{1}{\sqrt{12}}.$$

Theorem 3-8 (Property of the variance)

$$\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X) \qquad (a, b \text{ constants}).$$

Proof. [Supplement] Let $\mu = E(X)$. By Theorem 3-6(1), $E(aX+b) = a\mu+b$, so $(aX+b) - E(aX+b) = a(X-\mu)$ and

$$\mathrm{Var}(aX + b) = E[a^2(X-\mu)^2] = a^2E[(X-\mu)^2] = a^2\,\mathrm{Var}(X).$$

■

Subtracting the mean and dividing by the standard deviation is **standardization**:

$$Z = \frac{X - E(X)}{\mathrm{sd}(X)}.$$

By Theorems 3-6 and 3-8 (with $a = 1/\mathrm{sd}(X)$, $b = -E(X)/\mathrm{sd}(X)$), $E(Z) = 0$ and $\mathrm{Var}(Z) = \mathrm{Var}(X)/\mathrm{sd}(X)^2 = 1$.

3.5 Joint distribution of two random variables

Motivation. Related variables (height and weight) need a joint distribution; we treat the discrete case.

Illustration 3-5: same marginals, different relations

Toss coins A, B, C: $X = 1$ if A is heads; $Y = 1$ if A, B are heads; $Z = 1$ if B, C are heads (else 0). Y and Z both have $P(0) = \frac{3}{4}$, $P(1) = \frac{1}{4}$, but $X = 0$ forces $Y = 0$, whereas given $X = 0$, Z is 0 or 1 with probabilities $\frac{3}{4}, \frac{1}{4}$.

Definition 3-9 (Joint distribution of two random variables)

The description of how probability is spread over all ordered pairs of values of X and Y, by nonnegative numbers summing to 1, is the **joint probability distribution** (or **joint distribution**) of X and Y.

For discrete X, Y with values $x_1, x_2, \dots$ and $y_1, y_2, \dots$,

$$p(x,y) = \begin{cases} P(X = x,\ Y = y), & (x,y) = (x_i, y_j)\ (i,j = 1,2,\dots) \\ 0, & \text{otherwise} \end{cases}$$

is the **joint probability mass function**.

Definition 3-10 (Joint probability mass function)

$p(x,y)$ is the **joint probability mass function** of the discrete random variables X and Y if
(1) $0 \le p(x,y) \le 1$ and $\sum_{\text{all } x}\sum_{\text{all } y} p(x,y) = 1$;
(2) $P(a < X \le b,\ c < Y \le d) = \sum_{a<x\le b}\sum_{c<y\le d} p(x,y)$.

Example 3-14 (Joint distributions from three coins)

For the experiment of Illustration 3-5, find the joint distribution of
(a) X and Y; (b) X and Z.
Solution. Listing coins A, B, C:

Outcome	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
X	1	1	1	1	0	0	0	0
Y	1	1	0	0	0	0	0	0
Z	1	0	0	0	1	0	0	0

Each outcome has probability $\frac{1}{8}$; collecting them,

(a) X and Y

$y\backslash x$	0	1	Row sum
0	2/4	1/4	3/4
1	0	1/4	1/4
Col. sum	2/4	2/4	1

(b) X and Z

$z\backslash x$	0	1	Row sum
0	3/8	3/8	6/8
1	1/8	1/8	2/8
Col. sum	4/8	4/8	1

Row and column sums give the **marginal probability mass functions**

$$p_1(x) = \sum_{\text{all } y} p(x,y), \qquad p_2(y) = \sum_{\text{all } x} p(x,y),$$

since $\{X = x\} = \bigsqcup_y \{X = x, Y = y\}$.

Example 3-15 (Two games on one die)

A game pays according to one die roll. A receives 100 won for 1 or 2, 200 won for 3 or 4, 300 won for 5 or 6; B receives 100 won for an even number and (face $\times 100$) won for an odd number. Find the distribution of Z, the total received by A and B.
Solution. With X, Y the amounts of A, B (e.g. $(100, 100)$ for faces 1, 2):

$x\backslash y$	100	300	500	Row sum
100	2/6	0	0	2/6
200	1/6	1/6	0	2/6
300	1/6	0	1/6	2/6
Col. sum	4/6	1/6	1/6	1

$Z = X + Y$ takes the values 200, 300, 400, 500, 800 with probabilities $\frac{2}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$.

Theorem 3-9 (Expectation of a function of two random variables)

Let X, Y be discrete with joint p.m.f. $p(x, y)$. For any functions g, g_1, g_2:
(1) $E[g(X,Y)] = \sum_{\text{all } x} \sum_{\text{all } y} g(x,y)\, p(x,y)$;
(2) $E[c_1 g_1(X,Y) + c_2 g_2(X,Y)] = c_1 E[g_1(X,Y)] + c_2 E[g_2(X,Y)]$ (c_1, c_2 constants).

Proof. [Supplement] (1) For $W = g(X,Y)$, $p_W(w) = \sum_{(x,y):\, g(x,y)=w} p(x,y)$, so

$$E(W) = \sum_w w\, p_W(w) = \sum_w \sum_{(x,y):\, g(x,y)=w} g(x,y)\, p(x,y) = \sum_x \sum_y g(x,y)\, p(x,y),$$

(regrouping by absolute convergence).

(2) Apply (1) to $c_1 g_1 + c_2 g_2$ and split:

$$\sum_x \sum_y \{c_1 g_1 + c_2 g_2\}(x,y)\, p(x,y) = c_1 \sum_x \sum_y g_1(x,y) p(x,y) + c_2 \sum_x \sum_y g_2(x,y) p(x,y).$$

■

Definition 3-11 (Independence of two random variables)

Let $p(x,y)$ be the joint p.m.f. (joint p.d.f.) of discrete (continuous) X, Y and p_1, p_2 the marginals. X and Y are **independent** if for all x, y

$$p(x,y) = p_1(x)\, p_2(y).$$

Otherwise they are **dependent**.

Example 3-16 (Independence check)

A coin is tossed three times; X is the number of heads in the first two tosses and Y the number of heads on the third toss. Are X and Y independent?
Solution. The joint distribution is

$x\backslash y$	0	1	Row sum
0	1/8	1/8	1/4
1	2/8	2/8	1/2
2	1/8	1/8	1/4
Col. sum	1/2	1/2	1

Every cell is the product of its margins (e.g. $p_1(0)p_2(0) = \frac{1}{8} = p(0,0)$), so X and Y are independent.

3.6 Covariance and correlation

Motivation. The mean product of standardized deviations measures linear association.

Let X, Y have means μ_1, μ_2 and standard deviations σ_1, σ_2.

Definition 3-12 (Covariance and correlation coefficient)

(1) $\mathrm{Cov}(X,Y) = E[(X-\mu_1)(Y-\mu_2)]$.

(2) $\mathrm{Corr}(X,Y) = E\Big[\Big(\frac{X-\mu_1}{\sigma_1}\Big)\Big(\frac{Y-\mu_2}{\sigma_2}\Big)\Big] = \frac{\mathrm{Cov}(X,Y)}{\mathrm{sd}(X)\,\mathrm{sd}(Y)}$.

Proposition (Range of the correlation)

$-1 \le \mathrm{Corr}(X,Y) \le 1$; and if $Y = aX + b$ with $a \neq 0$, then $\mathrm{Corr}(X,Y) = 1$ for $a > 0$ and -1 for $a < 0$.

Proof. [Supplement]

Step 1. Let $U = (X-\mu_1)/\sigma_1$, $V = (Y-\mu_2)/\sigma_2$ and $\rho = \mathrm{Corr}(X,Y) = E(UV)$. Then $E(U^2) = E(V^2) = 1$, and by Theorem 3-9(2)

$$0 \le E[(U-\rho V)^2] = E(U^2) - 2\rho E(UV) + \rho^2 E(V^2) = 1 - \rho^2,$$

so $\rho^2 \le 1$.

Step 2. If $Y = aX + b$, then $\mu_2 = a\mu_1 + b$ and $\sigma_2 = |a|\sigma_1$ (Theorem 3-8), so $V = \frac{a}{|a|}U$ and $\rho = \frac{a}{|a|}E(U^2) = \pm 1$ according to the sign of a. ■

Theorem 3-10 (Shortcut formula for the covariance)

$$\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y).$$

Proof. By Theorem 3-9(2),

$$\begin{aligned}\mathrm{Cov}(X,Y) &= E[(X-\mu_1)(Y-\mu_2)] = E(XY - \mu_2 X - \mu_1 Y + \mu_1\mu_2)\\ &= E(XY) - \mu_2 E(X) - \mu_1 E(Y) + \mu_1\mu_2 = E(XY) - \mu_1\mu_2.\end{aligned}$$

■

Example 3-17 (Covariance and correlation)

X and Y have the joint distribution below. Find $\mathrm{Cov}(X,Y)$ and $\mathrm{Corr}(X,Y)$.

$x \backslash y$	0	1	2	3	Row sum
0	0.05	0.05	0.10	0.00	0.20
1	0.05	0.10	0.25	0.10	0.50
2	0.00	0.15	0.10	0.05	0.30
Col. sum	0.10	0.30	0.45	0.15	1.00

Solution. From the margins,

$$\mu_1 = 1.1, \qquad \sigma_1^2 = 1.7 - (1.1)^2 = 0.49, \qquad \sigma_1 = 0.7,$$

$$\mu_2 = 1.65, \qquad \sigma_2^2 = 3.45 - (1.65)^2 = 0.7275, \qquad \sigma_2 = \sqrt{0.7275} = 0.8529.$$

Summing $xy\,p(x,y)$ over the cells with $x, y, p \neq 0$,

$$E(XY) = 0.1 + 0.5 + 0.3 + 0.3 + 0.4 + 0.3 = 1.9.$$

Hence $\mathrm{Cov}(X,Y) = 1.90 - 1.65 \times 1.1 = 0.085$ and

$$\mathrm{Corr}(X,Y) = \frac{0.085}{0.70 \times 0.8529} = 0.1424.$$

Theorem 3-11 (Properties of covariance and correlation)

For constants a, b, c, d:
(1) $\mathrm{Cov}(aX+b,\, cY+d) = ac\,\mathrm{Cov}(X,Y)$;
(2) $\mathrm{Corr}(aX+b,\, cY+d) = \mathrm{Corr}(X,Y)$ if $ac > 0$, and $-\mathrm{Corr}(X,Y)$ if $ac < 0$.

Proof. [Supplement] (1) The centred variables are $a(X-\mu_1)$ and $c(Y-\mu_2)$, so

$$\mathrm{Cov}(aX+b, cY+d) = E[ac(X-\mu_1)(Y-\mu_2)] = ac\,\mathrm{Cov}(X,Y).$$

(2) By Theorem 3-8, $\mathrm{sd}(aX+b) = |a|\,\mathrm{sd}(X)$ and $\mathrm{sd}(cY+d) = |c|\,\mathrm{sd}(Y)$. With (1),

$$\mathrm{Corr}(aX+b, cY+d) = \frac{ac\,\mathrm{Cov}(X,Y)}{|a||c|\,\mathrm{sd}(X)\,\mathrm{sd}(Y)} = \frac{ac}{|ac|}\,\mathrm{Corr}(X,Y),$$

with $ac/|ac| = \pm 1$. ■

Theorem 3-12 (Variance of a sum of two random variables)

(1) $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$.
(2) $\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X,Y)$.

Proof. [Supplement] $(X \pm Y) - E(X \pm Y) = (X-\mu_1) \pm (Y-\mu_2)$; square and take expectations (Theorem 3-9(2)):

$$\mathrm{Var}(X \pm Y) = E[(X-\mu_1)^2] + E[(Y-\mu_2)^2] \pm 2E[(X-\mu_1)(Y-\mu_2)],$$

which is $\mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\,\mathrm{Cov}(X,Y)$. ■

Theorem 3-13 (Independent random variables)

If X and Y are independent, then
(1) $E(XY) = E(X)E(Y)$;
(2) $\mathrm{Cov}(X,Y) = 0$ and $\mathrm{Corr}(X,Y) = 0$;
(3) $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

Proof. (1) By Theorem 3-9(1) and $p(x,y) = p_1(x)p_2(y)$,

$$E(XY) = \sum_x \sum_y xy\,p(x,y) = \sum_x \sum_y xy\,p_1(x)p_2(y) = \sum_x x\,p_1(x) \sum_y y\,p_2(y) = E(X)E(Y).$$

(2) By Theorem 3-10, $\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y) = 0$, hence also $\mathrm{Corr}(X,Y) = 0$.

(3) Insert $\mathrm{Cov}(X,Y) = 0$ into Theorem 3-12. ■

Chapter 4. Sampling Distributions

4.1 The Bernoulli and normal distributions as population distributions

Motivation. An estimator is judged by its distribution, driven by the population model: Bernoulli or normal.

4.1.1 Bernoulli trials

An experiment with two outcomes (pass/fail, cured/not), labelled success s and failure f, is a **Bernoulli trial**.

Definition 4-1 (Bernoulli trial)

(a) A Bernoulli trial has exactly two outcomes, success s and failure f, so its sample space is the two-point set $S = \{s, f\}$.
(b) The success probability is $p = P(\{s\})$ and the failure probability is $q = P(\{f\}) = 1 - p$, where $0 \le p \le 1$.

Definition 4-2 (Bernoulli random variable)

$X(s) = 1$, $X(f) = 0$ on $S = \{s, f\}$ is a **Bernoulli random variable**, with the **Bernoulli distribution** $P(X = 1) = p$, $P(X = 0) = 1 - p$.

The Bernoulli distribution models a population whose characteristic is dichotomous (e.g. patients "cured"/"not cured" coded 1/0).

Example 4-1 (Mean and variance of the Bernoulli distribution)

Find the mean and variance of the Bernoulli distribution.
Solution. $E(X) = 1 \cdot p + 0 \cdot (1 - p) = p$ and $E(X^2) = 1^2 \cdot p + 0^2 \cdot (1 - p) = p$. Hence

$$\operatorname{Var}(X) = E(X^2) - \{E(X)\}^2 = p - p^2 = p(1 - p).$$

Bernoulli trials are **independent** if the result of one trial does not affect the others, and **repeated** if the success probability is the same in every trial.

Illustration 4-1: sampling with replacement

Two of 10 cards (3 winning) are drawn with replacement; X_1, X_2 are the Bernoulli variables of the draws. Then

$$P(X_2 = 1) = P(X_2 = 1 \mid X_1 = 1) = P(X_2 = 1 \mid X_1 = 0) = \tfrac{3}{10},$$

and $P(X_1 = 1) = \frac{3}{10}$, so X_1, X_2 are independent and identically distributed; likewise for n draws with replacement.

Illustration 4-2: sampling without replacement

Draw the two cards of Illustration 4-1 *without* replacement. Then $P(X_1 = 1) = \frac{3}{10}$, $P(X_2 = 1 \mid X_1 = 1) = \frac{2}{9}$, $P(X_2 = 1 \mid X_1 = 0) = \frac{3}{9}$, and by the law of total probability

$$P(X_2 = 1) = \tfrac{2}{9} \cdot \tfrac{3}{10} + \tfrac{3}{9} \cdot \tfrac{7}{10} = \tfrac{3}{10}.$$

So each draw wins with probability $\frac{3}{10}$, but $P(X_2 = 1) \neq P(X_2 = 1 \mid X_1 = 1)$: the trials are not independent.

Illustration 4-3: general finite deck

With N cards of which M are winning, sampling two without replacement gives

$$P(X_1 = 1) = P(X_2 = 1) = \frac{M}{N}, \qquad P(X_2 = 1 \mid X_1 = 1) = \frac{M-1}{N-1},$$

so again the draws are identically distributed but dependent. If N (and M) is large relative to the sample, $P(X_2 = 1) \approx P(X_2 = 1 \mid X_1 = 1)$ and the draws are nearly independent.

Sampling with vs. without replacement

From a finite population, sampling without replacement is practically indistinguishable from sampling with replacement when the population size N is large compared with the sample size.

4.1.2 The normal distribution

The normal distribution is the leading model for an infinite population whose characteristic is continuous. Its density curve

(a) is symmetric about a single point, the population mean μ;

(b) is highest at the centre μ;

(c) decreases as one moves away from μ;

(d) has inflection points at $\mu \pm \sigma$.

Definition 4-3 (Normal density with mean μ and standard deviation σ)

$$p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \qquad -\infty < x < \infty.$$

We write $X \sim N(\mu, \sigma^2)$.

μ shifts the curve; smaller σ makes it narrower and taller. An observation from a normal population has $X \sim N(\mu, \sigma^2)$.

Theorem 4-1 (Properties of the normal distribution 1)

If $X \sim N(\mu, \sigma^2)$, then for any constants $a \neq 0$ and b,

$$aX + b \sim N(a\mu + b,\ a^2\sigma^2).$$

Proof. [Supplement] We use moment generating functions (mgf), $M_X(t) = E(e^{tX})$; an mgf finite near 0 determines the distribution.

Step 1. For $X \sim N(\mu, \sigma^2)$, substituting $x = \mu + \sigma z$ and completing the square $tz\sigma - \frac{z^2}{2} = -\frac{1}{2}(z - \sigma t)^2 + \frac{\sigma^2 t^2}{2}$,

$$M_X(t) = e^{\mu t} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\sigma t z - z^2/2}\, dz = e^{\mu t + \sigma^2 t^2/2} \int_{-\infty}^{\infty} \frac{e^{-(z-\sigma t)^2/2}}{\sqrt{2\pi}}\, dz = e^{\mu t + \sigma^2 t^2/2}.$$

Step 2. Hence $M_{aX+b}(t) = e^{bt} M_X(at) = e^{(a\mu+b)t + a^2\sigma^2 t^2/2}$, which is the mgf of $N(a\mu + b, a^2\sigma^2)$. By uniqueness of mgfs the claim follows. ∎

(If $a = 0$ then $aX + b = b$ is constant, a "normal distribution with variance 0".)

In particular, taking $a = 1/\sigma$, $b = -\mu/\sigma$,

$$X \sim N(\mu, \sigma^2) \implies Z = \frac{X - \mu}{\sigma} \sim N(0, 1). \tag{4.1}$$

$N(0, 1)$ is the **standard normal distribution**; a standard normal variable is usually written Z. Appendix C tabulates $P(Z \le z)$.

Definition (Probabilities from the standard normal table)

(a) $P(a \le Z \le b) = P(Z \le b) - P(Z \le a)$.
(b) $P(Z \ge z) = 1 - P(Z < z)$.

Example 4-2 (Two tails)

Find $P(Z < -1.9 \text{ or } Z > 2.1)$.
Solution. The events are disjoint, so the probability is $P(Z < -1.9) + P(Z > 2.1) = 0.0287 + (1 - 0.9821) = 0.0287 + 0.0179 = 0.0466$.

Example 4-3 (A quantile)

Find z with $P(Z > z) = 0.025$.
Solution. $P(Z \le z) = 1 - 0.025 = 0.975$, so the table gives $z = 1.96$.

Definition (Upper quantile z_α)

For $0 < \alpha < 1$, z_α is the number with $P(Z > z_\alpha) = \alpha$, $Z \sim N(0, 1)$; it is the $100(1 - \alpha)$th percentile (quantile) of $N(0, 1)$. Frequently used values:

$$z_{0.005} = 2.58, \qquad z_{0.025} = 1.96, \qquad z_{0.05} = 1.645.$$

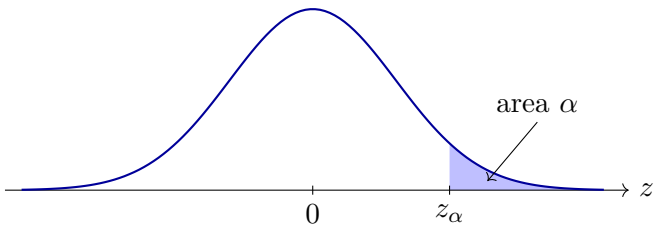


Figure 1: The upper quantile z_α of $N(0, 1)$.

If $X \sim N(\mu, \sigma^2)$, standardizing with (4.1) gives

$$P(X \le b) = P\Big(Z \le \frac{b - \mu}{\sigma}\Big),$$

$$P(a \le X \le b) = P\Big(\frac{a-\mu}{\sigma} \le Z \le \frac{b-\mu}{\sigma}\Big).$$

Example 4-4 (Proportion of A grades)

Grades in a statistics class are approximately $N(60, 10^2)$. Students scoring 75 or more get an A. Approximate the proportion of A grades.
Solution. $P(X \ge 75) = P(\frac{X-60}{10} \ge \frac{75-60}{10}) \approx P(Z \ge 1.5) = 0.0668$.

Theorem 4-2 (Properties of the normal distribution 2)

If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent, then for any constants a_1, a_2,

$$a_1X_1 + a_2X_2 \sim N(a_1\mu_1 + a_2\mu_2,\ a_1^2\sigma_1^2 + a_2^2\sigma_2^2).$$

Proof. [Supplement] By independence the mgf of a sum is the product of the mgfs. Using Step 1 of the proof of Theorem 4-1,

$$\begin{aligned} M_{a_1X_1+a_2X_2}(t) &= M_{X_1}(a_1t)\, M_{X_2}(a_2t) = e^{a_1\mu_1 t + a_1^2\sigma_1^2t^2/2}\, e^{a_2\mu_2 t + a_2^2\sigma_2^2t^2/2} \\ &= \exp\Big\{(a_1\mu_1 + a_2\mu_2)t + \tfrac{1}{2}(a_1^2\sigma_1^2 + a_2^2\sigma_2^2)t^2\Big\}, \end{aligned}$$

the mgf of the stated normal distribution. By induction the same holds for any finite number of independent normal variables: $\sum a_iX_i \sim N(\sum a_i\mu_i, \sum a_i^2\sigma_i^2)$. ■

Example 4-5 (Calories burned)

Calories burned running are approximately $N(200, 15^2)$ and cycling $N(80, 5^2)$, independently. Find the probability that the daily total is at least 300 kcal.
Solution. The total S is approximately normal with mean $200 + 80 = 280$ and variance $15^2 + 5^2 = 250$. So

$$P(S \ge 300) = P\Big(\frac{S-280}{\sqrt{250}} \ge \frac{300-280}{\sqrt{250}}\Big) \approx P(Z \ge 1.27) = 1 - 0.898 = 0.102.$$

4.2 Sampling distributions

Motivation. An estimation method is judged over *all* possible samples, by the distribution of the statistic.

4.2.1 Sampling distribution of a statistic

A survey estimates p by the sample proportion; accuracy is the probability, over random samples, that the error is at most a given amount.

Illustration: all samples of size 3

Population $\{\text{yes}, \text{no}, \text{yes}, \text{no}, \text{yes}\}$ (true $p = 0.6$), sample of size 3 without replacement.

Sample	Probability	Sample proportion	Error
{yes, yes, yes}	1/10	1	0.4
{yes, yes, no}	6/10	2/3	0.067
{yes, no, no}	3/10	1/3	−0.267

So $P(|\text{error}| \le 0.1) = 0.6$ and $P(|\text{error}| \le 0.3) = 0.9$.

Definition (Statistic and sampling distribution)

A **statistic** is a random variable computed from the sample. The probability distribution of a statistic is its **sampling distribution**; it quantifies how reliable an estimate is.

Example 4-6 (Sample mean from a small population)

From the finite population $\{2, 2, 4, 5\}$ draw a sample of size 2 without replacement. Find the distribution of the sample mean.
Solution. The six equally likely samples (probability $\frac{1}{6}$ each) and their means are

Sample	$\{2,2\}$	$\{2,4\}$	$\{2,5\}$	$\{2,4\}$	$\{2,5\}$	$\{4,5\}$
Mean	2	3	3.5	3	3.5	4.5

Hence $P(\bar{X} = 2) = \frac{1}{6}$, $P(\bar{X} = 3) = \frac{1}{3}$, $P(\bar{X} = 3.5) = \frac{1}{3}$, $P(\bar{X} = 4.5) = \frac{1}{6}$.

The sampling distribution depends on both the population distribution and the sampling method (with or without replacement).

4.2.2 Random samples

A simple random sample from a finite population is a **random sample** $X_1, \ldots, X_n$ (categorical answers coded 1/0). With replacement, $X_1, \ldots, X_n$

(a) each have the population distribution, and

(b) are mutually independent.

When the population is large, a random sample without replacement nearly has properties (a) and (b); this motivates the definition for infinite populations.

Definition 4-4 (Random sample)

(a) Finite population: a sample drawn by simple random sampling without replacement.
(b) Infinite population: random variables $X_1, \ldots, X_n$ such that (i) each X_i has the population distribution and (ii) $X_1, \ldots, X_n$ are independent.

Large finite populations are treated as infinite; from Chapter 5 on, all inference assumes a random sample from an infinite population.

4.3 The binomial distribution

Motivation. The number of successes in a random sample has the binomial distribution.

For a 0/1 population with proportion p, a random sample is n independent Bernoulli(p) trials, with number of successes

$$X = X_1 + \cdots + X_n.$$

Its distribution is the **binomial distribution** $B(n, p)$; $B(1, p)$ is the Bernoulli distribution.

Definition 4-5 (Binomial distribution)

With n trials and success probability p, the probability mass function is

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n.$$

$\binom{n}{x}$ is the coefficient of $a^{n-x}b^x$ in $(a+b)^n$ (Pascal's triangle). $B(n, 0.5)$ is symmetric; $p < 0.5$ gives **right skew**, $p > 0.5$ **left skew**.

Example 4-7 (Guessing on a multiple-choice test)

A test has 20 five-choice questions and every answer is guessed at random. Find the probability of
(a) no correct answer;
(b) 4 to 6 correct answers;
(c) at least 2 correct answers.
Solution. The number correct is $X \sim B(20, 0.2)$.
(a) $P(X = 0) = \binom{20}{0} 0.2^0\, 0.8^{20} = 0.012$.
(b) $P(4 \le X \le 6) = \sum_{x=4}^{6} \binom{20}{x} 0.2^x\, 0.8^{20-x} = 0.502$.
(c) $P(X \ge 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0.8^{20} - 20(0.2)(0.8^{19}) = 0.931$.

Example 4-8 (On-time deliveries)

A courier delivers on time 95% of the time. In a random sample of 20 deliveries, find the probability that the sample on-time rate is at most 90%.
Solution. The number of late deliveries X is approximately $B(20, 0.05)$. An on-time rate $\le 90\%$ means a late rate $X/20 \ge 0.1$, so

$$P\Big(\frac{X}{20} \ge 0.1\Big) = P(X \ge 2) = 1 - P(X = 0) - P(X = 1) = 0.264.$$

Theorem 4-3 (Mean and variance of $B(n, p)$)

If $X \sim B(n, p)$, then
(a) $E(X) = np$;
(b) $\mathrm{Var}(X) = np(1-p)$.

Proof. Write $X = X_1 + \cdots + X_n$ with X_i independent Bernoulli(p), so $E(X_i) = p$ and $\mathrm{Var}(X_i) = p(1-p)$ (Example 4-1).

(a) By linearity of expectation, $E(X) = \sum_i E(X_i) = np$.

(b) For independent variables the covariances vanish, so $\mathrm{Var}(X) = \sum_i \mathrm{Var}(X_i) = np(1-p)$. ■

4.4 The distribution of the sample mean

Motivation. The sample mean is exactly normal for normal populations and approximately normal for large n (CLT).

For a 0/1 population, $\bar{X}$ is the sample proportion (§4.3). Let $X_1, \ldots, X_n$ be a random sample from an infinite population with mean μ and variance σ^2, and

$$\bar{X} = \frac{1}{n}(X_1 + \cdots + X_n).$$

Theorem 4-4 (Distribution of the sample mean 1)

For a random sample of size n from a population with mean μ and variance σ^2,
(a) $E(\bar{X}) = \mu$;
(b) $\mathrm{Var}(\bar{X}) = \sigma^2/n$;
(c) $\mathrm{sd}(\bar{X}) = \sigma/\sqrt{n}$.

Proof.

(a) Since $E(X_i) = \mu$, $E(\bar{X}) = \frac{1}{n}\{E(X_1) + \cdots + E(X_n)\} = \frac{1}{n} \cdot n\mu = \mu$.

(b) Since the X_i are independent with $\mathrm{Var}(X_i) = \sigma^2$, $\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\{\mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n)\} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$.

(c) Take square roots in (b). ■

Finite population

For a sample of size n drawn without replacement from a finite population of size N with mean μ and variance σ^2,

$$E(\bar{X}) = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{N-n}{N-1} \cdot \frac{\sigma^2}{n}.$$

Thus $\bar{X}$ is centred at μ and concentrates around μ as n grows.

Theorem 4-5 (Distribution of the sample mean 2)

If the population is $N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \sigma^2/n)$.

Proof. By the extension of Theorem 4-2 to n variables, $\bar{X} = \sum_{i=1}^{n} \frac{1}{n} X_i$ is normal with mean $\sum \frac{1}{n}\mu = \mu$ and variance $\sum \frac{1}{n^2}\sigma^2 = \sigma^2/n$. ■

Theorem 4-6 (Distribution of the sample mean 3: central limit theorem)

For a random sample of size n from any population with mean μ and variance σ^2, when n is large $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \dot{\sim} N(0, 1).$$

Proof. [Supplement] We prove the precise statement $P(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le z) \to P(Z \le z)$ for every z, under the extra assumption that the mgf $M(t) = E(e^{t(X_1-\mu)/\sigma})$ is finite for $|t| < h$. We use the *continuity*

theorem: if $M_{W_n}(t) \to M_W(t)$ for all $|t| < h$, then the cdf of W_n converges to that of W at every continuity point.

Step 1. Let $Y_i = (X_i - \mu)/\sigma$, so the Y_i are i.i.d. with $E(Y_i) = 0$, $E(Y_i^2) = 1$, and

$$W_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Y_i, \qquad M_{W_n}(t) = \Big\{ M\Big(\frac{t}{\sqrt{n}}\Big) \Big\}^n .$$

Step 2. By Taylor's theorem, $M(s) = 1 + M'(0)s + \tfrac{1}{2}M''(0)s^2 + r(s)$ with $r(s)/s^2 \to 0$ as $s \to 0$. Here $M'(0) = E(Y_1) = 0$ and $M''(0) = E(Y_1^2) = 1$, so

$$M\Big(\frac{t}{\sqrt{n}}\Big) = 1 + \frac{t^2}{2n} + r\Big(\frac{t}{\sqrt{n}}\Big), \qquad n\, r\Big(\frac{t}{\sqrt{n}}\Big) = t^2 \cdot \frac{r(t/\sqrt{n})}{(t/\sqrt{n})^2} \to 0.$$

Step 3. Hence $M_{W_n}(t) = \big(1 + \frac{a_n}{n}\big)^n$ with $a_n = \frac{t^2}{2} + n\, r(t/\sqrt{n}) \to \frac{t^2}{2}$, and since $(1 + a_n/n)^n \to e^a$ whenever $a_n \to a$, $M_{W_n}(t) \to e^{t^2/2}$, the mgf of $N(0,1)$.

Step 4. By the continuity theorem, and since the cdf of $N(0,1)$ is continuous, $P(W_n \le z) \to P(Z \le z)$ for every z. (Without the mgf assumption the same argument works with characteristic functions $E(e^{itY})$.) ■

The striking point is that the conclusion holds whatever the shape of the population; this is why the normal distribution is central to inference.

Illustration 4-4: the CLT in action

From the uniform population on $0, 1, \ldots, 9$ ($\mu = 4.5$, $\sigma^2 = 8.25$), 100 samples of size 5 give a roughly bell-shaped histogram of $\bar{X}$ centred near 4.5 (relative frequencies 0.29, 0.27 in $[4,5)$, $[5,6)$, tapering to 0.03, 0.07), even for $n = 5$.

How large n must be depends on how far the population is from normal. For a normal population $\bar{X}$ is exactly normal; otherwise $n > 30$ usually gives a satisfactory normal approximation.

Example 4-9 (Calories of a new drink)

A new drink has mean 168 kcal and standard deviation 6 kcal. For a random sample of 100 drinks, find $P(167 \le \bar{X} \le 169)$.

Solution. With $n = 100$ the CLT applies; $\bar{X}$ has mean 168 and standard deviation $6/\sqrt{100} = 0.6$. So

$$\begin{aligned} P(167 \le \bar{X} \le 169) &\approx P\Big(\frac{167 - 168}{0.6} \le Z \le \frac{169 - 168}{0.6}\Big) \\ &= P(-1.67 \le Z \le 1.67) = 0.9525 - 0.0475 = 0.9050. \end{aligned}$$

4.4.1 Normal approximation to the binomial

For a random sample from $B(1,p)$, $X = \sum X_i \sim B(n,p)$, with mean p, variance $p(1-p)$ and $\bar{X} = \hat{p} = X/n$. The CLT gives:

Definition (Normal approximation to the binomial)

If $X \sim B(n,p)$ and n is large (rule of thumb: $np \geq 5$ and $n(1-p) \geq 5$), then X is approximately $N(np, np(1-p))$:

$$\frac{X - np}{\sqrt{np(1-p)}} = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \dot{\sim} N(0,1).$$

For example $B(15, 0.6)$ is already well approximated by $N(9, 3.6)$ since $np = 9 > 5$.

Example 4-10 (Normal approximation)

Let $X \sim B(100, \frac{1}{5})$. Approximate
(a) $P(X \leq 21)$;
(b) $P(15 \leq X \leq 19)$.
Solution. $np = 20$ and $\sqrt{np(1-p)} = \sqrt{100 \cdot \frac{1}{5} \cdot \frac{4}{5}} = 4$.
(a) $P(X \leq 21) \approx P(Z \leq \frac{21-20}{4}) = P(Z \leq 0.25) = 0.5987$.
(b) $P(15 \leq X \leq 19) \approx P(-1.25 \leq Z \leq -0.25) = 0.4013 - 0.1056 = 0.2957$.

Continuity correction

A binomial probability is a sum of bars of width 1 centred at the integers. For $X \sim B(15, 0.6)$, $P(7 \leq X \leq 10)$ is better approximated by the normal area from 6.5 to 10.5:

$$P(7 \leq X \leq 10) = P(6.5 \leq X \leq 10.5) \approx P\Big(\frac{6.5-9}{\sqrt{3.6}} \leq Z \leq \frac{10.5-9}{\sqrt{3.6}}\Big)$$
$$= P(-1.318 \leq Z \leq 0.791) = 0.692,$$

very close to the exact value 0.688. This adjustment is the **continuity correction**.

4.5 The chi-square, t and F distributions

Motivation. Statistics built from the sample variance follow chi-square, t and F distributions.

For a random sample $X_1, \ldots, X_n$ from $N(\mu, \sigma^2)$ the **sample variance** is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

4.5.1 The chi-square distribution

Definition 4-6 (Chi-square distribution)

If $Z_1, \ldots, Z_k$ is a random sample from $N(0,1)$, the distribution of $Z_1^2 + \cdots + Z_k^2$ is the **chi-square distribution with k degrees of freedom**:

$$Z_1^2 + \cdots + Z_k^2 \sim \chi^2(k).$$

The chi-square density has a long right tail. The upper quantile $\chi^2_\alpha(k)$ satisfies $P(V \geq \chi^2_\alpha(k)) = \alpha$; e.g. $\chi^2_{0.995}(4) = 0.21$, $\chi^2_{0.975}(4) = 0.48$, $\chi^2_{0.05}(4) = 9.49$.

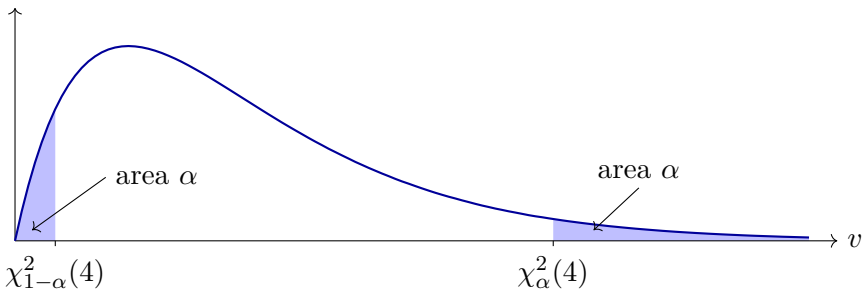


Figure 2: The $\chi^2(4)$ density $ve^{-v/2}/4$ with $\alpha = 0.05$: $\chi^2_{0.95}(4) = 0.71$, $\chi^2_{0.05}(4) = 9.49$.

Theorem 4-7 (Additivity of the chi-square distribution)

If $V_1 \sim \chi^2(k_1)$ and $V_2 \sim \chi^2(k_2)$ are independent, then $V_1 + V_2 \sim \chi^2(k_1 + k_2)$.

Proof. [Supplement]

Step 1. For $Z \sim N(0,1)$ and $t < \frac{1}{2}$, substituting $u = z\sqrt{1-2t}$,

$$E(e^{tZ^2}) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(1-2t)z^2/2}\, dz = (1-2t)^{-1/2}.$$

Step 2. Hence, by independence, $V \sim \chi^2(k)$ has mgf $M_V(t) = (1-2t)^{-k/2}$, $t < \frac{1}{2}$; conversely any variable with this mgf has the $\chi^2(k)$ distribution (uniqueness of mgfs).

Step 3. For independent V_1, V_2, $M_{V_1+V_2}(t) = (1-2t)^{-k_1/2}(1-2t)^{-k_2/2} = (1-2t)^{-(k_1+k_2)/2}$, so $V_1 + V_2 \sim \chi^2(k_1 + k_2)$. ■

Theorem 4-8 (Distribution of the sample variance from a normal population)

For a random sample $X_1, \ldots, X_n$ from $N(\mu, \sigma^2)$ with sample variance S^2,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

Moreover $\bar{X}$ and S^2 are independent.

Proof. The book's argument: since $(X_i - \mu)/\sigma \sim N(0,1)$ independently, $\sum_{i=1}^n \left(\frac{X_i-\mu}{\sigma}\right)^2 \sim \chi^2(n)$. Adding and subtracting $\bar{X}$ (the cross term vanishes because $\sum(X_i - \bar{X}) = 0$),

$$\sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma}\right)^2 = \frac{(n-1)S^2}{\sigma^2} + \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\right)^2,$$

and the last term is $\chi^2(1)$. "Subtracting" degrees of freedom gives $\chi^2(n-1)$. We make this rigorous.

[Supplement]

Step 1. Let $Z_i = (X_i - \mu)/\sigma$, independent $N(0,1)$, so $\mathbf{Z} = (Z_1, \ldots, Z_n)^\top$ has joint density $(2\pi)^{-n/2} e^{-\|\mathbf{z}\|^2/2}$. Let A be an $n \times n$ orthogonal matrix whose first row is $(1/\sqrt{n}, \ldots, 1/\sqrt{n})$ (complete this unit vector to an orthonormal basis), and $\mathbf{W} = A\mathbf{Z}$.

Step 2. The map $\mathbf{z} \mapsto A\mathbf{z}$ has Jacobian $|\det A| = 1$ and preserves $\|\mathbf{z}\|$, so $\mathbf{W}$ has the same density: $W_1, \ldots, W_n$ are independent $N(0,1)$.

Step 3. $W_1 = \sqrt{n}\,\bar{Z}$ and $\sum_{i=1}^n W_i^2 = \sum_{i=1}^n Z_i^2$, hence

$$\sum_{i=2}^{n} W_i^2 = \sum_{i=1}^{n} Z_i^2 - n\bar{Z}^2 = \sum_{i=1}^{n} (Z_i - \bar{Z})^2 = \frac{(n-1)S^2}{\sigma^2}.$$

Step 4. By Definition 4-6 this is $\chi^2(n-1)$. It is a function of $W_2, \dots, W_n$, while $\bar{X} = \mu + \sigma W_1/\sqrt{n}$ is a function of W_1; so $\bar{X}$ and S^2 are independent. ■

Next take independent random samples $X_1, \dots, X_{n_1}$ from $N(\mu_1, \sigma^2)$ and $Y_1, \dots, Y_{n_2}$ from $N(\mu_2, \sigma^2)$ (common variance), with

$$S_1^2 = \frac{1}{n_1 - 1}\sum_{i=1}^{n_1}(X_i - \bar{X})^2, \qquad S_2^2 = \frac{1}{n_2 - 1}\sum_{i=1}^{n_2}(Y_i - \bar{Y})^2.$$

Definition (Pooled sample variance)

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2},$$

the weighted average of S_1^2, S_2^2 with weights $\frac{n_j - 1}{(n_1-1)+(n_2-1)}$; it estimates the common variance σ^2.

Theorem 4-9 (Sample variances from two normal populations with equal variance)

Under the above assumptions,

$$\frac{(n_1 + n_2 - 2)S_p^2}{\sigma^2} \sim \chi^2(n_1 + n_2 - 2).$$

Proof. By Theorem 4-8, $(n_1 - 1)S_1^2/\sigma^2 \sim \chi^2(n_1 - 1)$ and $(n_2 - 1)S_2^2/\sigma^2 \sim \chi^2(n_2 - 1)$, and they are independent since they come from independent samples. Their sum is $(n_1 + n_2 - 2)S_p^2/\sigma^2$, which is $\chi^2(n_1 + n_2 - 2)$ by Theorem 4-7. ■

4.5.2 The t distribution

When σ is unknown we replace it by the sample standard deviation S and obtain the **studentized** sample mean $\frac{\bar{X} - \mu}{S/\sqrt{n}}$.

Definition 4-7 (t distribution)

If $Z \sim N(0,1)$ and $V \sim \chi^2(k)$ are independent, the distribution of

$$T = \frac{Z}{\sqrt{V/k}}$$

is the t **distribution with** k **degrees of freedom**, $T \sim t(k)$.

Like $N(0,1)$, $t(k)$ is symmetric about 0, but it has heavier tails; it is used especially for small samples. The upper quantile $t_\alpha(k)$ satisfies $P(T \geq t_\alpha(k)) = \alpha$; e.g. $t_{0.1}(4) = 1.533$, $t_{0.05}(4) = 2.132$, $t_{0.01}(4) = 3.747$.

Theorem 4-10 (Distribution of the studentized sample mean)

For a random sample $X_1, \dots, X_n$ from $N(\mu, \sigma^2)$,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1).$$

Proof. Dividing numerator and denominator by $\sigma/\sqrt{n}$,

$$\frac{\bar{X}-\mu}{S/\sqrt{n}} = \frac{(\bar{X}-\mu)/(\sigma/\sqrt{n})}{\sqrt{\dfrac{(n-1)S^2}{\sigma^2}\Big/(n-1)}}.$$

The numerator is $N(0,1)$ (Theorem 4-5), the root contains $\chi^2(n-1)/(n-1)$ (Theorem 4-8), independent of it (Theorem 4-8; the book omits this proof); so the ratio is $t(n-1)$ (Definition 4-7). ■

For two normal populations with equal variance, a short computation gives

$$\frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{\dfrac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{\sigma\sqrt{1/n_1+1/n_2}}}{\sqrt{\dfrac{(n_1+n_2-2)S_p^2}{\sigma^2}\Big/(n_1+n_2-2)}}.$$

Theorem 4-11 (t distribution for two normal populations with equal variance)

Under the assumptions of Theorem 4-9,

$$\frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t(n_1+n_2-2).$$

Proof.

Step 1. Numerator. $\bar{X} \sim N(\mu_1, \sigma^2/n_1)$ and $\bar{Y} \sim N(\mu_2, \sigma^2/n_2)$ are independent, so by Theorem 4-2 $\bar{X}-\bar{Y} \sim N(\mu_1-\mu_2, \sigma^2(\frac{1}{n_1}+\frac{1}{n_2}))$, and the standardized numerator in the display is $N(0,1)$.

Step 2. Denominator. By Theorem 4-9, $(n_1+n_2-2)S_p^2/\sigma^2 \sim \chi^2(n_1+n_2-2)$.

Step 3. Independence. By Theorem 4-8, $\bar{X}$ is independent of S_1^2 and $\bar{Y}$ of S_2^2; as the two samples are independent, $(\bar{X},\bar{Y})$ is independent of (S_1^2, S_2^2), hence of S_p^2.

Step 4. Definition 4-7 gives $t(n_1+n_2-2)$. ■

4.5.3 The F distribution

Definition 4-8 (F distribution)

If $V_1 \sim \chi^2(k_1)$ and $V_2 \sim \chi^2(k_2)$ are independent, the distribution of

$$F = \frac{V_1/k_1}{V_2/k_2}$$

is the F **distribution with** (k_1, k_2) **degrees of freedom**, $F \sim F(k_1, k_2)$.

The F density is right-skewed on $(0,\infty)$. Its upper quantile $F_\alpha(k_1,k_2)$ satisfies $P(F \geq F_\alpha(k_1,k_2)) = \alpha$; e.g. $F_{0.10}(3,4) = 4.19$, $F_{0.05}(3,4) = 6.59$.

Theorem 4-12 (Property of the F distribution)

If $F \sim F(k_1, k_2)$, then $1/F \sim F(k_2, k_1)$. Consequently

$$F_{1-\alpha}(k_2, k_1) = \frac{1}{F_\alpha(k_1, k_2)}.$$

Proof.

Step 1. Write $F = \frac{V_1/k_1}{V_2/k_2}$ as in Definition 4-8. Then $1/F = \frac{V_2/k_2}{V_1/k_1}$ is a ratio of the same form with the roles of (V_1, k_1) and (V_2, k_2) exchanged, so $1/F \sim F(k_2, k_1)$.

Step 2. Let $F' = 1/F \sim F(k_2, k_1)$. Since $F > 0$, $\alpha = P(F \geq F_\alpha(k_1, k_2)) = P(F' \leq 1/F_\alpha(k_1, k_2))$, so $P(F' \geq 1/F_\alpha(k_1, k_2)) = 1-\alpha$ (F' is continuous). By definition of the upper quantile, $F_{1-\alpha}(k_2, k_1) = 1/F_\alpha(k_1, k_2)$. ■

For example, $F_{0.95}(4, 3) = 1/F_{0.05}(3, 4) = 1/6.59 = 0.15$.

For independent samples from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, $(n_1 - 1)S_1^2/\sigma_1^2 \sim \chi^2(n_1 - 1)$ and $(n_2 - 1)S_2^2/\sigma_2^2 \sim \chi^2(n_2 - 1)$ independently.

Theorem 4-13 (Ratio of sample variances from two normal populations)

Under these assumptions,

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1).$$

Proof. Put $V_j = (n_j - 1)S_j^2/\sigma_j^2 \sim \chi^2(n_j - 1)$ (Theorem 4-8); V_1, V_2 are independent. Then $\frac{V_1/(n_1-1)}{V_2/(n_2-1)} = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$, which is $F(n_1 - 1, n_2 - 1)$ by Definition 4-8. ■

Theorem 4-14 (Relation between the F and t distributions)

If $T \sim t(k)$, then $T^2 \sim F(1, k)$.

Proof. Write $T = Z/\sqrt{V/k}$ with $Z \sim N(0, 1)$, $V \sim \chi^2(k)$ independent. Then $T^2 = \frac{Z^2/1}{V/k}$, where $Z^2 \sim \chi^2(1)$ (Definition 4-6) and Z^2 is independent of V. By Definition 4-8, $T^2 \sim F(1, k)$. ■

4.6 Normal quantile plots

Motivation. A normal quantile (Q–Q) plot checks the normality assumed later.

A q-quantile splits the area $q : (1-q)$. A **normal quantile plot** (Q–Q plot) plots standard normal quantiles against the data quantiles.

Definition (Normal quantile plot)

Order the data $x_{(1)} < x_{(2)} < \cdots < x_{(n)}$. The value $x_{(i)}$ is roughly the $\frac{i}{n+1}$-quantile of the data, whose standard normal counterpart is $z_{1-i/(n+1)}$. Plot the points

$$\left(z_{1-i/(n+1)},\, x_{(i)}\right), \qquad i = 1, \ldots, n.$$

If the points lie close to a straight line, the normality assumption is reasonable. Software often uses $\frac{i-3/8}{n+1/4}$ instead of $\frac{i}{n+1}$.

E.g. for $n = 99$ from $N(2, 3^2)$ the population quantiles $3z_{0.99} + 2, \ldots, 3z_{0.01} + 2$ are linear in the normal quantiles, so the points $(z_{0.99}, x_{(1)}), \ldots, (z_{0.01}, x_{(99)})$ lie near a line of slope 3, intercept 2.

Reading a Q–Q plot

Systematic departures from the line indicate a non-normal population:
(a) light tails: the extreme points bend toward the centre (an S-shape flatter than the line at both ends);
(b) heavy tails: the extreme points fall below the line on the left and above it on the right;
(c) long right tail: the points curve upward on the right;
(d) long left tail: the points curve downward on the left.

Chapter 5. Statistical Inference

5.1 Estimation

Motivation. Inference attaches a probability-based reliability to conclusions; estimation is explained through the mean.

Computing an estimate of a population characteristic (mean, s.d., proportion, correlation) *with its margin of error* is **estimation**.

5.1.1 Estimating a population mean

Illustration 5-1: tire lifetimes

A new tire process should raise the mean life μ, with $\sigma = 5000$ km unchanged. $n = 100$ tires give $\bar{x} = 38000$ km, so $\hat{\mu} = \bar{x} = 38000$ km.

How reliable is this method? By the CLT, $\bar{X}$ is approximately $N(\mu, \sigma^2/100)$ with standard deviation $\sigma/\sqrt{n} = 5000/10 = 500$ km. Hence

(a) with probability about 0.95, $\bar{X}$ lies within $1.96\,\sigma/\sqrt{n} = 980$ of μ;

(b) the event $|\bar{X} - \mu| < 980$ is the same as $\mu \in (\bar{X} - 980, \bar{X} + 980)$;

(c) if the experiment were repeated indefinitely, about 95% of the intervals $(\bar{x} - 980, \bar{x} + 980)$ would contain μ.

Here the interval is $(37020, 38980)$. Either it contains μ, or the sample is among the unusual 5%; the *method* is right 95 times in 100, so $37020 < \mu < 38980$ is a 95% **confidence interval** (**confidence level** 95%): $\hat{\mu} = 38000$, $|\mu - \hat{\mu}| \le 1.96\,\sigma/\sqrt{n} = 980$.

Definition (Estimate, estimator, standard error)

$\hat{\mu} = \bar{x}$ is an **estimate**, $1.96\,\sigma/\sqrt{n}$ its 95% **margin of error**; the statistic $\hat{\mu} = \bar{X}$ is the **estimator** (as a random variable to its observed value). The standard deviation of an estimator is its **standard error**:

$$\mathrm{SE}(\hat{\mu}) = \sqrt{\mathrm{Var}(\hat{\mu})} = \sqrt{\mathrm{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}}.$$

For large n,

$$P\Big(-z_{\alpha/2} \le \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\Big) \approx P(-z_{\alpha/2} \le Z \le z_{\alpha/2}) = 1 - \alpha.$$

Definition (Estimating μ when σ is known)

(a) Estimator and standard error: $\hat{\mu} = \bar{X}$, $\mathrm{SE}(\hat{\mu}) = \sigma/\sqrt{n}$.
(b) $100(1-\alpha)\%$ margin of error: $z_{\alpha/2}\,\mathrm{SE}(\hat{\mu}) = z_{\alpha/2}\,\sigma/\sqrt{n}$.
(c) $100(1-\alpha)\%$ confidence interval: $\Big(\bar{x} - z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}\Big)$.

These statements are exact for a normal population and approximate (large n) otherwise. Common confidence levels are 90%, 95%, 99%, with $z_{0.05} = 1.645$, $z_{0.025} = 1.96$, $z_{0.005} = 2.576$.

Illustration 5-2

From a normal population with $\sigma = 10$, a sample of $n = 25$ gives $\bar{x} = 40$. Then $\hat{\mu} = 40$, the 90% margin of error is $1.645 \cdot 10/\sqrt{25} = 3.29$, and the 90% confidence interval is (36.71, 43.29).

Meaning of the confidence level. In Illustration 5-2, twenty samples of size 25 give twenty intervals $\bar{x} \pm 3.29$, of which 18 contain μ in the book's simulation: 90% confidence means 90% coverage over repeated sampling.

Example 5-1 (Cholesterol of low-income residents)

A survey of $n = 49$ low-income Central American residents found mean blood cholesterol $\bar{x} = 157.02$ mg/L. Assume cholesterol is approximately normal with $\sigma = 30$ mg/L. Find a 95% confidence interval for μ.
Solution. $\bar{x} \pm 1.96\,\sigma/\sqrt{n} = 157.02 \pm 1.96 \cdot 30/7 = 157.02 \pm 8.40$, i.e. (148.62, 165.42) mg/L.

At 99% the interval is $\bar{x} \pm 2.576\,\sigma/\sqrt{n} = (145.98, 168.06)$: higher confidence gives a longer interval, and excessively high confidence makes the interval uselessly wide. The length of a $100(1-\alpha)\%$ interval is

$$\left(\bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) - \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 2z_{\alpha/2}\frac{\sigma}{\sqrt{n}},$$

twice the margin of error, so it can be controlled through n.

Definition (Sample size for estimating μ (σ known))

To make the $100(1-\alpha)\%$ margin of error at most d (interval length at most $2d$), take

$$n \geq \left(z_{\alpha/2}\frac{\sigma}{d}\right)^2.$$

(Indeed $z_{\alpha/2}\sigma/\sqrt{n} \leq d \iff \sqrt{n} \geq z_{\alpha/2}\sigma/d$.)

Example 5-2 (Required sample size)

In Example 5-1, how many residents must be surveyed for the 95% margin of error to be at most 5 mg/L?
Solution. With $\sigma = 30$, $n \geq (1.96 \cdot 30/5)^2 = 138.2976$, so at least 139 residents are needed.

5.1.2 Parameters, point and interval estimation

σ unknown is treated in Chapter 6. A numerical summary of a population is a **parameter** θ, with estimator (or estimate) $\hat{\theta}(X_1, \ldots, X_n)$, or $\hat{\theta}$.

Definition (Confidence interval for a parameter)

If statistics $L = L(X_1, \ldots, X_n)$ and $U = U(X_1, \ldots, X_n)$ satisfy

$$P(L < \theta < U) = 1 - \alpha,$$

then (L, U) is a $100(1-\alpha)\%$ confidence interval for θ. The confidence level has the same repeated-sampling meaning as before.

Illustration 5-3: estimating σ and σ^2

The usual estimator of σ is the sample standard deviation, and that of σ^2 the sample variance:

$$\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2} = S, \qquad \hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 = S^2.$$

Giving only an estimate of the parameter is **point estimation**; giving a confidence interval is **interval estimation**. Together they constitute **estimation of parameters**.

Example 5-3 (Thickness of plastic sheets)

The variability of thickness is a key quality characteristic. Ten sheets measured (mm):

226	228	226	225	232	228	227	229	225	230

Estimate the population standard deviation σ.
Solution. $\bar{x} = \frac{1}{10}(226 + 228 + \cdots + 230) = 227.6$ mm and

$$s^2 = \frac{1}{10-1}\{(226-227.6)^2 + (228-227.6)^2 + \cdots + (230-227.6)^2\} = 5.1556,$$

so $\hat{\sigma} = s = \sqrt{5.1556} = 2.27$ mm.

Other estimators exist (e.g. the trimmed mean for μ, more stable when there are outliers), but this book uses $\bar{X}$ and S.

5.2 Significance tests and hypothesis tests

Motivation. Significance testing measures evidence against a claim by a probability, the P-value.

5.2.1 Null and alternative hypotheses

Illustration 5-4: testing Mendel's law

Genetically modified peas gave 860 round-yellow, 340 round-green, 240 wrinkled-yellow, 160 wrinkled-green in the second generation. Mendel predicts $9:3:3:1$. Do the data suggest a new theory?

Science advances by finding strong evidence against established theories. Here the claim under scrutiny is the **hypothesis** "the four types occur in ratio $9:3:3:1$".

Definition (Null and alternative hypotheses; significance test)

The hypothesis we seek evidence *against* is the **null hypothesis** H_0; its alternative is the **alternative hypothesis** H_1. A **significance test** evaluates the evidence against H_0, as in proof by contradiction: under H_0 the data would be very unlikely.

Illustration 5-5: sprinkler temperature

Sprinklers are designed to activate at 55°C. The process is abnormal if the mean activation temperature μ satisfies $\mu < 55$ or $\mu > 55$. Since we seek clear evidence of abnormality,

$$H_0 : \mu = 55, \qquad H_1 : \mu \neq 55.$$

Illustration 5-6: paint drying time

A paint dries in 75 minutes on average ($\sigma = 9.4$). An additive is claimed to shorten this and will be adopted only on clear evidence. With μ the mean drying time with the additive,

$$H_0 : \mu = 75, \qquad H_1 : \mu < 75.$$

$H_1 : \mu \neq 55$ is **two-sided**, $H_1 : \mu < 75$ **one-sided**. The claim to be established goes into H_1.

Choice of the null value

The negation of $H_1 : \mu < 75$ is $H_0 : \mu \geq 75$; stating the boundary $H_0 : \mu = 75$ (hardest to distinguish from H_1) does not change the procedure.

Evidence for H_1 is evaluated through an estimator of the parameter: the farther the estimate lies from the value specified by H_0 in the direction of H_1, the stronger the evidence.

Illustration 5-7

For $H_0 : \mu = 55$, $H_1 : \mu \neq 55$, the evidence is sought in $\bar{X}$; since H_1 is two-sided, $\bar{X}$ far from 55 in *either* direction supports H_1.

Illustration 5-8

For $H_0 : \mu = 75$, $H_1 : \mu < 75$ (left-sided), values of $\bar{X}$ far *below* 75 support H_1.

A statistic such as $\bar{X}$ used in a test is a **test statistic**.

5.2.2 The P-value

Illustration 5-9: P-value for the sprinklers

For $H_0 : \mu = 55$, $H_1 : \mu \neq 55$, $n = 9$ sprinklers gave $\bar{x} = 55.63$°C; assume normality with $\sigma = 0.9$. The observed deviation is $\bar{x} - 55 = 0.63$. The probability, under H_0, of evidence at least as strong as observed is

$$P\Big(\Big|\frac{\bar{X} - 55}{\sigma/\sqrt{n}}\Big| \geq 2.1 \;\Big|\; \mu = 55\Big) = P(|Z| \geq 2.1) = 0.0357,$$

since $0.63/(0.9/3) = 2.1$. Repeating the survey, stronger evidence against H_0 would arise fewer than 4 times in 100 if H_0 were true; the data strongly suggest $\mu \neq 55$.

Definition 5-1 (Significance probability (P-value))

The ***P*-value** is the probability under H_0 that the test statistic is at least as extreme as observed, in the direction of H_1; smaller means stronger evidence against H_0.

Illustration 5-10: P-value for the paint

For $H_0 : \mu = 75$, $H_1 : \mu < 75$, $n = 100$ samples with the additive gave $\bar{x} = 73.5$ ($\sigma = 9.4$). By the CLT, $\bar{X} \approx N(\mu, 9.4^2/100)$, so

$$P = P(\bar{X} \le 73.5 \mid \mu = 75) = P\Big(\frac{\bar{X} - 75}{9.4/\sqrt{100}} \le -1.5957 \,\Big|\, \mu = 75\Big) \approx P(Z \le -1.5957) = 0.0553.$$

The data fairly strongly suggest a shorter drying time; the strength of evidence is about 5.5%.

5.2.3 Significance level

A **significance level** α (usually 0.05 or 0.01) is often fixed in advance; $P \le 0.05$ is reported as "statistically significant ($P \le 0.05$)".

Definition 5-2 (Statistical significance)

If the P-value is at most the significance level α, the result is **statistically significant at level** α; the evidence against H_0 is stronger than the prescribed level.

Definition (Procedure of a significance test)

(a) State H_0 and H_1.
(b) [Optional] Fix the significance level α.
(c) Compute the observed value of the test statistic (based on an estimator of the relevant parameter).
(d) Compute the P-value from the sampling distribution of the test statistic under H_0; it measures the evidence against H_0.
(e) [Optional] Compare the P-value with α: if $P \le \alpha$ the result is significant.

Steps (c) and (d) are mechanical (tables or software), but H_0, H_1 and the interpretation are the investigator's responsibility.

5.2.4 Tests for μ with σ known

For $H_0 : \mu = \mu_0$ against $H_1 : \mu < \mu_0$ the standardized statistic and its observed value are

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}, \qquad z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}},$$

and $P = P(\bar{X} \le \bar{x} \mid \mu = \mu_0) \approx P(Z \le z)$ (in Illustration 5-10, $z = -1.5957$ and $P = 0.0553$). For $H_1 : \mu > \mu_0$, $P \approx P(Z \ge z)$; for $H_1 : \mu \ne \mu_0$, deviations of $\bar{X}$ from μ_0 in both directions count, so

$$P(\bar{X} - \mu_0 \le -|\bar{x} - \mu_0| \text{ or } \bar{X} - \mu_0 \ge |\bar{x} - \mu_0| \mid \mu = \mu_0) \approx P(|Z| \ge |z|).$$

Definition (Significance test for μ (σ known))

For $H_0 : \mu = \mu_0$ use $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$. With observed value z, the P-value is
(a) $P = P(Z \ge z)$ if $H_1 : \mu > \mu_0$;
(b) $P = P(Z \le z)$ if $H_1 : \mu < \mu_0$;
(c) $P = P(|Z| \ge |z|)$ if $H_1 : \mu \ne \mu_0$.

The P-value is exact for a normal population and approximate for large n otherwise.

Example 5-4 (New tire process)

Current tires last 37,000 km on average ($\sigma = 5000$). 100 tires from a new process give $\bar{x} = 38000$ km. With σ unchanged, is the new process successful?
Solution. We want to establish $\mu > 37000$: $H_0 : \mu = 37000$, $H_1 : \mu > 37000$. The observed statistic is

$$z = \frac{38000 - 37000}{5000/\sqrt{100}} = 2.$$

Z is approximately standard normal ($n = 100$), and H_1 is right-sided, so $P = P(Z \geq 2) = 1 - 0.9772 = 0.0228$. The data strongly suggest that the new process increases tire life.

Example 5-5 (Concentration of an active ingredient)

A drug should contain 0.85 g/L of an ingredient; measurements are $N(\mu, 0.0123^2)$. Three measurements by a new method average $\bar{x} = 0.8404$ g/L. Is $\mu \neq 0.85$?
Solution. $H_0 : \mu = 0.85$, $H_1 : \mu \neq 0.85$, and

$$z = \frac{0.8404 - 0.85}{0.0123/\sqrt{3}} = -1.35.$$

The population is normal, so $Z \sim N(0,1)$ exactly, and $P = P(|Z| \geq 1.35) = 2P(Z \leq -1.35) = 2 \times 0.0885 = 0.177$. A P-value above 17% gives no clear evidence that the true concentration differs from 0.85 g/L.

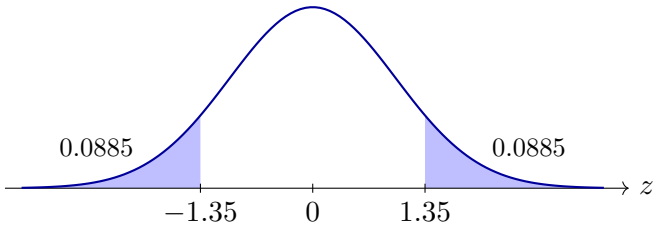


Figure 3: Two-sided P-value in Example 5-5: $P = P(|Z| \geq 1.35) = 0.177$.

5.2.5 Rejection regions and hypothesis testing

If the required strength of evidence α is fixed beforehand, the P-value need not be computed.

Illustration 5-11

In Example 5-4 ($H_0 : \mu = 37000$, $H_1 : \mu > 37000$) take $\alpha = 0.05$. We must decide whether $P(Z \geq z) \leq 0.05$. Two equivalent ways:
(a) compute the P-value from $z = 2$ as in Example 5-4;
(b) compare z with the upper 5% point $z_{0.05} = 1.645$. Since $z = 2 > 1.645$, i.e. $z \geq 1.645$ holds, the result is significant at 5%. Had $z < 1.645$, it would not be.

Definition (Rejection region; hypothesis test)

For fixed α, the set of statistic values declared significant (e.g. $z \geq 1.645$) is the **rejection region**; the conclusion reads "H_0 is rejected (cannot be rejected) at level α". This is **hypothesis**

testing:
(a) find the rejection region corresponding to α;
(b) compute the observed value of the test statistic;
(c) check whether it lies in the rejection region.

Definition (Hypothesis test for μ (σ known, level α))

For $H_0 : \mu = \mu_0$ with $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$ and observed value z, the rejection region is
(a) $z \geq z_\alpha$ if $H_1 : \mu > \mu_0$;
(b) $z \leq -z_\alpha$ if $H_1 : \mu < \mu_0$;
(c) $|z| \geq z_{\alpha/2}$ if $H_1 : \mu \neq \mu_0$.

Frequently $z_{0.05} = 1.645$, $z_{0.025} = 1.96$, $z_{0.01} = 2.326$, $z_{0.005} = 2.576$.

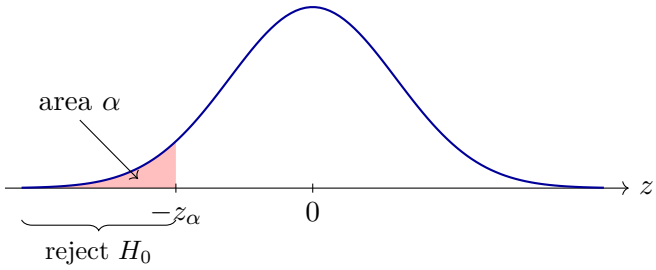


Figure 4: Rejection region $z \leq -z_\alpha$ for $H_1 : \mu < \mu_0$.

Example 5-6 (A faster pain reliever)

An existing painkiller acts in 30 minutes on average ($\sigma = 5$). For a new one, 50 patients give $\bar{x} = 28.5$ minutes (normal, $\sigma = 5$). Test at level 5% whether it acts faster.
Solution. $H_0 : \mu = 30$, $H_1 : \mu < 30$. The rejection region at 5% is

$$z = \frac{\bar{x} - 30}{5/\sqrt{50}} \leq -z_{0.05} = -1.645.$$

Here $z = \frac{28.5-30}{5/\sqrt{50}} = -2.12$ lies in the rejection region, so H_0 is rejected at 5%: the evidence that the new drug is faster exceeds the prescribed level.

Example 5-7 (Sprinklers at level 1%)

In Illustrations 5-5 and 5-9 ($n = 9$, $\bar{x} = 55.63$, $\sigma = 0.9$), test $H_0 : \mu = 55$ against $H_1 : \mu \neq 55$ at level 1%.
Solution. $z = \frac{55.63-55}{0.9/\sqrt{9}} = 2.1$, and the rejection region is $|z| \geq z_{0.005} = 2.576$. Since $2.1 < 2.576$, H_0 cannot be rejected at 1%: the evidence of a process abnormality is not stronger than the 1% level.

The conclusion depends on α: Example 5-6 has $P = P(Z \leq -2.12) = 0.0170$ (rejected at 0.05, not at 0.01), Example 5-7 $P = P(|Z| \geq 2.1) = 0.0357$ (likewise). The P-value reports evidence without fixing α; a decision needs α chosen for the context.

5.2.6 Two kinds of error

A level-α test is a rule for choosing between H_0 and H_1, and it can err in two ways.

Decision \ Truth	H_0 true	H_1 true
Do not reject H_0 (choose H_0)	correct	Type II error
Reject H_0 (choose H_1)	Type I error	correct

Illustration 5-12: Type I error probability

In Example 5-6 the level-α rejection region is $\frac{\bar{X}-30}{5/\sqrt{50}} \le -z_\alpha$, so

$$P(\text{Type I error}) = P\Big(\frac{\bar{X}-30}{5/\sqrt{50}} \le -z_\alpha \;\Big|\; \mu = 30\Big) = P(Z \le -z_\alpha) = \alpha.$$

The level-α rejection region has Type I error probability at most α (for $H_0 : \mu \ge \mu_0$ the maximum is at $\mu = \mu_0$).

Illustration 5-13: Type II error probability

In Example 5-6, a true mean of 28 minutes would be a notable improvement. If $\mu = 28$, a Type II error occurs when H_0 is not rejected:

$$\begin{aligned} P\Big(\frac{\bar{X}-30}{5/\sqrt{50}} > -1.645 \;\Big|\; \mu = 28\Big) &= P\Big(\frac{\bar{X}-28}{5/\sqrt{50}} + \frac{28-30}{5/\sqrt{50}} > -1.645 \;\Big|\; \mu = 28\Big) \\ &= P(Z > 2\sqrt{2} - 1.645) = P(Z > 1.183) \approx 0.1190. \end{aligned}$$

Just as α bounds the Type I error, one may require the Type II error probability at a specified alternative to be at most β; this is achieved by choosing n.

Example 5-8 (Sample size for given α and β)

In Example 5-6, find n such that the level-0.05 test has Type II error probability at most $\beta = 0.1$ when $\mu = 28$.

Solution. As in Illustration 5-13,

$$\begin{aligned} P\Big(\frac{\bar{X}-30}{5/\sqrt{n}} > -1.645 \;\Big|\; \mu = 28\Big) &= P\Big(\frac{\bar{X}-28}{5/\sqrt{n}} + \frac{28-30}{5/\sqrt{n}} > -1.645 \;\Big|\; \mu = 28\Big) \\ &= P\Big(Z > \frac{30-28}{5/\sqrt{n}} - 1.645\Big) \le 0.1. \end{aligned}$$

Since $z_{0.1} = 1.282$, this requires $\frac{30-28}{5/\sqrt{n}} - 1.645 \ge 1.282$, i.e.

$$n \ge \Big\{\frac{1.645 + 1.282}{(30-28)/5}\Big\}^2 = 53.546.$$

The required sample size is $n = 54$.

Definition (Sampling inspection; power)

Choosing n and the rejection region so that Type I error $\le \alpha$ and Type II error at $\mu = \mu_1$ is $\le \beta$ is **sampling inspection** (quality control). The probability of rejecting H_0 when $\mu = \mu_1$

is the **power** at μ_1.

Illustration 5-14: power

For the 5% test of Illustration 5-13, the power at $\mu = 28$ is

$$P\Big(\frac{\bar{X} - 30}{5/\sqrt{50}} \leq -1.645 \;\Big|\; \mu = 28\Big) = 1 - 0.1190 = 0.8810.$$

Conditions for valid inference

For the methods of this chapter to be valid:
(a) the sample must be drawn at random;
(b) the assumptions on the population distribution must be checked.

Chapter 6. Inference about Distributions

6.1 Inference about a population mean

Motivation. With σ unknown, replacing it by S leads to t intervals and t tests.

Replacing σ by S in $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ is **studentizing**. For a random sample from $N(\mu,\sigma^2)$, Theorem 4-10 gives

$$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1).$$

Hence

$$P\Big(-t_{\alpha/2}(n-1) \le \frac{\bar{X}-\mu}{S/\sqrt{n}} \le t_{\alpha/2}(n-1)\Big) = 1-\alpha, \quad \text{i.e.}$$
$$P\Big(\bar{X} - t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}} \le \mu \le \bar{X} + t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}}\Big) = 1-\alpha.$$

Definition (Confidence interval for μ (normal population, σ unknown))

A $100(1-\alpha)\%$ confidence interval for μ is

$$\Big(\bar{x} - t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}},\ \bar{x} + t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}}\Big).$$

For large n the normality assumption can be relaxed: by the CLT and since $t(n-1)$ approaches $N(0,1)$, the quantile $t_{\alpha/2}(n-1)$ may be replaced by $z_{\alpha/2}$.

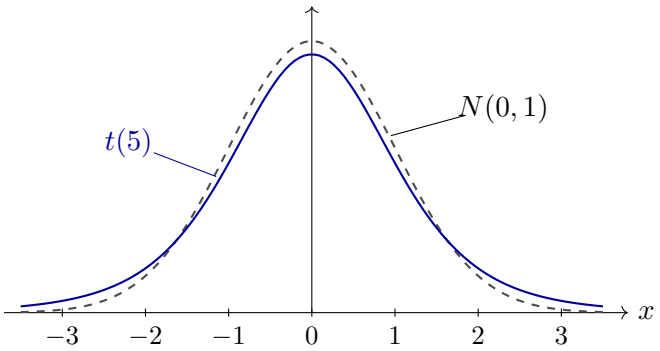


Figure 5: The $t(5)$ density (solid) has heavier tails than $N(0,1)$ (dashed).

Example 6-1 (Trees per 100 m^2)

A forest manager counted trees in 50 randomly chosen 100 m^2 plots of a 9 km^2 forest:

43	28	23	18	46	39	40	43	36	25
26	28	20	22	17	26	17	12	16	35
37	12	32	47	20	25	13	20	46	23
37	29	23	37	34	27	33	29	24	19
37	23	37	28	27	35	30	26	33	35

Find a 99% confidence interval for the mean number μ of trees per 100 m^2.

Solution. With $n = 50$,

$$\bar{x} = \frac{43 + 28 + \cdots + 35}{50} = 28.76, \qquad s = \sqrt{\frac{(43 - 28.76)^2 + \cdots + (35 - 28.76)^2}{49}} = 9.146.$$

The normal quantile plot is close to a line, so a normal population is reasonable. The table lacks 49 degrees of freedom; using the nearest value $t_{0.005}(50) = 2.678$,

$$\bar{x} \pm t_{0.005}(49)\frac{s}{\sqrt{n}} \approx 28.76 \pm 2.678 \times \frac{9.146}{\sqrt{50}} = 28.76 \pm 3.464,$$

so the 99% confidence interval is (25.296, 32.223).

Testing with the studentized mean is the t **test**.

Definition (t test for μ (normal population, σ unknown))

For $H_0 : \mu = \mu_0$ use $T = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$, which is $t(n-1)$ under H_0. With observed value t:

Alternative	P-value	Rejection region at level α
(a) $H_1 : \mu > \mu_0$	$P(T \geq t)$	$t \geq t_\alpha(n-1)$
(b) $H_1 : \mu < \mu_0$	$P(T \leq t)$	$t \leq -t_\alpha(n-1)$
(c) $H_1 : \mu \neq \mu_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n-1)$

(The rejection regions have Type I error exactly α because $T \sim t(n-1)$ when $\mu = \mu_0$; e.g. in (c), $P(|T| \geq t_{\alpha/2}(n-1)) = 2 \cdot \frac{\alpha}{2}$ by symmetry.)

Example 6-2 (Light bulb lifetimes)

Current bulbs last 1950 hours on average. Nine prototypes of a new bulb lasted (hours)

2000	1975	1900	2000	1950	1850	1950	2100	1975

(a) State suitable hypotheses.
(b) Assuming normal lifetimes, test them at level 5%.
(c) Find the P-value of (b).
Solution.
(a) $H_0 : \mu = 1950$, $H_1 : \mu > 1950$.
(b) $\bar{x} = 1966.7$, $s = 69.6$, so $t = \dfrac{1966.7 - 1950}{69.6/\sqrt{9}} = 0.718$. The rejection region is $t \geq t_{0.05}(8) = 1.860$; since $0.718 < 1.860$, H_0 is not rejected at 5%.
(c) $P = P(T \geq 0.718)$, $T \sim t(8)$: between the upper 25% (0.706) and 20% (0.889) points, so $0.20 < P < 0.25$ (software: 0.247). The evidence is weak.

The Q–Q plot looks roughly linear, but with so few observations the check is weak; subject knowledge or Chapter 10's nonparametric methods may help.

Example 6-3 (Cholesterol of immigrant women)

The national mean cholesterol for women is 190 mg/dL. For 100 randomly selected women who immigrated in the past year, $\bar{x} = 178.2$ and $s = 45.3$.
(a) State suitable hypotheses.

(b) Test them at level 5%.
(c) Find the P-value of (b).
Solution.
(a) $H_0 : \mu = 190$, $H_1 : \mu \neq 190$.
(b) $t = \dfrac{178.2 - 190}{45.3/\sqrt{100}} = -2.60$. With $n = 100$ the t test applies without assuming normality, and $z_{0.025} = 1.96$ can replace the $t(99)$ quantile: the rejection region is $|t| \geq 1.96$. Since $|t| = 2.60 \geq 1.96$, H_0 is rejected at 5%.
(c) $P = P(|T| \geq 2.60) \approx 2 \times 0.0047 = 0.0094$ (normal table; software 0.0107): the mean differs significantly from the national mean.

6.2 Comparing means by paired comparison

Motivation. Applying both treatments to each unit (or matched pair) removes unit variation and reduces to a one-sample problem.

Comparing two means by observing pairs and analysing the within-pair differences is called **paired comparison**.

Illustration 6-1: two sleeping pills

Patients' health varies greatly, so each of 6 randomly chosen patients took both pills A and B once, in an order decided by a coin toss, and sleeping hours were recorded:

Patient	1	2	3	4	5	6
Pill A	4.8	4.0	5.8	4.9	5.3	7.4
Pill B	4.0	4.2	5.2	4.9	5.6	7.1

What can be inferred about $\mu_A - \mu_B$?

Illustration 6-2: wearable walking robot

To see whether a gait-assist robot increases exercise effect, 15 rehabilitation patients each walked 10 minutes with and without the robot; calories burned (kcal):

Patient	1	2	3	4	5	6	7	8
Without	23.9	23.9	20.0	19.2	22.2	11.8	21.7	27.7
With	25.2	28.2	24.9	20.2	19.7	16.9	22.6	20.8
Patient	9	10	11	12	13	14	15	
Without	15.6	22.5	23.2	19.3	15.2	10.0	11.2	
With	23.8	18.3	23.7	22.5	16.2	12.1	22.5	

The patients are **experimental units**, the pills or robot **treatments**, the change produced the **treatment effect**; pairing homogeneous units sharpens the comparison. With $(X_1, Y_1), \ldots, (X_n, Y_n)$, $\mu_1 = E(X_i)$, $\mu_2 = E(Y_i)$, form

$$D_i = X_i - Y_i, \qquad E(D_i) = \mu_1 - \mu_2, \qquad i = 1, \ldots, n,$$

and apply §6.1 to the sample $D_1, \ldots, D_n$ with mean $\bar{D}$ and standard deviation S_D.

Definition (Paired comparison of two means)

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$\Big(\bar{d} - t_{\alpha/2}(n-1)\frac{s_D}{\sqrt{n}},\ \bar{d} + t_{\alpha/2}(n-1)\frac{s_D}{\sqrt{n}}\Big).$$

(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ the test statistic is $T = \dfrac{\bar{D} - \delta_0}{S_D/\sqrt{n}}$; with observed value t:

Alternative	P-value	Rejection region at level α
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq t_\alpha(n-1)$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -t_\alpha(n-1)$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n-1)$

These are exact for a normal random sample $D_1, \ldots, D_n$ (Theorem 4-10) and approximate for large n; units should be chosen at random and the treatment order randomized (**randomization**).

Example 6-4 (Sleeping pills: confidence interval)

Using Illustration 6-1, find a 95% confidence interval for $\mu_A - \mu_B$.
Solution. The differences A−B are 0.8, −0.2, 0.6, 0.0, −0.3, 0.3, so $\bar{d} = 0.20$ and $s_D = 0.443$. With $t_{0.025}(5) = 2.571$,

$$\bar{d} \pm t_{0.025}(5)\frac{s_D}{\sqrt{n}} = 0.20 \pm 2.571 \times \frac{0.443}{\sqrt{6}} = 0.20 \pm 0.46,$$

so the 95% confidence interval is $(-0.26,\ 0.66)$.

A normal quantile plot can check normality here, but more important is randomization in planning and observation.

Example 6-5 (Walking robot: test)

Using Illustration 6-2:
(a) test at level 1% whether wearing the robot increases calorie consumption;
(b) find the P-value of (a).
Solution.
(a) With μ_1, μ_2 the mean calories without and with the robot, $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 < \mu_2$. The $n = 15$ differences (without − with) are

$$-1.3,\ -4.3,\ -4.9,\ -1.0,\ 2.5,\ -5.1,\ -0.9,\ 6.9,\ -8.2,\ 4.2,\ -0.5,\ -3.2,\ -1.0,\ -2.1,\ -11.3,$$

with $\bar{d} = -2.01$ and $s_D = 4.58$, so $t = \dfrac{-2.01 - 0}{4.58/\sqrt{15}} = -1.70$. Since $t_{0.01}(14) = 2.624$, the rejection region is $t \leq -2.624$; as $-1.70 > -2.624$, H_0 is not rejected at 1%.
(b) $P = P(T \leq -1.70)$, $T \sim t(14)$: the table gives $0.05 < P < 0.1$, and software $P = 0.056$. At the 1% level there is no clear evidence that the robot increases calorie consumption.

A normal quantile plot of these differences supports the normality assumption.

6.3 Comparing means with two independent samples

Motivation. With two separate groups, inference on $\mu_1 - \mu_2$ uses the difference of independent sample means.

Definition (Data structure for two-sample comparison)

(a) The observations in each group are a random sample from that group's population.
(b) Observations in different groups are independent.

Random samples from two populations satisfy this automatically; when one pool of units receives two treatments, the units must be split into the groups at random (randomization).

Label the populations $1, 2$ with means μ_1, μ_2 and standard deviations σ_1, σ_2, and let $X_{11}, \ldots, X_{1n_1}$ and $X_{21}, \ldots, X_{2n_2}$ be the samples, with

$$\bar{X}_1 = \frac{X_{11} + \cdots + X_{1n_1}}{n_1}, \quad S_1 = \sqrt{\frac{1}{n_1 - 1}\sum_{j=1}^{n_1}(X_{1j} - \bar{X}_1)^2},$$

and $\bar{X}_2, S_2$ likewise. We estimate $\mu_1 - \mu_2$ by $\widehat{\mu_1 - \mu_2} = \bar{X}_1 - \bar{X}_2$. By independence,

$$\mathrm{Var}(\bar{X}_1 - \bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}, \qquad \text{so} \qquad \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

is the standardized difference, which is exactly $N(0,1)$ for normal populations (Theorem 4-2). With σ_1^2, σ_2^2 unknown we use the studentized difference

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}.$$

6.3.1 Large samples

When n_1, n_2 are both large, $S_1^2 \approx \sigma_1^2$ and $S_2^2 \approx \sigma_2^2$, so T is approximately $N(0,1)$ by the CLT.

Definition (Two-sample comparison of means (large n_1, n_2))

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ use $T = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$, $T \dot{\sim} N(0,1)$; with observed value t:

Alternative	P-value	Rejection region
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq z_\alpha$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -z_\alpha$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq z_{\alpha/2}$

Example 6-6 (Tennis rackets and tennis elbow)

A new racket should reduce elbow stress. Of 220 players, 100 were randomly assigned the new racket and 120 the old; the elbow force (kgf) in a forehand stroke was:

	n_i	$\bar{x}_i$	s_i
New racket	100	25.2	5.1
Old racket	120	27.9	5.6

(a) Test at level 5% whether the mean forces differ.
(b) Find the P-value of (a).
(c) Find a 95% confidence interval for the difference of means.
Solution.
(a) $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$. The samples are large, and

$$t = \frac{(25.2 - 27.9) - 0}{\sqrt{5.1^2/100 + 5.6^2/120}} = -3.74.$$

The rejection region is $|t| \geq z_{0.025} = 1.96$; since $3.74 \geq 1.96$, H_0 is rejected at 5%.
(b) $P = P(|T| \geq 3.74)$ with $T \dot{\sim} N(0,1)$: $P < 0.0006$ from the table, and $P = 0.00018$ by software.
(c) $(25.2 - 27.9) \pm 1.96\sqrt{5.1^2/100 + 5.6^2/120} = -2.7 \pm 1.42$, i.e. $(-4.12, -1.28)$.

6.3.2 Normal populations: Welch's approximation

Even when n_1, n_2 are not large, T is approximately a t distribution with suitably chosen degrees of freedom. For normal populations with $n_1, n_2 \geq 5$,

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \dot{\sim} t(df), \qquad df = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{S_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{S_2^2}{n_2}\right)^2}.$$

Definition (Two-sample comparison of means (normal populations))

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2}(df)\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

with df computed from s_1^2, s_2^2 by the formula above.
(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ use $T = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \dot{\sim} t(df)$; with observed value t:

Alternative	P-value	Rejection region
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq t_\alpha(df)$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -t_\alpha(df)$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\|T\| \geq \|t\|)$	$\|t\| \geq t_{\alpha/2}(df)$

Normality matters little for moderate n_1, n_2 unless Q–Q plots contradict it. For non-integer df use the nearest tabulated value, interpolate, or use software.

Example 6-7 (Grocery spending offline vs. online)

Is grocery spending lower offline than online? 18 and 20 random people bought two weeks' groceries for a family of four offline and online (thousand won):

Offline	169	206	257	294	252	283	240	207	230	183
	298	269	256	277	300	126	318	184		
Online	289	334	278	268	336	438	388	388	394	394
	425	386	356	342	305	365	355	312	209	458

(a) Test at level 5% whether offline spending is lower.
(b) Find the P-value of (a).
Solution.
(a) $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 < \mu_2$ (1 = offline, 2 = online). Summary: $n_1 = 18$, $\bar{x}_1 = 241.6$, $s_1 = 52.82$; $n_2 = 20$, $\bar{x}_2 = 351.0$, $s_2 = 61.92$. The Q–Q plots of both groups are close to lines. Then

$$t = \frac{(241.6 - 351.0) - 0}{\sqrt{52.82^2/18 + 61.92^2/20}} = -5.875, \qquad df = \frac{\left(\frac{52.82^2}{18} + \frac{61.92^2}{20}\right)^2}{\frac{1}{17}\left(\frac{52.82^2}{18}\right)^2 + \frac{1}{19}\left(\frac{61.92^2}{20}\right)^2} = 35.91.$$

The rejection region is $t \le -t_{0.05}(35.91)$, where $t_{0.05}(35.91) \approx 1.620$ by interpolation in the table (1.688 by software). In either case $t = -5.875$ lies in the rejection region, so H_0 is rejected at 5%.
(b) $P = P(T \le -5.875)$, $T \sim t(35.91)$; the table gives $P < 0.0005$. There is clear evidence that offline grocery spending is lower.

6.3.3 Normal populations with a common variance

If $\sigma_1^2 = \sigma_2^2 = \sigma^2$, both S_1^2 and S_2^2 estimate σ^2, and combining them is more efficient. Weighting by degrees of freedom gives the **pooled sample variance**

$$S_p^2 = \frac{n_1 - 1}{(n_1 - 1) + (n_2 - 1)} S_1^2 + \frac{n_2 - 1}{(n_1 - 1) + (n_2 - 1)} S_2^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2},$$

with $n_1 + n_2 - 2$ degrees of freedom (Theorem 4-9).

	Population 1	Population 2	Pooled
Variance	σ^2	σ^2	
Sample variance	S_1^2	S_2^2	S_p^2
Degrees of freedom	$n_1 - 1$	$n_2 - 1$	$n_1 + n_2 - 2$

Under a common variance the standardized difference is $\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n_1 + 1/n_2}}$; replacing σ by S_p,

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2)$$

for normal populations, by Theorem 4-11.

Definition (Two-sample comparison (normal populations, common variance))

(a) A $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2}(n_1+n_2-2)\, s_p \sqrt{\frac{1}{n_1}+\frac{1}{n_2}}.$$

(b) For $H_0 : \mu_1 - \mu_2 = \delta_0$ use $T = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{S_p\sqrt{1/n_1 + 1/n_2}}$; with observed value t:

Alternative	P-value	Rejection region
$H_1 : \mu_1 - \mu_2 > \delta_0$	$P(T \geq t)$	$t \geq t_\alpha(n_1+n_2-2)$
$H_1 : \mu_1 - \mu_2 < \delta_0$	$P(T \leq t)$	$t \leq -t_\alpha(n_1+n_2-2)$
$H_1 : \mu_1 - \mu_2 \neq \delta_0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n_1+n_2-2)$

This is the **pooled two-sample t test**; the previous one, with adjusted degrees of freedom and no common-variance assumption, is the **unpooled two-sample t test**.

Pooled or unpooled?

The pooled test needs equal variances and degrades badly otherwise, especially for very different n_1, n_2; since equal variances are hard to verify, also examine the unpooled test.

Example 6-8 (Semaglutide and weight loss)

Twelve similar volunteers were split at random into two groups of 6: one received weekly semaglutide injections, the other a placebo, for 12 weeks with diet and exercise controlled. Weight loss (kg):

Semaglutide	5.12	4.87	5.64	5.01	5.49	5.25
Placebo	3.62	3.98	4.14	3.25	4.42	3.60

(a) Test at level 5% whether the mean weight losses differ.
(b) Find the P-value of (a).
(c) Find a 95% confidence interval for the difference of means.
Solution.
(a) $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$. Summary: $n_1 = n_2 = 6$, $\bar{x}_1 = 5.23$, $s_1 = 0.29$; $\bar{x}_2 = 3.84$, $s_2 = 0.42$. The Q–Q plots support normality and the standard deviations are similar, so we assume a common variance:

$$s_p^2 = \frac{(6-1)\times 0.29^2 + (6-1)\times 0.42^2}{6+6-2} = 0.13025, \qquad s_p = \sqrt{0.13025} = 0.361,$$

$$t = \frac{(5.23-3.84)-0}{0.361\sqrt{\frac{1}{6}+\frac{1}{6}}} = 6.67.$$

The rejection region is $|t| \geq t_{0.025}(10) = 2.228$; since $6.67 > 2.228$, H_0 is rejected at 5%.
(b) $P = P(|T| \geq 6.67)$, $T \sim t(10)$: $P < 0.001$. There is strong evidence that the mean weight losses differ.
(c) $(5.23-3.84) \pm 2.228 \times 0.361 \times \sqrt{\frac{1}{6}+\frac{1}{6}} = 1.39 \pm 0.46$, i.e. (0.93, 1.85).

The two-sample t tests and intervals are not very sensitive to the normality assumption and are approximately valid for large samples. What really matters is that the experiment was randomized.

6.4 Inference about population variances

Motivation. Variability itself matters (parts of the right mean size can still be defective); normal variances use chi-square and F.

6.4.1 One variance

For a random sample of size n from $N(\mu, \sigma^2)$ with μ, σ unknown, Theorem 4-8 gives

$$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2(n-1).$$

With the quantiles $\chi^2_{1-\alpha/2}(n-1)$ and $\chi^2_{\alpha/2}(n-1)$,

$$P\Big(\chi^2_{1-\alpha/2}(n-1) \le (n-1)\frac{S^2}{\sigma^2} \le \chi^2_{\alpha/2}(n-1)\Big) = 1-\alpha,$$

and solving the inequalities for σ^2,

$$P\Big((n-1)\frac{S^2}{\chi^2_{\alpha/2}(n-1)} \le \sigma^2 \le (n-1)\frac{S^2}{\chi^2_{1-\alpha/2}(n-1)}\Big) = 1-\alpha.$$

Definition (Confidence interval for σ^2 (normal population))

A $100(1-\alpha)\%$ confidence interval for σ^2 is

$$\Big((n-1)\frac{s^2}{\chi^2_{\alpha/2}(n-1)},\ (n-1)\frac{s^2}{\chi^2_{1-\alpha/2}(n-1)}\Big).$$

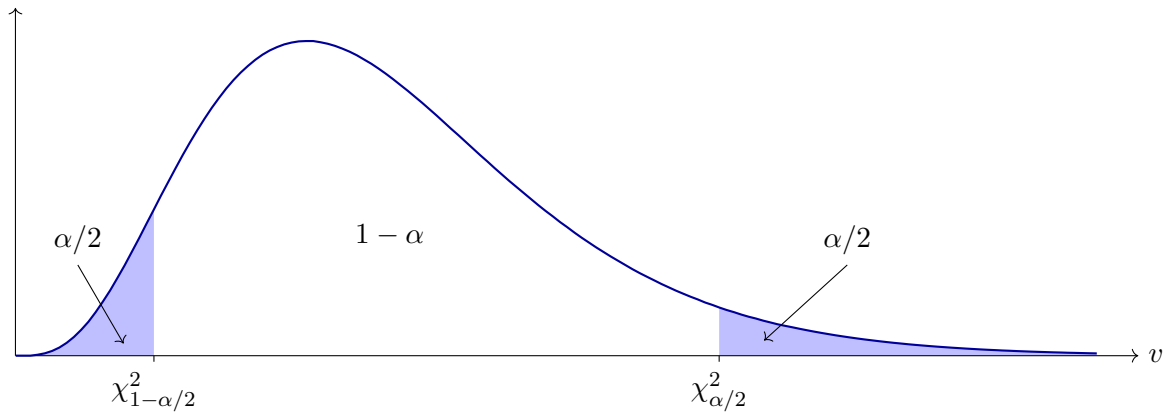


Figure 6: Quantiles of $\chi^2(9)$ used for a 90% interval for σ^2 ($\alpha = 0.1$): $\chi^2_{0.95}(9) = 3.33$, $\chi^2_{0.05}(9) = 16.92$.

The test of a variance based on the chi-square distribution is the **chi-square test**.

Definition (Chi-square test for σ^2 (normal population))

For $H_0 : \sigma^2 = \sigma_0^2$ use $\chi^2 = (n-1)\dfrac{S^2}{\sigma_0^2}$, which is $\chi^2(n-1)$ under H_0. With observed value χ_0^2:

Alternative	P-value	Rejection region at level α
$H_1:\sigma^2>\sigma_0^2$	$P(\chi^2\ge\chi_0^2)$	$\chi_0^2\ge\chi_\alpha^2(n-1)$
$H_1:\sigma^2<\sigma_0^2$	$P(\chi^2\le\chi_0^2)$	$\chi_0^2\le\chi_{1-\alpha}^2(n-1)$
$H_1:\sigma^2\ne\sigma_0^2$	$2P(\chi^2\ge\chi_0^2)$ or $2P(\chi^2\le\chi_0^2)$	$\chi_0^2\le\chi_{1-\alpha/2}^2(n-1)$ or $\chi_0^2\ge\chi_{\alpha/2}^2(n-1)$

For the two-sided alternative, compute both $2P(\chi^2\ge\chi_0^2)$ and $2P(\chi^2\le\chi_0^2)$ and use the one smaller than 1.

Example 6-9 (Coffee grinder)

A café uses 10 g of beans per Americano, but the grinder dose varies; a standard deviation above 1.5 g is considered abnormal. Ten doses (g):

11.15	9.62	9.83	10.85	12.66	9.51	11.23	9.38	9.11	8.56

Past records indicate a normal distribution.
(a) Test at level 5% whether the grinder is abnormal.
(b) Find the P-value of (a).
(c) Find a 90% confidence interval for the standard deviation of the dose.
Solution.
(a) $H_0:\sigma^2=1.5^2$, $H_1:\sigma^2>1.5^2$. Here $\bar{x}=10.19$, $s=1.24$, and

$$\chi_0^2=(n-1)\frac{s^2}{\sigma_0^2}=9\times\frac{1.24^2}{1.5^2}=6.15.$$

The rejection region is $\chi_0^2\ge\chi_{0.05}^2(9)=16.92$; since $6.15<16.92$, H_0 is not rejected at 5%.
(b) $P=P(\chi^2\ge6.15)$, $\chi^2\sim\chi^2(9)$; software gives $P=0.72$. There is no clear evidence of abnormal variation.
(c) With $\chi_{0.05}^2(9)=16.92$ and $\chi_{0.95}^2(9)=3.33$, the 90% interval for σ^2 is

$$\Big(9\times\frac{1.24^2}{16.92},\ 9\times\frac{1.24^2}{3.33}\Big)=(0.818,\ 4.156),$$

so the interval for σ is $(\sqrt{0.818},\sqrt{4.156})=(0.90,\ 2.04)$.

Sensitivity to normality

Unlike t methods, chi-square inference on a variance depends heavily on normality; check it first (the Q–Q plot for Example 6-9 is roughly linear).

6.4.2 Ratio of two variances

For independent random samples $X_{11},\dots,X_{1n_1}$ and $X_{21},\dots,X_{2n_2}$ from normal populations, with sample variances S_1^2,S_2^2, Theorem 4-13 gives

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}\sim F(n_1-1,n_2-1).$$

Therefore

$$P\Big(F_{1-\alpha/2}(n_1-1,n_2-1)\le\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}\le F_{\alpha/2}(n_1-1,n_2-1)\Big)=1-\alpha,$$

and solving for σ_1^2/σ_2^2,

$$P\Big(\frac{S_1^2}{S_2^2}\cdot\frac{1}{F_{\alpha/2}(n_1-1,n_2-1)}\le\frac{\sigma_1^2}{\sigma_2^2}\le\frac{S_1^2}{S_2^2}\cdot\frac{1}{F_{1-\alpha/2}(n_1-1,n_2-1)}\Big)=1-\alpha.$$

By Theorem 4-12, $F_{1-\alpha/2}(n_1-1,n_2-1)=1/F_{\alpha/2}(n_2-1,n_1-1)$.

Definition (Confidence interval for σ_1^2/σ_2^2 (normal populations))

A $100(1-\alpha)\%$ confidence interval for σ_1^2/σ_2^2 is

$$\Big(\frac{s_1^2}{s_2^2}\times\frac{1}{F_{\alpha/2}(n_1-1,n_2-1)},\ \frac{s_1^2}{s_2^2}\times F_{\alpha/2}(n_2-1,n_1-1)\Big).$$

The test of σ_1^2/σ_2^2 based on the F distribution is the F **test**.

Definition (F test for the ratio of variances (normal populations))

For $H_0:\sigma_1^2=\sigma_2^2$ use $F=S_1^2/S_2^2$, which is $F(n_1-1,n_2-1)$ under H_0. With observed value f:

Alternative	P-value	Rejection region at level α
$H_1:\sigma_1^2>\sigma_2^2$	$P(F\ge f)$	$f\ge F_{\alpha}(n_1-1,n_2-1)$
$H_1:\sigma_1^2<\sigma_2^2$	$P(F\le f)$	$f\le 1/F_{\alpha}(n_2-1,n_1-1)$
$H_1:\sigma_1^2\ne\sigma_2^2$	$2P(F\ge f)$ or $2P(F\le f)$	$f\le 1/F_{\alpha/2}(n_2-1,n_1-1)$ or $f\ge F_{\alpha/2}(n_1-1,n_2-1)$

Example 6-10 (Polymer injection pressure)

Cracks in concrete are repaired by injecting a polymer. Injection pressures (10^6 psi) for two polymers:

Epoxy	1.75	2.12	2.05	1.97
MMA prepolymer	1.77	1.59	1.70	1.69

(a) Test at level 5% whether the variabilities of the injection pressures differ.
(b) Find a 95% confidence interval for the ratio of standard deviations.
Solution.
(a) $H_0:\sigma_1^2=\sigma_2^2$, $H_1:\sigma_1^2\ne\sigma_2^2$. Summary: $n_1=n_2=4$, $\bar{x}_1=1.97$, $s_1=0.160$; $\bar{x}_2=1.69$, $s_2=0.074$. The Q–Q plots do not contradict normality. The observed statistic is $f=0.160^2/0.074^2=4.675$. Since $F_{0.025}(3,3)=15.44$, the 5% rejection region is $f\le 1/15.44$ or $f\ge 15.44$. As $1/15.44<4.675<15.44$, H_0 is not rejected at 5%.
(b) The 95% interval for σ_1^2/σ_2^2 is

$$\Big(\frac{0.160^2}{0.074^2}\times\frac{1}{15.44},\ \frac{0.160^2}{0.074^2}\times 15.44\Big)=(0.301,\ 72.182),$$

so that for σ_1/σ_2 is $(\sqrt{0.301},\sqrt{72.182})=(0.549,\ 8.496)$.

Chapter 7. Analysis of Discrete Data

7.1 Estimation and testing of a population proportion

Motivation. Sample counts of an attribute are binomial; normal approximation gives intervals and tests for proportions.

Let p be the population proportion with an attribute and X the count in a random sample of size n; $X \sim B(n,p)$, and p is estimated by the **sample proportion**

$$\hat{p} = \frac{X}{n}, \qquad \mathrm{Var}(\hat{p}) = \frac{\mathrm{Var}(X)}{n^2} = \frac{p(1-p)}{n}.$$

The standard error $\mathrm{SE}(\hat{p}) = \sqrt{p(1-p)/n}$ depends on the unknown p, so it is estimated by plugging in $\hat{p}$:

$$\widehat{\mathrm{SE}}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

By the normal approximation to the binomial, $X \approx N(np, np(1-p))$ for large n, and

$$\frac{X-np}{\sqrt{n\hat{p}(1-\hat{p})}} = \frac{\hat{p}-p}{\sqrt{\hat{p}(1-\hat{p})/n}} \;\approx\; N(0,1), \qquad \text{so} \qquad P\Big(-z_{\alpha/2} \le \frac{\hat{p}-p}{\sqrt{\hat{p}(1-\hat{p})/n}} \le z_{\alpha/2}\Big) \approx 1-\alpha.$$

Definition (Estimating p (large sample))

(i) Estimator and standard error: $\hat{p} = X/n$, $\widehat{\mathrm{SE}}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n}$.
(ii) $100(1-\alpha)\%$ margin of error: $z_{\alpha/2}\,\widehat{\mathrm{SE}}(\hat{p})$.
(iii) $100(1-\alpha)\%$ confidence interval:

$$\Big(\hat{p} - z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}\,,\;\; \hat{p} + z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}\Big).$$

Rule of thumb

The normal approximation is usually considered adequate when $n\hat{p} \ge 5$ and $n(1-\hat{p}) \ge 5$.

Example 7-1 (Homecoming rate in Seoul)

To estimate the proportion p of Seoul households that will travel home for Chuseok, 500 households were sampled by simple random sampling; 79 said they would go. Find a 95% CI for p.
Solution. The population is huge, so treat the sample as a random sample from an infinite population. $\hat{p} = 79/500 = 0.158$, $\widehat{\mathrm{SE}}(\hat{p}) = \sqrt{0.158(0.842)/500} = 0.016$. Since $n\hat{p} \ge 5$ and $n(1-\hat{p}) \ge 5$, the 95% CI is

$$(0.158 - 1.96 \times 0.016,\; 0.158 + 1.96 \times 0.016) = (0.126,\; 0.190).$$

7.1.1 Choosing the sample size

To keep the $100(1-\alpha)\%$ margin of error at most d we need $z_{\alpha/2}\sqrt{p(1-p)/n} \le d$. Since p is unknown and $p(1-p) \le 1/4$ for all p, it suffices that $z_{\alpha/2}\sqrt{1/4}/\sqrt{n} \le d$.

Definition (Sample size for estimating p)

To make the margin of error at most d (equivalently, the CI length at most $2d$):
(i) with a prior estimate p^* of p: $n \geq p^*(1-p^*)(z_{\alpha/2}/d)^2$;
(ii) with no prior information: $n \geq \frac{1}{4}(z_{\alpha/2}/d)^2$.

Why (ii) suffices. $p(1-p) = \frac{1}{4} - (p-\frac{1}{2})^2 \leq \frac{1}{4}$, so if $n \geq \frac{1}{4}(z_{\alpha/2}/d)^2$ then $z_{\alpha/2}\sqrt{p(1-p)/n} \leq z_{\alpha/2}\sqrt{1/(4n)} \leq d$ whatever the true p. ■

Example 7-2 (Poll for a district election)

Two candidates A and B run for district mayor. With no prior information, how large a sample is needed so that the 95% margin of error for A's support rate is within 3%?
Solution. $n \geq \frac{1}{4}(1.96/0.03)^2 = 1067.1$, so sample $n = 1068$ people.

7.1.2 Tests for p

Consider $H_0 : p = p_0$ vs. $H_1 : p > p_0$. Large observed counts x are evidence against H_0, so the P-value is $P = P(X \geq x)$ with $X \sim B(n, p_0)$, computed exactly from the binomial; reject at level α iff $P \leq \alpha$.

Definition (Small-sample test for p)

Test statistic: the count X, observed value x; $X \sim B(n, p_0)$ under $H_0 : p = p_0$.

H_1	P-value	reject at level α
(a) $p > p_0$	$P(X \geq x)$	$P \leq \alpha$
(b) $p < p_0$	$P(X \leq x)$	$P \leq \alpha$
(c) $p \neq p_0$	$P(X \leq y(x)) + P(X \geq x)$ if $x \geq np_0$	$P \leq \alpha$
	$P(X \leq x) + P(X \geq y(x))$ if $x < np_0$	

In (c), $y(x)$ is the integer on the opposite side of np_0 from x that is the largest ($x \geq np_0$) or smallest ($x < np_0$) nonnegative integer k with $P(X = k) \leq P(X = x)$.

Since the exact P-value is practical only for small n, this is called the **small-sample test** for a proportion.

Example 7-3 (A cheaper cutting tool)

With the current tool the defect rate is 5%. A cheaper tool may raise it, so 20 items were produced with it; 3 were defective. Test at level 5%.
Solution. $H_0 : p = 0.05$ vs. $H_1 : p > 0.05$, $n = 20$, $x = 3$. $P = P(X \geq 3) = 1 - P(X \leq 2) = 1 - 0.925 = 0.075$ for $X \sim B(20, 0.05)$. As $0.075 > 0.05$, H_0 is not rejected: this trial does not show that the cheaper tool raises the defect rate.

For large n, under H_0,

$$Z = \frac{X - np_0}{\sqrt{np_0(1-p_0)}} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \approx N(0,1),$$

usable when $np_0 \geq 5$ and $n(1-p_0) \geq 5$. Note that the null value p_0 (not $\hat{p}$) is used in the standard error.

Definition (Large-sample test for p ($np_0 \geq 5$, $n(1-p_0) \geq 5$))

$Z = (\hat{p} - p_0)/\sqrt{p_0(1-p_0)/n}$ with observed value z.

H_1	P-value	rejection region
(a) $p > p_0$	$P(Z \geq z)$	$z \geq z_\alpha$
(b) $p < p_0$	$P(Z \leq z)$	$z \leq -z_\alpha$
(c) $p \neq p_0$	$P(\|Z\| \geq \|z\|)$	$\|z\| \geq z_{\alpha/2}$

Example 7-4 (Has the homecoming rate dropped?)

In the past the Chuseok homecoming rate was 20%. Using the data of Example 7-1, test at level 1% whether the rate has decreased.
Solution. $H_0 : p = 0.2$ vs. $H_1 : p < 0.2$; $n = 500$, $X = 79$, $np_0 = 100 \geq 5$, $n(1-p_0) = 400 \geq 5$.

$$z = \frac{79/500 - 0.2}{\sqrt{0.2 \times 0.8/500}} = -2.348 \;\leq\; -z_{0.01} = -2.327,$$

so H_0 is rejected at level 1%: the homecoming rate is significantly lower than before.

7.2 Comparing two population proportions

Motivation. Two independent samples give inference on $p_1 - p_2$ by the same normal approximation.

Independent random samples of sizes n_1, n_2 are drawn from two populations; $X_1 \sim B(n_1, p_1)$ and $X_2 \sim B(n_2, p_2)$ are independent. The estimator of $p_1 - p_2$ is

$$\widehat{p_1 - p_2} = \hat{p}_1 - \hat{p}_2 = \frac{X_1}{n_1} - \frac{X_2}{n_2}, \qquad \mathrm{Var}(\hat{p}_1 - \hat{p}_2) = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2},$$

by independence. For large n_1, n_2, $\hat{p}_i \approx N(p_i, p_i(1-p_i)/n_i)$ independently, so $\hat{p}_1 - \hat{p}_2$ is approximately normal and

$$\frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}} \;\approx\; N(0,1).$$

Definition (Estimating $p_1 - p_2$ (large n_1, n_2))

(i) $\widehat{p_1 - p_2} = \hat{p}_1 - \hat{p}_2$, $\widehat{\mathrm{SE}} = \sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}$.
(ii) $100(1-\alpha)\%$ CI for $p_1 - p_2$: $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\,\widehat{\mathrm{SE}}(\widehat{p_1 - p_2})$.

Example 7-5 (1954 Salk polio vaccine trial)

In a double-blind trial, a control group received placebo and a treatment group the vaccine. Let p_1, p_2 be the polio rates in the control and treatment groups.

	sample size	children with polio
Control	201,229	115
Treatment	200,745	33

Estimate $p_1 - p_2$ and give a 95% CI.
Solution. $\hat{p}_1 = 115/201229 = 5.71 \times 10^{-4}$, $\hat{p}_2 = 33/200745 = 1.64 \times 10^{-4}$, so $\hat{p}_1 - \hat{p}_2 =$

4.07×10^{-4} with $\widehat{\mathrm{SE}} = 0.60 \times 10^{-4}$. The 95% CI is $4.07 \times 10^{-4} \pm 1.96 \times 0.60 \times 10^{-4}$, i.e.

$$2.89 \times 10^{-4} < p_1 - p_2 < 5.25 \times 10^{-4}.$$

7.2.1 Testing $H_0 : p_1 = p_2$

Under H_0 write $p = p_1 = p_2$; then $\mathrm{Var}(\hat{p}_1 - \hat{p}_2) = p(1-p)(1/n_1 + 1/n_2)$. Pooling the two samples, $X_1 + X_2$ of the $n_1 + n_2$ units have the attribute, so p is estimated by the **pooled sample proportion**

$$\hat{p} = \frac{X_1 + X_2}{n_1 + n_2}, \qquad \widehat{\mathrm{Var}}(\hat{p}_1 - \hat{p}_2) = \hat{p}(1-\hat{p})\Big(\frac{1}{n_1} + \frac{1}{n_2}\Big).$$

Definition (Large-sample test of $H_0 : p_1 = p_2$)

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(n_1^{-1} + n_2^{-1})}}, \qquad \hat{p} = \frac{X_1 + X_2}{n_1 + n_2}.$$

With observed value z and $Z \approx N(0,1)$ under H_0:
(a) $H_1 : p_1 > p_2$: $P = P(Z \geq z)$, reject if $z \geq z_\alpha$;
(b) $H_1 : p_1 < p_2$: $P = P(Z \leq z)$, reject if $z \leq -z_\alpha$;
(c) $H_1 : p_1 \neq p_2$: $P = P(|Z| \geq |z|)$, reject if $|z| \geq z_{\alpha/2}$.

Example 7-6 (Is the vaccine effective?)

With the data of Example 7-5, test whether the vaccine is effective.
Solution. $H_0 : p_1 = p_2$ vs. $H_1 : p_1 > p_2$ (p_1 placebo, p_2 vaccine). $\hat{p} = (115 + 33)/(201229 + 200745) = 3.68 \times 10^{-4}$,

$$z = \frac{5.71 \times 10^{-4} - 1.64 \times 10^{-4}}{\sqrt{\hat{p}(1-\hat{p})(n_1^{-1} + n_2^{-1})}} = 6.727, \qquad P = P(Z \geq 6.727) = 8.66 \times 10^{-12}.$$

H_0 is rejected decisively: the 1954 trial strongly indicates that the vaccine prevents polio.

Example 7-7 (Parole success of murderers)

Random samples of 42 impulsive and 40 premeditated murderers released on parole were followed; success means no reoffence. Test at level 5% whether the success rates differ.

	success	failure	size
Impulsive	13	29	$n_1 = 42$
Premeditated	22	18	$n_2 = 40$
Total	35	47	$n = 82$

Solution. $H_0 : p_1 = p_2$ vs. $H_1 : p_1 \neq p_2$. $\hat{p}_1 = 13/42 = 0.310$, $\hat{p}_2 = 22/40 = 0.550$, $\hat{p} = 35/82 = 0.427$, and

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(n_1^{-1} + n_2^{-1})}} = -2.20, \qquad |z| \geq z_{0.025} = 1.96,$$

so H_0 is rejected: the parole success rates differ significantly.

7.3 Comparing several populations with categorical data

Motivation. For more populations or categories, comparing observed with expected counts gives the chi-square statistic.

The idea: reject H_0 (all populations have the same category proportions) when the **observed frequencies** differ much from the **estimated expected frequencies** under H_0, where

$$\text{estimated expected frequency} = (\text{sample size}) \times (\text{estimated probability under } H_0).$$

Illustration 7-1: expected counts for Example 7-7

Under $H_0 : p_1 = p_2$ the common success rate is estimated by $\hat{p} = 0.427$, so

$$\hat{E}_{11} = n_1\hat{p} = 42 \times 0.427 = 17.93, \qquad \hat{E}_{12} = n_1(1-\hat{p}) = 24.07,$$
$$\hat{E}_{21} = n_2\hat{p} = 40 \times 0.427 = 17.08, \qquad \hat{E}_{22} = n_2(1-\hat{p}) = 22.92.$$

Observed (expected): impulsive 13 (17.93), 29 (24.07); premeditated 22 (17.08), 18 (22.92). The discrepancy statistic is

$$\sum_{i=1}^{2}\sum_{j=1}^{2}\frac{(O_{ij}-\hat{E}_{ij})^2}{\hat{E}_{ij}} = \frac{(13-17.93)^2}{17.93} + \cdots + \frac{(18-22.92)^2}{22.92} = 4.84 = (-2.20)^2 = z^2.$$

The identity $\sum\sum(O_{ij}-\hat{E}_{ij})^2/\hat{E}_{ij} = z^2$ holds in general for 2×2 tables; since $Z \approx N(0,1)$ under H_0, the statistic is approximately $\chi^2(1)$. It is called the **chi-square statistic**,

$$\chi^2 = \sum_i\sum_j \frac{(O_{ij}-\hat{E}_{ij})^2}{\hat{E}_{ij}}.$$

In Example 7-7 the 5% critical region is $\chi^2 \geq \chi^2_{0.05}(1) = 3.84$, equivalently $z^2 \geq z^2_{0.025} = 1.96^2 = 3.84$.

Fact ($\chi^2 = z^2$ for a 2×2 table)

For two samples with pooled proportion $\hat{p}$, the chi-square statistic of the 2×2 table equals the square of the two-sample z-statistic.

Proof. [Supplement] Write $q = 1 - \hat{p}$ and $D = X_1 - n_1\hat{p}$. Because $X_1 + X_2 = (n_1 + n_2)\hat{p}$, we have $X_2 - n_2\hat{p} = -D$, and for the failures $(n_i - X_i) - n_iq = -(X_i - n_i\hat{p})$. Hence every $|O_{ij} - \hat{E}_{ij}| = |D|$, and

$$\chi^2 = D^2\Big(\frac{1}{n_1\hat{p}} + \frac{1}{n_1q} + \frac{1}{n_2\hat{p}} + \frac{1}{n_2q}\Big) = \frac{D^2}{\hat{p}q}\Big(\frac{1}{n_1} + \frac{1}{n_2}\Big).$$

On the other hand $\hat{p}_1 - \hat{p}_2 = D/n_1 + D/n_2 = D(n_1^{-1} + n_2^{-1})$, so

$$z^2 = \frac{D^2(n_1^{-1}+n_2^{-1})^2}{\hat{p}q(n_1^{-1}+n_2^{-1})} = \frac{D^2}{\hat{p}q}\Big(\frac{1}{n_1} + \frac{1}{n_2}\Big) = \chi^2.$$

■

7.3.1 Several binomial populations

Observing r binomial populations gives an $r \times 2$ **contingency table** with counts O_{i1} (success), O_{i2} (failure), row totals n_i, column totals $O_{\cdot 1}, O_{\cdot 2}$ and grand total n. We test

$$H_0 : p_1 = p_2 = \cdots = p_r \qquad \text{vs.} \qquad H_1 : \text{not all } p_i \text{ are equal.}$$

Under H_0 the common rate is estimated by the pooled proportion $\hat{p} = (O_{11} + \cdots + O_{r1})/(n_1 + \cdots + n_r) = O_{\cdot 1}/n$, and

$$\hat{E}_{i1} = n_i \hat{p} = n_i \frac{O_{\cdot 1}}{n}, \qquad \hat{E}_{i2} = n_i(1-\hat{p}) = n_i \frac{O_{\cdot 2}}{n} \qquad (i = 1, \ldots, r).$$

For large samples, $\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{2} (O_{ij} - \hat{E}_{ij})^2 / \hat{E}_{ij} \approx \chi^2(r-1)$ under H_0 (this is a known result; it rests on the multivariate normal approximation).

Definition (Comparing r binomial populations ($\hat{E}_{ij} \geq 5$))

For $H_0 : p_1 = \cdots = p_r$ vs. H_1: at least one differs, use

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{2} \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}}, \qquad \hat{E}_{ij} = n_i \frac{O_{\cdot j}}{n}.$$

With observed value χ_0^2: $P = P(\chi^2 \geq \chi_0^2)$ for $\chi^2 \sim \chi^2(r-1)$; reject at level α iff $\chi_0^2 \geq \chi_\alpha^2(r-1)$.

Example 7-8 (Siting an LNG storage base)

Residents of three candidate regions were sampled (400, 350, 350 people) and asked whether they favour an LNG storage base. Do the approval rates differ?

	for	against	size
Region 1	198	202	400
Region 2	140	210	350
Region 3	133	217	350
Total	471	629	1100

Solution. $H_0 : p_1 = p_2 = p_3$. Expected counts: $\hat{E}_{11} = 400 \cdot 471/1100 = 171.27$, $\hat{E}_{12} = 228.73$, $\hat{E}_{21} = \hat{E}_{31} = 149.86$, $\hat{E}_{22} = \hat{E}_{32} = 200.14$. Then

$$\chi_0^2 = \frac{(198 - 171.27)^2}{171.27} + \cdots + \frac{(217 - 200.14)^2}{200.14} = 11.75,$$

and $P = P(\chi^2(2) \geq 11.75) = 0.003$ (from the table, $0.0025 < P < 0.005$). The approval rates differ significantly across the regions.

7.3.2 Several multinomial populations

Populations classified into c **categories** are **multinomial populations**; r samples give an $r \times c$ table of **categorical data**, with proportions $p_{i1}, \ldots, p_{ic}$ ($\sum_j p_{ij} = 1$) in population i. The test of

$$H_0 : p_{1j} = p_{2j} = \cdots = p_{rj} \ (j = 1, \ldots, c) \qquad \text{vs.} \qquad H_1 : \text{not } H_0$$

is the **test of homogeneity**. Under H_0 the common proportion p_j of category j is estimated by $\hat{p}_j = O_{\cdot j}/n$, so $\hat{E}_{ij} = n_i \hat{p}_j = n_i O_{\cdot j}/n$, and for large samples

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}} \approx \chi^2((r-1)(c-1)).$$

Counting degrees of freedom

There are $r(c-1)$ free proportions in general and $c-1$ under H_0, so $(r-1)(c-1)$ degrees of freedom ($r-1$ for $c=2$).

Definition (Test of homogeneity ($\hat{E}_{ij} \geq 5$))

$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}(O_{ij}-\hat{E}_{ij})^2/\hat{E}_{ij}$ with $\hat{E}_{ij} = n_i O_{\cdot j}/n$. With observed value χ_0^2: $P = P(\chi^2 \geq \chi_0^2)$, $\chi^2 \sim \chi^2((r-1)(c-1))$; reject iff $\chi_0^2 \geq \chi_\alpha^2((r-1)(c-1))$.

Example 7-9 (Odour from an industrial complex)

In three regions near an industrial complex, 97, 95 and 99 people reported how often they notice a bad odour (1 daily, 2 weekly, 3 monthly, 4 less, 5 never). Test at level 1% whether the frequency differs by region.

	1	2	3	4	5	size
Region 1	20	28	23	14	12	97
Region 2	14	34	21	14	12	95
Region 3	4	12	10	20	53	99
Total	38	74	54	48	77	291

Solution. $\hat{p}_j = 38/291, 74/291, 54/291, 48/291, 77/291$; $\hat{E}_{1j} = 97\hat{p}_j$, $\hat{E}_{2j} = 95\hat{p}_j$, $\hat{E}_{3j} = 99\hat{p}_j$:

	1	2	3	4	5
Region 1	12.67	24.67	18.00	16.00	25.67
Region 2	12.41	24.16	17.63	15.67	25.14
Region 3	12.93	25.18	18.37	16.33	26.20

All expected counts exceed 5. $\chi_0^2 = (20-12.67)^2/12.67 + \cdots + (53-26.20)^2/26.20 = 71.36 \geq \chi_{0.01}^2(8) = 20.09$, so H_0 is rejected: odour perception differs by region.

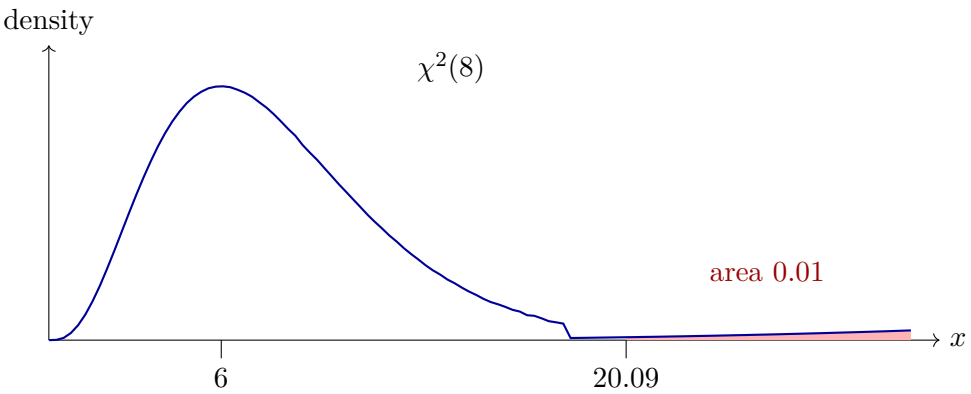


Figure 7: The $\chi^2(8)$ density with the 1% rejection region of Example 7-9; the observed $\chi_0^2 = 71.36$ lies far to the right.

7.4 Test of independence with categorical data

Motivation. For one sample cross-classified by two attributes, the same statistic tests independence, with expected counts from the margins.

Units are classified by A $(A_1, \dots, A_r)$ and B $(B_1, \dots, B_c)$, with cell probabilities p_{ij}, margins $p_{i\cdot}$, $p_{\cdot j}$, and sample counts O_{ij} with margins $O_{i\cdot}, O_{\cdot j}$. Independence means

$$H_0 : p_{ij} = p_{i\cdot}\, p_{\cdot j} \quad \text{for all } i, j, \qquad H_1 : p_{ij} \neq p_{i\cdot}\, p_{\cdot j} \text{ for at least one } (i, j).$$

Under H_0 we estimate each margin separately and multiply:

$$\hat{p}_{ij} = \hat{p}_{i\cdot}\, \hat{p}_{\cdot j} = \frac{O_{i\cdot}}{n} \cdot \frac{O_{\cdot j}}{n}, \qquad \hat{E}_{ij} = n\hat{p}_{ij} = \frac{O_{i\cdot}\, O_{\cdot j}}{n}.$$

For large n, $\chi^2 = \sum_i \sum_j (O_{ij} - \hat{E}_{ij})^2 / \hat{E}_{ij} \approx \chi^2((r-1)(c-1))$ under H_0: in general there are $rc - 1$ free cell probabilities, under H_0 only $(r-1) + (c-1)$, and $rc - 1 - (r-1) - (c-1) = (r-1)(c-1)$.

Definition (Test of independence ($\hat{E}_{ij} \geq 5$))

For $H_0 : p_{ij} = p_{i\cdot} p_{\cdot j}$ $(i = 1, \dots, r,\ j = 1, \dots, c)$ vs. H_1 : not H_0,

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}}, \qquad \hat{E}_{ij} = \frac{O_{i\cdot} O_{\cdot j}}{n}.$$

$P = P(\chi^2 \geq \chi_0^2)$ with $\chi^2 \sim \chi^2((r-1)(c-1))$; reject iff $\chi_0^2 \geq \chi_\alpha^2((r-1)(c-1))$.

Example 7-10 (Car size and commuting distance)

250 solo car commuters in a metropolitan area were classified by car size and commuting distance. Test at level 5% whether they are related.

	<15 km	15–30 km	≥ 30 km	total
Light	6	27	19	52
Small	8	36	17	61
Mid-size	21	45	33	99
Large	14	18	6	38
Total	49	126	75	250

Solution. $\hat{E}_{ij} = O_{i\cdot} O_{\cdot j}/250$: row 1: 10.19, 26.21, 15.60; row 2: 11.96, 30.74, 18.30; row 3: 19.40, 49.90, 29.70; row 4: 7.45, 19.15, 11.40 — all ≥ 5. Then

$$\chi_0^2 = \frac{(6 - 10.19)^2}{10.19} + \cdots + \frac{(6 - 11.40)^2}{11.40} = 14.16 \ \geq \ \chi_{0.05}^2(6) = 12.59,$$

so H_0 is rejected: car size and commuting distance are related.

Homogeneity vs. independence

Independence and homogeneity (§7.3) use the same statistic and procedure but different sampling: for homogeneity the row totals n_i are fixed (separate samples), for independence only n (one cross-classified sample).

Chapter 8. Correlation and Regression Analysis

8.1 Introduction

Motivation. Paired data (IQ and grades, temperature and strength): correlation and regression make their relation precise.

Correlation analysis infers the strength of a linear relation via r; **regression analysis** models one variable as a function of another for prediction: **simple linear regression** for a line, **multiple regression** for several explanatory variables.

The terms go back to Francis Galton, who studied the inheritance of sweet-pea seed size and of heights of parents and children; "regression" comes from his phrase *regression toward the mean.*

8.2 Correlation analysis

Motivation. r estimates ρ; under bivariate normality a t-test decides whether $\rho \neq 0$.

$\rho = \mathrm{Corr}(X, Y) \in [-1, 1]$ is the **population correlation coefficient**; values near ± 1 mean a strong linear relation, near 0 almost none. It is estimated by the **sample correlation coefficient**

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\,\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

$r \to \pm 1$ as points cluster around a line; $r \approx 0$ means a weak *linear* relation only (a parabola can have $r = 0$).

Illustration 8-1: cycling distance and calories

Twenty randomly chosen adult men cycled for 10 minutes; distance x (km) and calories burned y (kcal) were recorded.

x	3.4	3.0	4.3	4.5	3.8	4.3	3.2	3.4	3.1	3.1
y	96	84	112	123	111	104	95	105	91	79
x	3.2	3.5	3.2	3.2	3.1	3.4	3.2	3.1	3.9	4.6
y	98	103	96	82	97	87	83	86	100	126

The scatter plot shows an increasing linear trend, and

$$r = \frac{110.88}{\sqrt{5.10}\,\sqrt{3258.55}} = 0.860.$$

Inference on ρ assumes the population is **bivariate normal**, i.e. (X, Y) has density

$$f(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\Big\{-\frac{1}{2(1-\rho^2)}\Big[\Big(\tfrac{x-\mu_X}{\sigma_X}\Big)^2 - 2\rho\,\tfrac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \Big(\tfrac{y-\mu_Y}{\sigma_Y}\Big)^2\Big]\Big\},$$

whose marginals are $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$.

Definition (Test for correlation (bivariate normal population))

For $H_0 : \rho = 0$ use $T = \sqrt{n-2}\,\dfrac{r}{\sqrt{1-r^2}}$, which is $t(n-2)$ under H_0. With observed value t:

H_1	P-value	rejection region
(a) $\rho > 0$	$P(T \geq t)$	$t \geq t_\alpha(n-2)$
(b) $\rho < 0$	$P(T \leq t)$	$t \leq -t_\alpha(n-2)$
(c) $\rho \neq 0$	$P(\lvert T\rvert \geq \lvert t\rvert)$	$\lvert t\rvert \geq t_{\alpha/2}(n-2)$

Why $T \sim t(n-2)$

[Supplement] Regress y on x (§8.3). One checks algebraically that $\sqrt{n-2}\,r/\sqrt{1-r^2} = \hat{\beta}/(\hat{\sigma}/\sqrt{S_{(xx)}})$, using $r^2 = \text{SSR/SST}$ and $1 - r^2 = \text{SSE/SST}$. Under bivariate normality with $\rho = 0$, conditionally on the x_i the y_i follow the simple linear regression model with $\beta = 0$, so by §8.4 this ratio is $t(n-2)$ given the x's; since this conditional law does not depend on the x's, it is also the unconditional law.

Example 8-1 (Is distance correlated with calories?)

In Illustration 8-1 assume bivariate normality and test at level 5% whether the two variables are correlated.

Solution. $H_0 : \rho = 0$ vs. $H_1 : \rho \neq 0$; $n = 20$, $r = 0.860$:

$$t = \sqrt{20-2}\,\frac{0.860}{\sqrt{1-0.860^2}} = 7.150, \qquad |t| \geq t_{0.025}(18) = 2.101.$$

H_0 is rejected; the P-value is 0.000001, strong evidence of correlation.

8.3 The simple linear regression model and its fit

Motivation. A straight-line model for the mean response is fitted by least squares.

The variable that influences is the **explanatory variable**; the one influenced is the **response variable**. Regression first examines the scatter plot to choose a tentative model.

Illustration 8-2: temperature and hot-drink sales

A coffee shop recorded the daily maximum temperature x (°C) and hot-drink sales y (10,000 won) on 12 random days.

x	27.1	9.1	19.2	29.1	22.0	34.9
y	46.8	63.7	46.0	46.1	50.3	34.7
x	16.5	7.6	13.1	2.6	23.5	5.6
y	57.8	63.2	52.7	62.9	45.3	69.9

The points scatter around an unknown line $y = \alpha + \beta x$ with negative slope (Figure 8).

We regard y as an observed value of $Y = \alpha + \beta x + e$, where $E(e) = 0$, $\text{Var}(e) = \sigma^2$; equivalently $E(Y \mid X = x) = \alpha + \beta x$ and $\text{Var}(Y \mid X = x) = \sigma^2$. This is the **simple linear regression model**; the responses $y_1, \ldots, y_n$ at $x_1, \ldots, x_n$ are observed independently.

Definition (Assumptions of the simple linear regression model)

$Y_i = \alpha + \beta x_i + e_i$ $(i = 1, \dots, n)$, where
(i) linearity: $E(e_i) = 0$, i.e. $E(Y_i \mid X = x_i) = \alpha + \beta x_i$ $(-\infty < \alpha, \beta < \infty)$;
(ii) equal variance: $\mathrm{Var}(e_1) = \cdots = \mathrm{Var}(e_n) = \sigma^2 > 0$;
(iii) independence: $e_1, \dots, e_n$ are independent.

$y = \alpha + \beta x$ is the **population regression line**, α, β the **population regression coefficients**, σ^2 the **error variance**; they are estimated by the **method of least squares**.

Definition 8-1 (Method of least squares)

In the model $Y_i = \alpha + \beta x_i + e_i$, the estimators of α, β are the values minimizing

$$\sum_{i=1}^{n} \{Y_i - (\alpha + \beta x_i)\}^2.$$

Setting the partial derivatives in α and β to zero gives the **normal equations**

$$\begin{cases} -2\sum(y_i - \hat\alpha - \hat\beta x_i) = 0 \\ -2\sum x_i(y_i - \hat\alpha - \hat\beta x_i) = 0 \end{cases} \quad \text{i.e.} \quad \begin{cases} n\hat\alpha + (\sum x_i)\hat\beta = \sum y_i \\ (\sum x_i)\hat\alpha + (\sum x_i^2)\hat\beta = \sum x_i y_i. \end{cases}$$

The solutions are the **least squares estimators** (LSE), and $\hat y = \hat\alpha + \hat\beta x$ is the **least squares regression line**; $\hat y_i$ estimates the *mean* response $E(Y_i \mid X = x_i)$. With

$$S_{(xy)} = \sum(x_i - \bar x)(y_i - \bar y) = \sum x_i y_i - \tfrac{1}{n}(\sum x_i)(\sum y_i), \qquad S_{(xx)} = \sum(x_i - \bar x)^2 = \sum x_i^2 - \tfrac{1}{n}(\sum x_i)^2,$$

and $S_{(yy)}$ defined likewise, the solution is as follows.

Proposition (Least squares estimators and line)

(i) $\hat\alpha = \bar y - \hat\beta \bar x, \quad \hat\beta = S_{(xy)}/S_{(xx)}$.
(ii) The least squares line is $\hat y = \hat\alpha + \hat\beta x = \bar y + \hat\beta(x - \bar x)$.
(iii) The minimum value of the sum of squares is $\mathrm{SSE} = S_{(yy)} - \hat\beta^2 S_{(xx)} = S_{(yy)} - S_{(xy)}^2/S_{(xx)}$.

Proof. [Supplement] Assume $S_{(xx)} > 0$ (not all x_i equal). Put $Q(a, b) = \sum(y_i - a - bx_i)^2$.

Step 1. Write $y_i - a - bx_i = \{(y_i - \bar y) - b(x_i - \bar x)\} + (\bar y - a - b\bar x)$. The cross term vanishes because $\sum(y_i - \bar y) = \sum(x_i - \bar x) = 0$, so

$$Q(a, b) = \sum\{(y_i - \bar y) - b(x_i - \bar x)\}^2 + n(\bar y - a - b\bar x)^2.$$

Step 2. Expanding the first sum and completing the square in b,

$$\sum\{(y_i - \bar y) - b(x_i - \bar x)\}^2 = S_{(yy)} - 2bS_{(xy)} + b^2 S_{(xx)} = S_{(xx)}\Big(b - \frac{S_{(xy)}}{S_{(xx)}}\Big)^2 + S_{(yy)} - \frac{S_{(xy)}^2}{S_{(xx)}}.$$

Step 3. Both squares in Q are ≥ 0, and both vanish exactly when $b = S_{(xy)}/S_{(xx)}$ and $a = \bar y - b\bar x$. Hence this is the unique minimizer, giving (i) and (ii), and the minimum is $S_{(yy)} - S_{(xy)}^2/S_{(xx)}$; since $\hat\beta^2 S_{(xx)} = S_{(xy)}^2/S_{(xx)}$, (iii) follows. The minimizer of course also solves the normal equations. ■

Example 8-2 (Fitting the hot-drink data)

For Illustration 8-2, with temperature as X and sales as Y, find the LSE of α, β and draw the fitted line.

Solution. $\bar{x} = 210.3/12 = 17.525$, $\bar{y} = 639.4/12 = 53.283$,

$$S_{(xy)} = 10134.21 - \tfrac{1}{12} \times 210.3 \times 639.4 = -1071.275,$$
$$S_{(xx)} = 4826.67 - \tfrac{1}{12} \times 210.3^2 = 1141.162.$$

So $\hat{\beta} = S_{(xy)}/S_{(xx)} = -0.939$, $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 69.739$, and $\hat{y} = 69.739 - 0.939x$ (Figure 8).

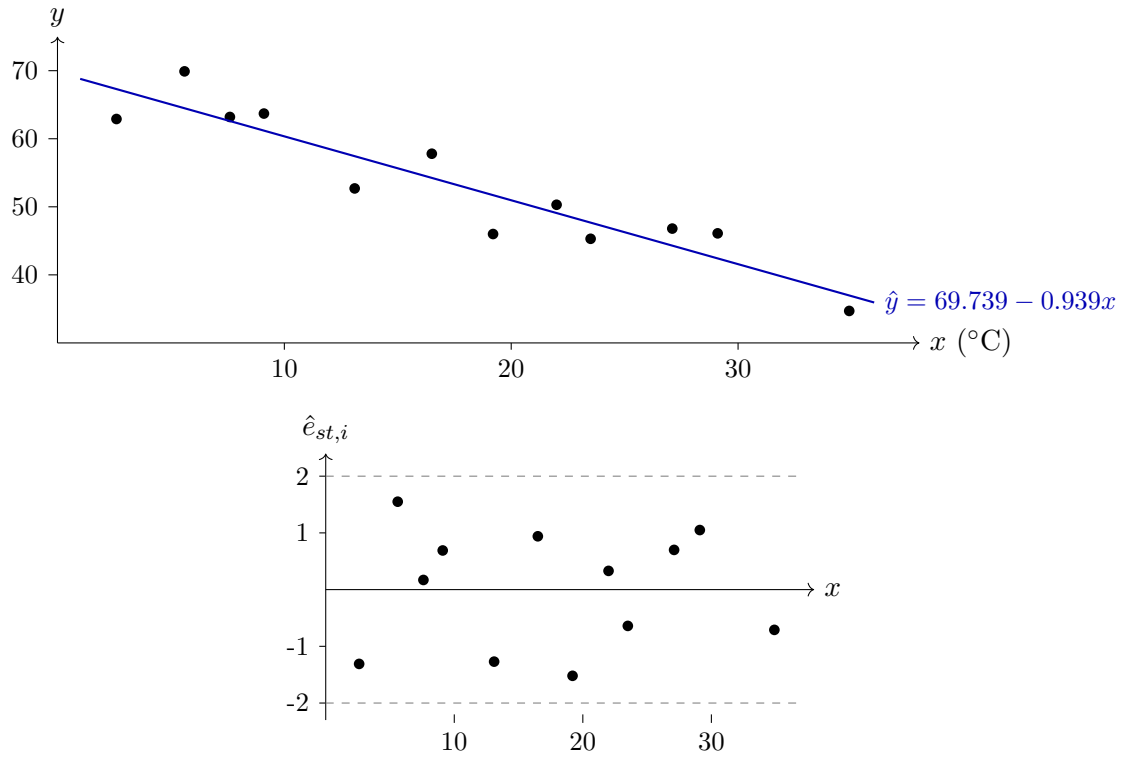


Figure 8: Top: hot-drink data with least squares line (Example 8-2). Bottom: studentized residuals (§8.5), patternless and within ±2.

8.3.1 Residuals and the estimate of σ^2

The **residuals** $\hat{e}_i = y_i - \hat{y}_i$ behave like observed errors. The **residual (error) sum of squares** is $\text{SSE} = \sum(y_i - \hat{y}_i)^2$, and σ^2 is estimated by

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n-2} = \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$$

The divisor $n-2$ is the degrees of freedom of SSE, and $\hat{\sigma}^2$ is called the **mean squared error**, MSE.

Definition (Residual sum of squares and the estimate of σ^2)

(i) $\text{SSE} = \sum(y_i - \hat{y}_i)^2 = \sum\{y_i - \bar{y} - \hat{\beta}(x_i - \bar{x})\}^2 = S_{(yy)} - \hat{\beta}^2 S_{(xx)} = S_{(yy)} - S^2_{(xy)}/S_{(xx)}$.
(ii) $\hat{\sigma}^2 = \text{MSE} = \text{SSE}/(n-2)$.

Proposition (MSE is unbiased)

Under the model assumptions, $E(\text{SSE}) = (n-2)\sigma^2$, so $E(\text{MSE}) = \sigma^2$.

Proof. [Supplement] Treat the x_i as fixed.

Step 1. $\sum(Y_i - \bar{Y})^2 = \sum Y_i^2 - n\bar{Y}^2$ with $E(Y_i^2) = \sigma^2 + \mu_i^2$, $\mu_i = \alpha + \beta x_i$, and $E(\bar{Y}^2) = \sigma^2/n + \bar{\mu}^2$. Hence $E\,S_{(YY)} = (n-1)\sigma^2 + \sum(\mu_i - \bar{\mu})^2 = (n-1)\sigma^2 + \beta^2 S_{(xx)}$.

Step 2. By the next proposition, $E(\hat{\beta}) = \beta$ and $\text{Var}(\hat{\beta}) = \sigma^2/S_{(xx)}$, so $E(\hat{\beta}^2)S_{(xx)} = \sigma^2 + \beta^2 S_{(xx)}$.

Step 3. Subtracting, $E(\text{SSE}) = E\,S_{(YY)} - E(\hat{\beta}^2)S_{(xx)} = (n-2)\sigma^2$. ■

Example 8-3 (Estimating σ^2)

For the hot-drink data, estimate the error variance.
Solution. $S_{(xy)} = -1071.275$, $S_{(xx)} = 1141.162$, $S_{(yy)} = 1160.837$, so

$$\text{SSE} = 1160.837 - \frac{(-1071.275)^2}{1141.162} = 155.169, \qquad \hat{\sigma}^2 = \text{MSE} = \frac{155.169}{10} = 15.517.$$

8.3.2 Decomposition of the total sum of squares

How much of the variation of y does the fitted line explain? The deviation of y_i from $\bar{y}$ splits as

$$y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y}), \tag{8.1}$$

splitting the **total deviation** $y_i - \bar{y}$ into residual and explained parts (**decomposition** (8.1)). Squaring and summing gives a Pythagoras-type identity, with **total sum of squares** $\text{SST} = \sum(y_i - \bar{y})^2$ (d.f. $n-1$) and **regression sum of squares** $\text{SSR} = \sum(\hat{y}_i - \bar{y})^2$ (d.f. 1).

Proposition (Decomposition of the total sum of squares)

$$\underbrace{\sum_{i=1}^{n}(y_i - \bar{y})^2}_{\text{SST},\ n-1} = \underbrace{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}_{\text{SSE},\ n-2} + \underbrace{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}_{\text{SSR},\ 1}.$$

Moreover $\text{SSR} = \hat{\beta}^2 S_{(xx)} = S_{(xy)}^2/S_{(xx)}$ and $\text{SST} = S_{(yy)}$.

Proof. [Supplement]

Step 1. The normal equations say $\sum \hat{e}_i = 0$ and $\sum x_i \hat{e}_i = 0$, where $\hat{e}_i = y_i - \hat{y}_i$.

Step 2. Since $\hat{y}_i - \bar{y} = \hat{\beta}(x_i - \bar{x})$, the cross term is

$$\sum(y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) = \hat{\beta}\sum \hat{e}_i(x_i - \bar{x}) = \hat{\beta}\Big(\sum x_i\hat{e}_i - \bar{x}\sum \hat{e}_i\Big) = 0.$$

Step 3. Squaring (8.1) and summing therefore gives $\text{SST} = \text{SSE} + \text{SSR}$, and $\text{SSR} = \hat{\beta}^2\sum(x_i - \bar{x})^2 = \hat{\beta}^2 S_{(xx)} = S_{(xy)}^2/S_{(xx)}$. The degrees of freedom $n-1$ and $n-2$ reflect the one and two estimated parameters $(\bar{y};\ \hat{\alpha}, \hat{\beta})$, and $1 = (n-1) - (n-2)$. ■

The larger the share of SSR in SST, the better the model explains the data. This share is the **coefficient of determination**

$$r^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}} = \frac{S_{(xy)}^2}{S_{(xx)}S_{(yy)}} = \Big(\frac{S_{(xy)}}{\sqrt{S_{(xx)}}\sqrt{S_{(yy)}}}\Big)^2,$$

which is exactly the square of the sample correlation coefficient. Values near 1 mean the points cluster tightly around the line.

Example 8-4 (Coefficient of determination)

Compute and interpret r^2 for the hot-drink data.
Solution. $r^2 = (-1071.275)^2/(1141.162 \times 1160.837) = 0.866$: the line explains 86.6% of the variation in sales, and together with the scatter plot the model describes the data well.

Statistics alone can mislead (Anscombe)

Both data sets have $\hat{y} = 3 + 0.5x$, SST $= 41.29$, SSR $= 27.51$, SSE $= 13.78$, $\hat{\sigma}^2 = 1.53$, $r^2 = 0.67$; yet A is linear with scatter and B a smooth curve. Always inspect scatter plot and residuals (§8.5).

A	x	10	8	13	9	11	14	6	4	12	7	5
	y	8.04	6.95	7.58	8.81	8.33	9.96	7.24	4.26	10.84	4.82	5.68
B	x	10	8	13	9	11	14	6	4	12	7	5
	y	9.14	8.14	8.74	8.77	9.26	8.10	6.13	3.10	9.13	7.26	4.74

8.4 Inference in simple linear regression

Motivation. Normal errors give intervals and tests for slope, intercept and mean response.

From now on assume, in addition,

$$Y_i = \alpha + \beta x_i + e_i, \qquad e_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2) \qquad (i = 1, \ldots, n).$$

8.4.1 Significance of the regression line

The line is useful iff $\beta \neq 0$, so we test $H_0 : \beta = 0$ vs. $H_1 : \beta \neq 0$. A significant line makes SSR large relative to SSE. Under H_0,

$$F = \frac{\text{SSR}/1}{\text{SSE}/(n-2)} \sim F(1, n-2),$$

with P-value $P(F \geq f)$ and critical region $f \geq F_\alpha(1, n-2)$, summarized in an **analysis of variance table** (a **mean square** is a sum of squares over its d.f.).

Definition (ANOVA table for the significance of the regression line)

Source	SS	d.f.	Mean square	F	P-value
Regression	SSR	1	MSR $=$ SSR$/1$	$f =$ MSR$/$MSE	$P(F \geq f)$
Residual	SSE	$n-2$	MSE $=$ SSE$/(n-2)$		
Total	SST	$n-1$			

Example 8-5 (Is the regression significant?)

Test the significance of the regression line for the hot-drink data.
Solution. SST $= S_{(yy)} = 1160.837$, SSR $= (-1071.275)^2/1141.162 = 1005.668$, SSE $=$ 155.169, so MSR $= 1005.668$, MSE $= 15.517$ and $f = 64.811$. Since $P(F(1, 10) \geq 64.811) =$

$0.00001 < 0.001$, the line is highly significant.

Source	SS	d.f.	MS	F	P
Regression	1005.668	1	1005.668	64.811	0.00001
Residual	155.169	10	15.517		
Total	1160.837	11			

8.4.2 Inference on the slope β

Proposition (Distribution of $\hat{\beta}$)

$E(\hat{\beta}) = \beta$, $\mathrm{Var}(\hat{\beta}) = \sigma^2/S_{(xx)}$, $\hat{\beta}$ is normal, and

$$\frac{\hat{\beta} - \beta}{\hat{\sigma}/\sqrt{S_{(xx)}}} \sim t(n-2).$$

Proof. [Supplement]

Step 1. Let $c_i = (x_i - \bar{x})/S_{(xx)}$. Since $\sum (x_i - \bar{x})\bar{Y} = 0$, $\hat{\beta} = \sum c_i Y_i$, a linear combination of independent normal variables, hence normal. Note $\sum c_i = 0$, $\sum c_i x_i = \sum c_i (x_i - \bar{x}) = 1$, $\sum c_i^2 = 1/S_{(xx)}$.

Step 2. $E(\hat{\beta}) = \sum c_i(\alpha + \beta x_i) = \alpha \sum c_i + \beta \sum c_i x_i = \beta$.

Step 3. By independence, $\mathrm{Var}(\hat{\beta}) = \sum c_i^2 \sigma^2 = \sigma^2/S_{(xx)}$. Thus $Z = (\hat{\beta} - \beta)/(\sigma/\sqrt{S_{(xx)}}) \sim N(0,1)$.

Step 4. It is a standard fact of normal linear models (proved via Cochran's theorem) that $\mathrm{SSE}/\sigma^2 \sim \chi^2(n-2)$ and that SSE is independent of $(\hat{\alpha}, \hat{\beta})$. Then $(\hat{\beta} - \beta)/(\hat{\sigma}/\sqrt{S_{(xx)}}) = Z/\sqrt{(\mathrm{SSE}/\sigma^2)/(n-2)}$ is $t(n-2)$ by definition of the t distribution. (Independence is at least easy to see at the level of covariances: $\mathrm{Cov}(\hat{e}_j, \hat{\beta}) = \sigma^2(c_j - \frac{1}{n}\sum_i c_i - (x_j - \bar{x})\sum_i c_i^2) = 0$, and jointly normal uncorrelated variables are independent.) ■

Definition (Inference on β)

(i) $100(1-\alpha)\%$ CI: $\hat{\beta} \pm t_{\alpha/2}(n-2)\,\hat{\sigma}/\sqrt{S_{(xx)}}$.

(ii) For $H_0 : \beta = \beta_0$: $T = (\hat{\beta} - \beta_0)/(\hat{\sigma}/\sqrt{S_{(xx)}}) \sim t(n-2)$. With observed t:
- (a) $H_1 : \beta > \beta_0$: $P = P(T \geq t)$, reject if $t \geq t_\alpha(n-2)$;
- (b) $H_1 : \beta < \beta_0$: $P = P(T \leq t)$, reject if $t \leq -t_\alpha(n-2)$;
- (c) $H_1 : \beta \neq \beta_0$: $P = P(|T| \geq |t|)$, reject if $|t| \geq t_{\alpha/2}(n-2)$.

Fact ($F = T^2$)

For $H_0 : \beta = 0$, the ANOVA statistic equals the square of the t-statistic, so the two-sided t-test and the F-test are the same test.

Proof. Using $\mathrm{SSR} = \hat{\beta}^2 S_{(xx)}$ and $\mathrm{SSE}/(n-2) = \mathrm{MSE} = \hat{\sigma}^2$,

$$F = \frac{\mathrm{SSR}/1}{\mathrm{SSE}/(n-2)} = \frac{\hat{\beta}^2 S_{(xx)}}{\mathrm{MSE}} = \frac{\hat{\beta}^2}{\mathrm{MSE}/S_{(xx)}} = \left(\frac{\hat{\beta} - 0}{\hat{\sigma}/\sqrt{S_{(xx)}}}\right)^2 = T^2.$$

Since $t_{\alpha/2}(n-2)^2 = F_\alpha(1, n-2)$, the rejection regions coincide. ■

Example 8-6 (Inference on the slope)

For the hot-drink data ($n = 12$, $\hat\beta = -0.939$, $S_{(xx)} = 1141.162$, $\hat\sigma^2 = 15.517$):
(a) test the significance of the line by a t-test at level 1% and relate it to the F-test;
(b) test at level 5% whether a 10°C rise lowers mean sales by more than 90,000 won;
(c) give a 95% CI for β and interpret it.
Solution.
(a) $H_0 : \beta = 0$ vs. $H_1 : \beta \neq 0$: $t = -0.939/\sqrt{15.517/1141.162} = -8.053$, and $|t| \geq t_{0.005}(10) = 3.169$, so the line is significant at 1%. Also $F = T^2$ (here $8.053^2 \approx 64.8$), so the F-test is the same test.
(b) Sales are in units of 10,000 won, so the claim is a drop of more than 0.9 per 1°C: $H_0 : \beta = -0.9$ vs. $H_1 : \beta < -0.9$. $t = (-0.939 + 0.9)/\sqrt{15.517/1141.162} = -0.334 > -t_{0.05}(10) = -1.812$, so H_0 is not rejected; there is no clear evidence of a drop exceeding 90,000 won per 10°C.
(c) $t_{0.025}(10) = 2.228$: $-0.939 \pm 2.228\sqrt{15.517/1141.162} = -0.939 \pm 0.260$, i.e. $(-1.199, -0.679)$. With 95% confidence, each 1°C rise lowers mean hot-drink sales by between 6,790 and 11,990 won.

8.4.3 Inference on the mean response $\alpha + \beta x$

The mean response $E(Y \mid X = x) = \alpha + \beta x$ is estimated by $\hat\alpha + \hat\beta x$.

Proposition (Distribution of $\hat\alpha + \hat\beta x$)

$E(\hat\alpha + \hat\beta x) = \alpha + \beta x$, $\operatorname{Var}(\hat\alpha + \hat\beta x) = \left(\frac{1}{n} + \frac{(x-\bar x)^2}{S_{(xx)}}\right)\sigma^2$, it is normal, and

$$\frac{(\hat\alpha + \hat\beta x) - (\alpha + \beta x)}{\hat\sigma\sqrt{n^{-1} + (x-\bar x)^2/S_{(xx)}}} \sim t(n-2).$$

Proof. [Supplement]

Step 1. $\hat\alpha + \hat\beta x = \bar Y + \hat\beta(x - \bar x) = \sum(\frac{1}{n} + (x-\bar x)c_i)Y_i$ is a linear combination of independent normals, hence normal, with mean $(\alpha + \beta\bar x) + \beta(x - \bar x) = \alpha + \beta x$ (using $E\hat\beta = \beta$).

Step 2. $\operatorname{Cov}(\bar Y, \hat\beta) = \sum_i \frac{1}{n} c_i \sigma^2 = 0$ since $\sum c_i = 0$. Therefore

$$\operatorname{Var}(\hat\alpha + \hat\beta x) = \operatorname{Var}(\bar Y) + (x-\bar x)^2 \operatorname{Var}(\hat\beta) = \frac{\sigma^2}{n} + \frac{(x-\bar x)^2\sigma^2}{S_{(xx)}}.$$

Step 3. Standardizing and replacing σ by $\hat\sigma$, the $t(n-2)$ law follows exactly as in Step 4 of the previous proof (SSE is independent of $(\bar Y, \hat\beta)$). ■

Definition (Inference on the mean response $E(Y \mid X = x) = \alpha + \beta x$)

(i) $100(1-\alpha)\%$ CI: $(\hat\alpha + \hat\beta x) \pm t_{\alpha/2}(n-2)\,\hat\sigma\sqrt{n^{-1} + (x-\bar x)^2/S_{(xx)}}$.

(ii) For $H_0 : \alpha + \beta x = \mu_0$: $T = \dfrac{(\hat\alpha + \hat\beta x) - \mu_0}{\hat\sigma\sqrt{n^{-1} + (x-\bar x)^2/S_{(xx)}}} \sim t(n-2)$, with the usual one- or two-sided P-values and critical regions $t \ge t_\alpha(n-2)$, $t \le -t_\alpha(n-2)$, $|t| \ge t_{\alpha/2}(n-2)$.

Example 8-7 (Mean sales at given temperatures)

For the hot-drink data ($n = 12$, $\bar x = 17.525$, $S_{(xx)} = 1141.162$, $\hat y = 69.739 - 0.939x$, $\hat\sigma^2 = 15.517$):
(a) at $x = 25$°C, test at level 1% whether mean sales exceed 400,000 won;
(b) find 95% CIs for mean sales at $x = 10, 20, 30$°C and plot them.
Solution.
(a) $H_0 : \alpha + 25\beta = 40$ vs. $H_1 : \alpha + 25\beta > 40$; $t = 4.372 \ge t_{0.01}(10) = 2.764$, so H_0 is rejected: mean sales at 25°C clearly exceed 400,000 won.
(b) At $x = 10$:

$$(69.739 - 0.939 \times 10) \pm 2.228\sqrt{15.517}\sqrt{\tfrac{1}{12} + \tfrac{(10-17.525)^2}{1141.162}} = 60.349 \pm 3.200,$$

i.e. (57.149, 63.549). Similarly (48.345, 53.573) at $x = 20$ and (37.455, 45.683) at $x = 30$. Joining the upper and lower limits over all x gives a band, narrowest at $x = \bar x$ (a "bow-tie" shape).

The CI for the mean response is shorter when x is close to $\bar x$ and when $S_{(xx)}$ is large; so it pays to observe the response at widely spread x values around the x of interest.

8.4.4 Inference on the intercept α

Taking $x = 0$ above:

Definition (Inference on α)

(i) $100(1-\alpha)\%$ CI: $\hat\alpha \pm t_{\alpha/2}(n-2)\,\hat\sigma\sqrt{n^{-1} + \bar x^2/S_{(xx)}}$.
(ii) For $H_0 : \alpha = \alpha_0$: $T = (\hat\alpha - \alpha_0)/(\hat\sigma\sqrt{n^{-1} + \bar x^2/S_{(xx)}}) \sim t(n-2)$, with the usual P-values and critical regions.

Example 8-8 (CI for the intercept)

Find a 95% CI for α in the hot-drink model.
Solution. $\hat\alpha \pm t_{0.025}(10)\hat\sigma\sqrt{1/12 + 17.525^2/1141.162} = 69.739 \pm 5.210$, i.e. (64.529, 74.949).

8.5 Residual analysis in simple linear regression

Motivation. Residuals, proxies for the errors, are plotted to check the model assumptions of §8.4.

The full model is $Y_i = \alpha + \beta x_i + e_i$ with (i) linearity $E(e_i) = 0$;

(ii) equal variance $\mathrm{Var}(e_i) = \sigma^2$;

(iii) independence of $e_1, \dots, e_n$;

(iv) normality $e_i \sim N(0, \sigma^2)$. If they hold, $e_i/\mathrm{SD}(e_i) = (Y_i - (\alpha + \beta x_i))/\sigma$ are independent $N(0,1)$. Replacing e_i by $\hat e_i$ and dividing by an estimate of its standard deviation gives the **studentized**

residuals

$$\hat{e}_{st,i} = \frac{\hat{e}_i}{\widehat{\mathrm{SD}}(\hat{e}_i)} = \frac{y_i - \hat{y}_i}{\widehat{\mathrm{SD}}(Y_i - \hat{Y}_i)} \qquad (i = 1, \ldots, n),$$

which should look like n independent standard normal values ($\mathrm{Var}(\hat{e}_i) = \sigma^2\{1 - \frac{1}{n} - (x_i - \bar{x})^2/S_{(xx)}\}$ with $\hat{\sigma}^2$ plugged in). This is **residual analysis**, usually via a **residual plot** of $(x_i, \hat{e}_{st,i})$.

Definition (Residual analysis)

Compute $\hat{e}_{st,i} = \hat{e}_i/\widehat{\mathrm{SD}}(\hat{e}_i)$ and judge whether they look like n independent $N(0,1)$ observations. The model is adequate if the residuals
(i) are roughly symmetric about 0;
(ii) have constant spread across x;
(iii) show no systematic pattern;
(iv) lie about 95% within ± 2.

The hot-drink residual plot (Figure 8, bottom) satisfies (i)–(iv), so the model and §8.4 apply.

Recommended workflow

(i) Draw a scatter plot to see the rough linear trend.
(ii) Fit the least squares line and MSE, then check the studentized residuals (symmetric about 0, mostly within ± 2, no pattern).
(iii) If nothing is wrong, carry out CIs and tests.
Violations: a curve (linearity), a funnel (equal variance), runs or waves (independence), skew or outliers (normality).

Example 8-9 (Electricity bills and temperature)

Monthly mean temperature x (°C) and the monthly electricity bill y (1,000 won) of an apartment household:

x	y	x	y	x	y	x	y	x	y
−2.2	14.0	0.8	12.2	8.2	12.9	15.2	11.9	21.8	10.3
−1.9	13.1	1.8	13.0	9.2	11.3	16.3	10.9	22.3	9.1
−1.7	14.4	3.9	11.3	14.2	10.8	21.1	9.6	23.6	7.5
−1.3	13.1	7.0	10.5	14.5	10.0	21.1	8.1	23.6	10.5
−0.7	14.8	8.0	9.7	14.9	9.9	21.5	9.2	24.8	10.0

(a) Check by residual analysis whether simple linear regression is appropriate.
(b) Test the significance of the regression line at level 5%.
(c) Give a 95% CI for the mean bill when the mean temperature is 10°C.
Solution. The scatter plot shows a rough decreasing linear relation.
(a) $n = 25$, $\bar{x} = 11.44$, $\bar{y} = 11.124$, $S_{(xx)} = 2210.44$, $S_{(yy)} = 88.8856$, $S_{(xy)} = -374.9945$. So $\hat{\beta} = -0.17$, $\hat{\alpha} = 13.06$, $\hat{y} = 13.06 - 0.17x$, and $\hat{\sigma}^2 = \{S_{(yy)} - S_{(xy)}^2/S_{(xx)}\}/(n-2) = 1.0987$. The studentized residuals are symmetric about 0, have constant spread, show no pattern and all lie within ± 2: the model is adequate. Also $r^2 = S_{(xy)}^2/(S_{(xx)}S_{(yy)}) = 0.7157$, so about 72% of the variation is explained.
(b) SST = 88.8856, SSR $= S_{(xy)}^2/S_{(xx)} = 63.6165$, SSE = 25.2691 (d.f. 24, 1, 23); $f = (\mathrm{SSR}/1)/(\mathrm{SSE}/23) = 57.902 \geq F_{0.05}(1, 23) = 4.28$, $P = 0.0000001$: the line is significant.

Source	SS	d.f.	MS	F	P
Regression	63.6165	1	63.6165	57.902	0.0000001
Residual	25.2691	23	1.0987		
Total	88.8856	24			

(c) $\hat{\alpha} + 10\hat{\beta} = 13.06 - 1.7 = 11.36$ with standard error $\hat{\sigma}\sqrt{1/n + (10-\bar{x})^2/S_{(xx)}} = 0.2121$; $t_{0.025}(23) = 2.069$ gives 11.36 ± 0.44, i.e. (10.92, 11.80).

8.6 Multiple regression analysis

Motivation. Several explanatory variables in one model usually explain and predict better.

A regression model with two or more explanatory variables is a **multiple regression model**. With p explanatory variables, the **multiple linear regression model** is

$$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_p x_{pi} + e_i, \qquad e_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2) \quad (i = 1, \ldots, n),$$

with unknown real coefficients $\beta_0, \ldots, \beta_p$ and error variance $\sigma^2 > 0$.

8.6.1 Two explanatory variables

For $Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + e_i$, least squares minimizes $\sum\{Y_i - (\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})\}^2$. Setting the three partial derivatives to zero gives the normal equations

$$\begin{aligned} -2\sum(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i}) &= 0, \\ -2\sum x_{1i}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i}) &= 0, \\ -2\sum x_{2i}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i}) &= 0. \end{aligned}$$

The first gives $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}_1 - \hat{\beta}_2\bar{x}_2$; substituting into the other two yields

$$S_{(11)}\hat{\beta}_1 + S_{(12)}\hat{\beta}_2 = S_{(1y)}, \qquad S_{(12)}\hat{\beta}_1 + S_{(22)}\hat{\beta}_2 = S_{(2y)},$$

where $S_{(11)} = S_{(x_1x_1)}$, $S_{(22)} = S_{(x_2x_2)}$, $S_{(12)} = S_{(x_1x_2)}$, $S_{(1y)} = S_{(x_1y)}$, $S_{(2y)} = S_{(x_2y)}$.

Definition (LSE in multiple linear regression with two explanatory variables)

$\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$ solve

$$\begin{cases} S_{(11)}\hat{\beta}_1 + S_{(12)}\hat{\beta}_2 = S_{(1y)} \\ S_{(12)}\hat{\beta}_1 + S_{(22)}\hat{\beta}_2 = S_{(2y)} \\ \hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}_1 - \hat{\beta}_2\bar{x}_2 \end{cases}$$

and the regression plane $E(Y \mid x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ is estimated by $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$.

Derivation of the reduced equations. [Supplement] Substitute $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}_1 - \hat{\beta}_2\bar{x}_2$ into the second normal equation: $\sum x_{1i}\{(y_i - \bar{y}) - \hat{\beta}_1(x_{1i} - \bar{x}_1) - \hat{\beta}_2(x_{2i} - \bar{x}_2)\} = 0$. Since each bracketed deviation sums to zero over i, x_{1i} may be replaced by $x_{1i} - \bar{x}_1$, giving $S_{(1y)} - \hat{\beta}_1 S_{(11)} - \hat{\beta}_2 S_{(12)} = 0$. The third equation gives the other one in the same way. (As in §8.3, the sum of squares is a convex quadratic, so when $S_{(11)}S_{(22)} - S_{(12)}^2 > 0$ this unique stationary point is the minimum.) ■

Example 8-10 (Standing long jump, age and body fat)

For 10 adult men at a fitness centre, Y = standing long jump (cm), X_1 = age, X_2 = body-fat percentage. Fit the multiple linear regression model.

age	43	64	29	22	31	20	20	26	20	19
body fat	22.7	31.9	12.3	18.4	16.4	26.9	16.0	17.5	22.5	14.5
jump	206	120	250	224	219	195	260	190	226	222

Solution. $n = 10$, $\bar{x}_1 = 29.4$, $\bar{x}_2 = 19.91$, $\bar{y} = 211.2$; $S_{(11)} = 1824.4$, $S_{(22)} = 329.989$, $S_{(12)} = 472.56$, $S_{(1y)} = -3809.8$, $S_{(2y)} = -1723.02$. The normal equations

$$\begin{aligned} 1824.4\,\hat{\beta}_1 + 472.56\,\hat{\beta}_2 &= -3809.8, \\ 472.56\,\hat{\beta}_1 + 329.989\,\hat{\beta}_2 &= -1723.02, \\ \hat{\beta}_0 &= 211.2 - 29.4\hat{\beta}_1 - 19.91\hat{\beta}_2 \end{aligned}$$

give $\hat{\beta}_0 = 316.198$, $\hat{\beta}_1 = -1.1696$, $\hat{\beta}_2 = -3.5465$, so $\hat{y} = 316.198 - 1.1696x_1 - 3.5465x_2$.

In practice multiple regression is computed by software; understanding the model, the procedure and the concepts matters more than the arithmetic.

8.6.2 ANOVA in multiple regression

As before $y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})$ and

$$\underbrace{\sum (y_i - \bar{y})^2}_{\text{SST}} = \underbrace{\sum (y_i - \hat{y}_i)^2}_{\text{SSE}} + \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{\text{SSR}},$$

with degrees of freedom $n-1$, $n-p-1$ and p. ([Supplement] The proof is the same as in §8.3: the normal equations say that the residuals satisfy $\sum \hat{e}_i = 0$ and $\sum x_{ki}\hat{e}_i = 0$ for each k, so $\sum \hat{e}_i(\hat{y}_i - \bar{y}) = 0$ because $\hat{y}_i - \bar{y}$ is a linear combination of $1, x_{1i}, \ldots, x_{pi}$.) Define

$$\text{MSE} = \frac{\text{SSE}}{n-p-1}, \qquad \text{MSR} = \frac{\text{SSR}}{p}, \qquad \hat{\sigma}^2 = \text{MSE}, \qquad R^2 = \frac{\text{SSR}}{\text{SST}}.$$

(R^2 is the multiple-regression analogue of r^2.) To test the regression function,

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \qquad \text{vs.} \qquad H_1 : \text{not all } \beta_k \text{ are } 0,$$

use $F = \text{MSR}/\text{MSE}$, which is $F(p, n-p-1)$ under H_0; reject at level α if $f \geq F_\alpha(p, n-p-1)$, $P = P(F \geq f)$.

Definition (ANOVA table for multiple regression)

Source	SS	d.f.	MS	F	P-value
Regression	SSR	p	MSR	$f = \text{MSR}/\text{MSE}$	$P(F \geq f)$
Residual	SSE	$n-p-1$	MSE		
Total	SST	$n-1$			

Example 8-11 (Significance of the long-jump model)

For Example 8-10, SST = 13503.6 and SSE = 2936.9. Test the regression function at level 5% and compute R^2.

Solution. SSR = SST − SSE = 10566.7; with $p = 2$ the d.f. are $9, 7, 2$. MSR = 10566.7/2 = 5283.35, MSE = 2936.9/7 = 419.557, $f = 12.593 \geq F_{0.05}(2, 7) = 4.737$ ($P = 0.0048$): the regression function is significant.

Source	SS	d.f.	MS	F	P
Regression	10566.7	2	5283.35	12.593	0.0048
Residual	2936.9	7	419.557		
Total	13503.6	9			

$R^2 = \text{SSR}/\text{SST} = 0.7825$: about 78% of the variation is explained by the regression function.

Check the model first with studentized residuals $\hat{e}_{st,i} = (y_i - \hat{y}_i)/\widehat{\text{SD}}(Y_i - \hat{Y}_i)$, plotted against fitted values $(\hat{y}_i, \hat{e}_{st,i})$; for Example 8-10 no problem appears, so Example 8-11 is trustworthy.

Chapter 9. Analysis of Variance

9.1 What is analysis of variance?

Motivation. ANOVA asks whether a characteristic differs across conditions by splitting total variation into factor and error parts.

Comparing tensile strengths by production method, a difference may come from the method or from other sources (worker skill). Repeating each method splits the variation into a between-method part and a part from other sources; if the former is large relative to the latter, the method matters.

Analysis of variance (ANOVA, R. A. Fisher), the main tool of the **design of experiments** (DOE), decomposes the sum of squares into parts attributable to the factors and compares each factor's *mean square* (sum of squares over d.f.) with the error mean square. Here

$$\text{total sum of squares} = \text{sum of squares due to the factor} + \text{sum of squares due to error.}$$

ANOVA generalizes the two-sample test (Chapter 6) to three or more means. We treat *fixed* factors: one-way ANOVA (equal and unequal replication) and two-way ANOVA (without and with replication).

9.2 One-way analysis of variance

Motivation. One factor at $k \geq 3$ levels: are the k treatment means equal?

In a **one-way layout** factor A has k levels (treatments), treatment i replicated n_i times (typically $k = 3$–5, $n_i = 3$–10), all runs in random order: a **completely randomized design** (e.g. 15 numbered units for methods A, B, C with 5 runs each, ordered by lot). Statistically, these are independent random samples from k populations; $k = 2$ is the two-sample comparison.

9.2.1 Equal replication

With n replicates per treatment the data are y_{ij} ($i = 1, \ldots, k$ treatments, $j = 1, \ldots, n$), with treatment means and grand mean

$$\bar{y}_{i\cdot} = \frac{1}{n}\sum_{j=1}^{n} y_{ij}, \qquad \bar{y}_{\cdot\cdot} = \frac{1}{kn}\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij} \qquad (i = 1, \ldots, k).$$

Under treatment i, $Y_{ij} = \mu_i + e_{ij}$ with $e_{ij} \sim N(0, \sigma^2)$ independent. With overall mean $\mu = \frac{1}{k}\sum \mu_i$, $\alpha_i = \mu_i - \mu$ is the ith **treatment effect**, $\sum \alpha_i = 0$.

Definition (One-way ANOVA model (equal replication))

$$Y_{ij} = \mu + \alpha_i + e_{ij}, \qquad e_{ij} \sim N(0, \sigma^2) \text{ independent}, \qquad \sum_{i=1}^{k} \alpha_i = 0,$$

$i = 1, \ldots, k$, $j = 1, \ldots, n$. Equivalently:
(i) linearity $E(e_{ij}) = 0$, i.e. $E(Y_{ij}) = \mu + \alpha_i$;
(ii) equal variance $\mathrm{Var}(e_{ij}) = \sigma^2 > 0$;
(iii) independence of the e_{ij};
(iv) normality $e_{ij} \sim N(0, \sigma^2)$.

Testing whether the treatment effects differ is the same as testing whether the k population means differ. Decompose the total deviation:

$$y_{ij} - \bar{y}_{\cdot\cdot} = (y_{ij} - \bar{y}_{i\cdot}) + (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}). \tag{9.1}$$

$y_{ij} - \bar{y}_{i\cdot}$ is the **residual** and $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ the **deviation of the treatment effect**. Squaring and summing decomposes the **total sum of squares** SST into the **treatment** (**between**) **sum of squares** SS_{tr} and the **residual** (**within**) **sum of squares** SSE.

Proposition (Decomposition of the total sum of squares, one-way)

With $N = kn$ observations,

$$\underbrace{\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{\cdot\cdot})^2}_{\mathrm{SST},\ N-1} = \underbrace{n\sum_{i=1}^{k}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2}_{\mathrm{SS}_{tr},\ k-1} + \underbrace{\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i\cdot})^2}_{\mathrm{SSE},\ N-k}.$$

Proof. [Supplement] Square (9.1) and sum over i, j. The cross term is

$$2\sum_{i=1}^{k}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i\cdot}) = 0,$$

because the inner sum is $n\bar{y}_{i\cdot} - n\bar{y}_{i\cdot} = 0$. The middle square does not depend on j, so it contributes $n\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$. As for degrees of freedom: SST has $N-1$ (the deviations sum to 0), SS_{tr} has $k-1$ (k treatment means minus one constraint), and SSE has $N-k$ ($n-1$ in each of k groups); indeed $N-1 = (k-1)+(N-k)$. ∎

A sum of squares divided by its degrees of freedom is a **mean square**:

$$\mathrm{MS}_{tr} = \frac{\mathrm{SS}_{tr}}{k-1}, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{N-k}.$$

We test

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_k = 0 \qquad \text{vs.} \qquad H_1 : \text{at least one } \alpha_i \neq 0,$$

which is the same as $H_0 : \mu_1 = \cdots = \mu_k$. If the treatments matter, SS_{tr} takes a large share of SST, so large values of $F = \mathrm{MS}_{tr}/\mathrm{MSE}$ are evidence against H_0.

Proposition (Expected mean squares, one-way)

$E(\mathrm{MSE}) = \sigma^2$ and $E(\mathrm{MS}_{tr}) = \sigma^2 + \dfrac{n}{k-1}\sum_{i=1}^{k}\alpha_i^2$. Under H_0, $F = \mathrm{MS}_{tr}/\mathrm{MSE} \sim F(k-1, N-k)$.

Proof. [Supplement]

Step 1. For each i, $Y_{i1}, \ldots, Y_{in}$ is a random sample from $N(\mu_i, \sigma^2)$, so its sample variance has mean σ^2: $E\sum_j(Y_{ij} - \bar{Y}_{i\cdot})^2 = (n-1)\sigma^2$. Summing over i, $E(\mathrm{SSE}) = k(n-1)\sigma^2 = (N-k)\sigma^2$.

Step 2. The treatment means $\bar{Y}_{i\cdot} \sim N(\mu + \alpha_i, \sigma^2/n)$ are independent. For independent Z_i with means m_i and common variance v, $E\sum(Z_i - \bar{Z})^2 = (k-1)v + \sum(m_i - \bar{m})^2$. With $v = \sigma^2/n$, $m_i = \mu + \alpha_i$ and $\bar{m} = \mu$ (as $\sum\alpha_i = 0$),

$$E(\mathrm{SS}_{tr}) = n\Big\{(k-1)\frac{\sigma^2}{n} + \sum\alpha_i^2\Big\} = (k-1)\sigma^2 + n\sum\alpha_i^2.$$

Step 3. Distribution. $\mathrm{SSE}/\sigma^2 = \sum_i(n-1)S_i^2/\sigma^2$ is a sum of k independent $\chi^2(n-1)$ variables, hence $\chi^2(N-k)$. Each S_i^2 is independent of $\bar{Y}_{i\cdot}$ (normal sampling theory), and the groups are independent,

so SSE is independent of $(\bar{Y}_{1\cdot},\dots,\bar{Y}_{k\cdot})$, hence of SS_{tr}. Under H_0 the $\bar{Y}_{i\cdot}$ are i.i.d. $N(\mu,\sigma^2/n)$, so $\mathrm{SS}_{tr}/\sigma^2 = \sum(\bar{Y}_{i\cdot}-\bar{Y}_{\cdot\cdot})^2/(\sigma^2/n) \sim \chi^2(k-1)$. Therefore $F = \dfrac{(\mathrm{SS}_{tr}/\sigma^2)/(k-1)}{(\mathrm{SSE}/\sigma^2)/(N-k)} \sim F(k-1,N-k)$. ■

Thus MSE always estimates σ^2, while MS_{tr} is inflated by treatment effects — which is why a large F indicates H_1.

Definition (Test of treatment effects (one-way, equal replication))

For $H_0:\alpha_1=\cdots=\alpha_k=0$ use $F=\mathrm{MS}_{tr}/\mathrm{MSE}$. With observed value f:
(i) level-α critical region: $f \ge F_\alpha(k-1,N-k)$;
(ii) P-value: $P=P(F\ge f)$, $F\sim F(k-1,N-k)$.

Source	SS	d.f.	MS	F	P
Treatment	$\mathrm{SS}_{tr}=n\sum_i(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})^2$	$k-1$	$\mathrm{MS}_{tr}=\frac{\mathrm{SS}_{tr}}{k-1}$	$f=\frac{\mathrm{MS}_{tr}}{\mathrm{MSE}}$	$P(F\ge f)$
Residual	$\mathrm{SSE}=\sum_i\sum_j(y_{ij}-\bar{y}_{i\cdot})^2$	$N-k$	$\mathrm{MSE}=\frac{\mathrm{SSE}}{N-k}$		
Total	$\mathrm{SST}=\sum_i\sum_j(y_{ij}-\bar{y}_{\cdot\cdot})^2$	$N-1$			

Example 9-1 (Concentration of a treatment solution)

Four concentrations of a fabric treatment (A_1: 3.0%, A_2: 3.5%, A_3: 4.0%, A_4: 4.5%) were each run 5 times in random order. Build the ANOVA table and test at level 5% whether concentration affects tensile strength.

	A_1	A_2	A_3	A_4
	47	51	50	22
	58	62	38	23
	51	31	47	28
	61	46	27	42
	46	49	23	25
mean	52.6	47.8	37.0	28.0

grand mean 41.35

Solution. Model $Y_{ij}=\mu+\alpha_i+e_{ij}$ with $k=4$, $n=5$; $H_0:\alpha_1=\cdots=\alpha_4=0$.

$$\mathrm{SST} = (47-41.35)^2+\cdots+(25-41.35)^2 = 3334.55,$$
$$\mathrm{SS}_{tr} = 5\{(52.6-41.35)^2+\cdots+(28.0-41.35)^2\} = 1826.55,$$
$$\mathrm{SSE} = (47-52.6)^2+\cdots+(25-28.0)^2 = 1508.00,$$

with d.f. 19, 3, 16. So $\mathrm{MS}_{tr}=608.85$, $\mathrm{MSE}=94.25$, $f=6.46$, and $P=P(F(3,16)\ge 6.46)=0.0045$ (table: between 0.001 and 0.01). Concentration has a significant effect on strength at level 5%.

Source	SS	d.f.	MS	F	P
Treatment	1826.55	3	608.85	6.46	0.0045
Residual	1508.00	16	94.25		
Total	3334.55	19			

9.2.2 Unequal replication

In practice the number of replicates often varies by treatment (e.g. more runs for a treatment of special interest). The analysis is almost the same. With n_i replicates of treatment i and $N = \sum n_i$,

$$\bar{y}_{i\cdot} = \frac{1}{n_i}\sum_{j=1}^{n_i} y_{ij}, \qquad \bar{y}_{\cdot\cdot} = \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}, \qquad \mu = \frac{1}{N}\sum_{i=1}^{k} n_i\mu_i,$$

and $\alpha_i = \mu_i - \mu$ now satisfies $\sum n_i\alpha_i = 0$.

Definition (One-way ANOVA model (unequal replication))

$Y_{ij} = \mu + \alpha_i + e_{ij}$, $e_{ij} \sim N(0,\sigma^2)$ independent, $\sum_{i=1}^{k} n_i\alpha_i = 0$ $(i = 1,\dots,k,\ j = 1,\dots,n_i)$.

Proposition (Decomposition and expected mean squares, unequal replication)

$$\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{\cdot\cdot})^2 = \sum_{i=1}^{k} n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 + \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i\cdot})^2,$$

with d.f. $N-1 = (k-1)+(N-k)$. Moreover $E(\mathrm{MSE}) = \sigma^2$, $E(\mathrm{MS}_{tr}) = \sigma^2 + \sum n_i\alpha_i^2/(k-1)$, and under $H_0 : \alpha_1 = \cdots = \alpha_k = 0$, $F = \mathrm{MS}_{tr}/\mathrm{MSE} \sim F(k-1, N-k)$.

Proof. [Supplement]

Step 1. The cross term is $2\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})\sum_j(y_{ij} - \bar{y}_{i\cdot}) = 0$, exactly as before; the middle square summed over j gives $n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

Step 2. $E(\mathrm{SSE}) = \sum(n_i - 1)\sigma^2 = (N-k)\sigma^2$ as before.

Step 3. Write $\mathrm{SS}_{tr} = \sum n_i\bar{Y}_{i\cdot}^2 - N\bar{Y}_{\cdot\cdot}^2$. Since $E\bar{Y}_{i\cdot}^2 = \sigma^2/n_i + \mu_i^2$ and $\bar{Y}_{\cdot\cdot} \sim N(\mu, \sigma^2/N)$,

$$E(\mathrm{SS}_{tr}) = k\sigma^2 + \sum n_i\mu_i^2 - \sigma^2 - N\mu^2 = (k-1)\sigma^2 + \sum n_i(\mu_i - \mu)^2,$$

using $\sum n_i\mu_i = N\mu$. This gives $E(\mathrm{MS}_{tr})$.

Step 4. The independence argument of the equal-replication case is unchanged. Under H_0, $\sqrt{n_i}(\bar{Y}_{i\cdot} - \mu)/\sigma$ are i.i.d. $N(0,1)$, and $\mathrm{SS}_{tr}/\sigma^2$ is the squared length of the projection of this vector onto the $(k-1)$-dimensional subspace orthogonal to $(\sqrt{n_1},\dots,\sqrt{n_k})$, hence $\chi^2(k-1)$. The F law follows. ■

Definition (Test of treatment effects (one-way, unequal replication))

$F = \mathrm{MS}_{tr}/\mathrm{MSE}$ with $\mathrm{MS}_{tr} = \mathrm{SS}_{tr}/(k-1)$, $\mathrm{MSE} = \mathrm{SSE}/(N-k)$, $N = \sum n_i$. Reject H_0 at level α if $f \geq F_\alpha(k-1, N-k)$; $P = P(F \geq f)$. The ANOVA table is as before with $\mathrm{SS}_{tr} = \sum_i n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

Example 9-2 (Ratings on three review sites)

A film's ratings (out of 10) were sampled randomly: 4 from site 1, 5 from site 2, 6 from site 3. Test at level 5% whether the ratings differ by site.

	ratings	mean
Site 1	2, 3, 4, 5	3.5
Site 2	4, 5, 6, 4, 3	4.4
Site 3	6, 5, 7, 4, 6, 8	6.0

grand mean 4.8

Solution. $k = 3$, $n_1 = 4$, $n_2 = 5$, $n_3 = 6$; $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = 0$.

$$\begin{aligned}\text{SST} &= (2-4.8)^2 + \cdots + (8-4.8)^2 = 36.4,\\ \text{SS}_{tr} &= 4(3.5-4.8)^2 + 5(4.4-4.8)^2 + 6(6.0-4.8)^2 = 16.2,\\ \text{SSE} &= (2-3.5)^2 + \cdots + (8-6.0)^2 = 20.2,\end{aligned}$$

with d.f. 14, 2, 12. $\text{MS}_{tr} = 8.1$, $\text{MSE} = 1.68$, $f = 4.82$, $P = P(F(2,12) \geq 4.82) = 0.0292$. The ratings differ significantly by site.

Source	SS	d.f.	MS	F	P
Treatment	16.2	2	8.1	4.82	0.0292
Residual	20.2	12	1.68		
Total	36.4	14			

9.3 Two-way analysis of variance without replication

Motivation. A two-way layout examines two factors at once; with one observation per cell this requires no interaction.

Two-way ANOVA studies factors A $(A_1, \ldots, A_p)$ and B $(B_1, \ldots, B_q)$; the pq level combinations are the treatments, run in random order (running through B at fixed A violates randomization). All factors are fixed. With one observation y_{ij} per cell, define

$$\bar{y}_{i\cdot} = \frac{1}{q}\sum_{j=1}^{q} y_{ij}, \qquad \bar{y}_{\cdot j} = \frac{1}{p}\sum_{i=1}^{p} y_{ij}, \qquad \bar{y}_{\cdot\cdot} = \frac{1}{pq}\sum_{i=1}^{p}\sum_{j=1}^{q} y_{ij}.$$

Definition (Two-way ANOVA model (no replication))

$$Y_{ij} = \mu + \alpha_i + \beta_j + e_{ij}, \qquad e_{ij} \sim N(0,\sigma^2) \text{ independent}, \qquad \sum_{i=1}^{p}\alpha_i = 0, \quad \sum_{j=1}^{q}\beta_j = 0,$$

$i = 1, \ldots, p$, $j = 1, \ldots, q$; α_i is the effect of level A_i and β_j that of B_j.

The hypotheses of interest are

$$H_0 : \alpha_1 = \cdots = \alpha_p = 0 \text{ vs. } H_1 : \text{some } \alpha_i \neq 0; \qquad H_0 : \beta_1 = \cdots = \beta_q = 0 \text{ vs. } H_1 : \text{some } \beta_j \neq 0.$$

The total deviation decomposes as

$$y_{ij} - \bar{y}_{\cdot\cdot} = (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) + (\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}) + (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}),$$

into the deviation due to A_i, that due to B_j, and the residual.

Proposition (Decomposition, two-way without replication)

$\text{SST} = \text{SS}_A + \text{SS}_B + \text{SSE}$, where

$$\begin{aligned}&\text{SST} = \textstyle\sum_i\sum_j (y_{ij} - \bar{y}_{\cdot\cdot})^2 \quad (pq-1), \quad \text{SS}_A = q\textstyle\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 \quad (p-1),\\ &\text{SS}_B = p\textstyle\sum_j(\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2 \quad (q-1), \qquad \text{SSE} = \textstyle\sum_i\sum_j (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2 \quad ((p-1)(q-1)),\end{aligned}$$

with degrees of freedom in parentheses.

Proof. [Supplement] Write $a_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, $b_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$, $r_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$. Then $\sum_i a_i = \sum_j b_j = 0$, and for each fixed i, $\sum_j r_{ij} = q\bar{y}_{i\cdot} - q\bar{y}_{i\cdot} - q\bar{y}_{\cdot\cdot} + q\bar{y}_{\cdot\cdot} = 0$; likewise $\sum_i r_{ij} = 0$ for each j.

Step 1. $\sum_i \sum_j a_i b_j = (\sum_i a_i)(\sum_j b_j) = 0$.

Step 2. $\sum_i \sum_j a_i r_{ij} = \sum_i a_i \sum_j r_{ij} = 0$ and $\sum_i \sum_j b_j r_{ij} = \sum_j b_j \sum_i r_{ij} = 0$.

Step 3. Squaring $y_{ij} - \bar{y}_{\cdot\cdot} = a_i + b_j + r_{ij}$ and summing, all cross terms vanish, leaving $q\sum a_i^2 + p\sum b_j^2 + \sum\sum r_{ij}^2$. The residuals satisfy $p+q-1$ independent linear constraints, so SSE has $pq-(p+q-1) = (p-1)(q-1)$ d.f.; and $(pq-1) = (p-1) + (q-1) + (p-1)(q-1)$. ■

Mean squares and F-statistics:

$$\mathrm{MS}_A = \frac{\mathrm{SS}_A}{p-1}, \quad \mathrm{MS}_B = \frac{\mathrm{SS}_B}{q-1}, \quad \mathrm{MSE} = \frac{\mathrm{SSE}}{(p-1)(q-1)}, \qquad F_A = \frac{\mathrm{MS}_A}{\mathrm{MSE}}, \quad F_B = \frac{\mathrm{MS}_B}{\mathrm{MSE}}.$$

Proposition (Expected mean squares, two-way without replication)

$E(\mathrm{MSE}) = \sigma^2$, $E(\mathrm{MS}_A) = \sigma^2 + \frac{q}{p-1}\sum \alpha_i^2$, $E(\mathrm{MS}_B) = \sigma^2 + \frac{p}{q-1}\sum \beta_j^2$. Under the respective null hypotheses $F_A \sim F(p-1, (p-1)(q-1))$ and $F_B \sim F(q-1, (p-1)(q-1))$.

Proof. [Supplement]

Step 1. $\bar{Y}_{i\cdot} = \mu + \alpha_i + \bar{e}_{i\cdot}$ (since $\sum_j \beta_j = 0$), and the $\bar{e}_{i\cdot}$ are independent $N(0, \sigma^2/q)$. As in the one-way proof, $E\sum_i (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2 = (p-1)\sigma^2/q + \sum \alpha_i^2$, so $E(\mathrm{SS}_A) = (p-1)\sigma^2 + q\sum \alpha_i^2$. Symmetrically for B.

Step 2. $\mathrm{SST} = \sum\sum (Y_{ij} - \bar{Y}_{\cdot\cdot})^2$ has expectation $(pq-1)\sigma^2 + \sum\sum(\alpha_i + \beta_j)^2 = (pq-1)\sigma^2 + q\sum\alpha_i^2 + p\sum\beta_j^2$ (the cross term vanishes by the constraints). Subtracting, $E(\mathrm{SSE}) = (pq - 1 - (p-1) - (q-1))\sigma^2 = (p-1)(q-1)\sigma^2$.

Step 3. The F laws follow from Cochran's theorem: $\mathrm{SS}_A, \mathrm{SS}_B, \mathrm{SSE}$ are quadratic forms in the errors with ranks $p-1, q-1, (p-1)(q-1)$ adding to $pq-1$, so under the null hypotheses they are independent $\sigma^2\chi^2$ variables with these d.f. ■

Definition (Tests of treatment effects (two-way, no replication))

(i) Factor A: $H_0 : \alpha_1 = \cdots = \alpha_p = 0$; $F_A = \mathrm{MS}_A/\mathrm{MSE}$; reject if $f_A \geq F_\alpha(p-1, (p-1)(q-1))$; $P_A = P(F_A \geq f_A)$.
(ii) Factor B: $H_0 : \beta_1 = \cdots = \beta_q = 0$; $F_B = \mathrm{MS}_B/\mathrm{MSE}$; reject if $f_B \geq F_\alpha(q-1, (p-1)(q-1))$; $P_B = P(F_B \geq f_B)$.

Source	SS	d.f.	MS	F	P
Factor A	SS_A	$p-1$	MS_A	$f_A = \mathrm{MS}_A/\mathrm{MSE}$	$P(F_A \geq f_A)$
Factor B	SS_B	$q-1$	MS_B	$f_B = \mathrm{MS}_B/\mathrm{MSE}$	$P(F_B \geq f_B)$
Residual	SSE	$(p-1)(q-1)$	MSE		
Total	SST	$pq-1$			

Example 9-3 (Diet and exercise)

Twenty similar men in their twenties were randomly assigned to one of 5 diets $(A_1, \ldots, A_5)$ and one of 4 exercises (B_1 running, B_2 weights, B_3 swimming, B_4 yoga). Weight loss (kg) after 4 weeks:

	A_1	A_2	A_3	A_4	A_5	mean
B_1	7.9	7.2	5.1	5.8	6.8	6.56
B_2	7.5	6.6	4.8	5.6	6.5	6.20
B_3	6.9	6.4	4.4	5.4	6.4	5.90
B_4	6.5	6.2	4.1	4.5	5.8	5.42
mean	7.20	6.60	4.60	5.33	6.38	6.02

Test at level 5% whether diet and exercise affect weight loss.
Solution. $H_0 : \alpha_1 = \cdots = \alpha_5 = 0$ and $H_0 : \beta_1 = \cdots = \beta_4 = 0$.

$$\begin{aligned}
\mathrm{SS}_A &= 4\{(7.20-6.02)^2 + \cdots + (6.38-6.02)^2\} = 17.417,\\
\mathrm{SS}_B &= 5\{(6.56-6.02)^2 + \cdots + (5.42-6.02)^2\} = 3.492,\\
\mathrm{SST} &= (7.9-6.02)^2 + \cdots + (5.8-6.02)^2 = 21.232,\\
\mathrm{SSE} &= \mathrm{SST} - \mathrm{SS}_A - \mathrm{SS}_B = 0.323,
\end{aligned}$$

with d.f. $4, 3, 19, 12$. Then $\mathrm{MS}_A = 4.354$, $\mathrm{MS}_B = 1.164$, $\mathrm{MSE} = 0.027$, $f_A = 161.77$, $f_B = 43.24$, both with $P < 0.000001$. Both the diet and the exercise type have significant effects at level 5%.

Source	SS	d.f.	MS	F	P
Factor A	17.417	4	4.354	161.77	< 0.000001
Factor B	3.492	3	1.164	43.24	< 0.000001
Residual	0.323	12	0.027		
Total	21.232	19			

9.4 Two-way analysis of variance with replication

Motivation. Replication separates an *interaction*, an effect of A varying with B, from pure error.

Each of the pq combinations is replicated r times, all pqr runs in random order (running the r replicates of each combination consecutively is not completely randomized).

Replication lets us estimate the **interaction**: the effect of A depends on the level of B. Without it, $E(Y_{ij}) = \mu + \alpha_i + \beta_j$, so the difference between two levels of A is the same at every level of B, e.g.

$$(\mu + \alpha_1 + \beta_j) - (\mu + \alpha_3 + \beta_j) = \alpha_1 - \alpha_3 \quad \text{for all } j.$$

Plotting mean response against B, one line per level of A, gives parallel segments without interaction (Figure 9, left) and crossing lines with interaction (right), when the unreplicated analysis fails.

The data are y_{ijk} $(i = 1, \ldots, p;\ j = 1, \ldots, q;\ k = 1, \ldots, r)$ with means

$$\bar{y}_{ij\cdot} = \frac{1}{r}\sum_k y_{ijk}, \quad \bar{y}_{i\cdot\cdot} = \frac{1}{qr}\sum_j\sum_k y_{ijk}, \quad \bar{y}_{\cdot j\cdot} = \frac{1}{pr}\sum_i\sum_k y_{ijk}, \quad \bar{y}_{\cdots} = \frac{1}{pqr}\sum_i\sum_j\sum_k y_{ijk}.$$

Let γ_{ij} be the interaction effect of A_i and B_j.

Definition (Two-way ANOVA model (with replication))

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}, \qquad e_{ijk} \sim N(0, \sigma^2) \text{ independent},$$

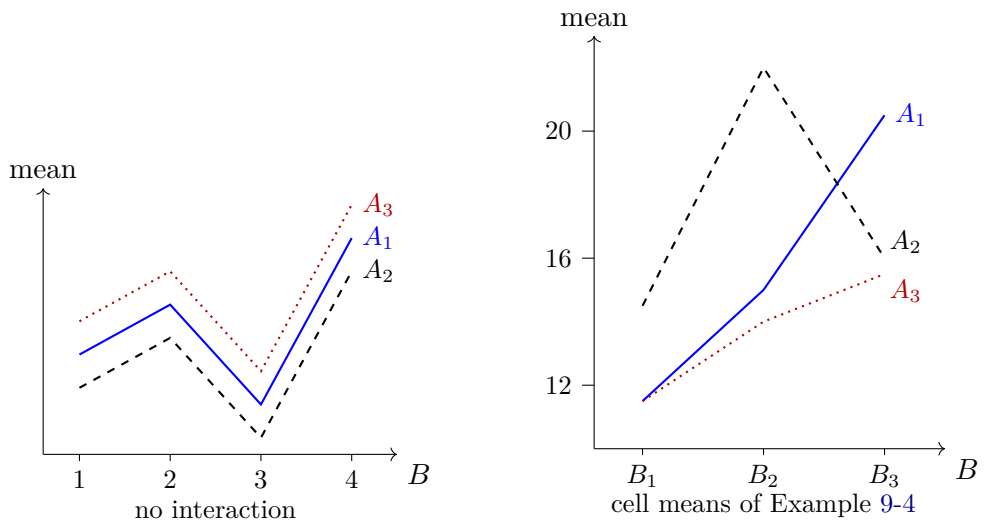


Figure 9: Interaction plots. Left: parallel profiles (no interaction). Right: crossing profiles, here the cell means $\bar{y}_{ij\cdot}$ of Example 9-4.

$$\sum_{i=1}^{p}\alpha_i = 0, \quad \sum_{j=1}^{q}\beta_j = 0, \quad \sum_{i=1}^{p}\gamma_{ij} = \sum_{j=1}^{q}\gamma_{ij} = 0,$$

$i = 1,\dots,p$, $j = 1,\dots,q$, $k = 1,\dots,r$.

Tests of the main effects proceed as in §9.3; the new question is the interaction:

$$H_0 : \gamma_{ij} = 0 \ \text{ (all } i,j) \qquad \text{vs.} \qquad H_1 : \text{at least one } \gamma_{ij} \neq 0.$$

The total deviation decomposes as

$$y_{ijk} - \bar{y}_{\cdots} = (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots}) + (y_{ijk} - \bar{y}_{ij\cdot}),$$

into deviations due to A_i, to B_j, to the interaction, and the residual.

Proposition (Decomposition, two-way with replication)

$\mathrm{SST} = \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SS}_{A\times B} + \mathrm{SSE}$, where

$$\begin{aligned}
\mathrm{SST} &= \textstyle\sum_i\sum_j\sum_k (y_{ijk} - \bar{y}_{\cdots})^2 && (pqr-1),\\
\mathrm{SS}_A &= qr\textstyle\sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 && (p-1),\\
\mathrm{SS}_B &= pr\textstyle\sum_j (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots})^2 && (q-1),\\
\mathrm{SS}_{A\times B} &= r\textstyle\sum_i\sum_j (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2 && ((p-1)(q-1)),\\
\mathrm{SSE} &= \textstyle\sum_i\sum_j\sum_k (y_{ijk} - \bar{y}_{ij\cdot})^2 && (pq(r-1)).
\end{aligned}$$

Proof. [Supplement]

Step 1. Split $y_{ijk} - \bar{y}_{\cdots} = (\bar{y}_{ij\cdot} - \bar{y}_{\cdots}) + (y_{ijk} - \bar{y}_{ij\cdot})$. This is a one-way layout with the pq cells as treatments, each replicated r times, so by the one-way decomposition

$$\mathrm{SST} = r\textstyle\sum_i\sum_j (\bar{y}_{ij\cdot} - \bar{y}_{\cdots})^2 + \mathrm{SSE}.$$

Step 2. The cell means $\bar{y}_{ij\cdot}$ form a $p \times q$ table whose row means are $\bar{y}_{i\cdot\cdot}$, column means $\bar{y}_{\cdot j\cdot}$ and grand mean $\bar{y}_{\cdots}$. Applying the two-way decomposition of §9.3 to this table and multiplying by r,

$$r\textstyle\sum_i\sum_j (\bar{y}_{ij\cdot} - \bar{y}_{\cdots})^2 = \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SS}_{A\times B}.$$

Step 3. Combining gives the identity. D.f.: $pq(r-1)$ for SSE ($r-1$ per cell), and $pq-1 = (p-1)+(q-1)+(p-1)(q-1)$ for the cell-mean part; the total is $pqr-1$. ■

Mean squares and the interaction test statistic:

$$\mathrm{MS}_{A\times B} = \frac{\mathrm{SS}_{A\times B}}{(p-1)(q-1)}, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{pq(r-1)}, \qquad F = \frac{\mathrm{MS}_{A\times B}}{\mathrm{MSE}}.$$

Proposition (Expected mean squares, two-way with replication)

$E(\mathrm{MSE}) = \sigma^2$, $E(\mathrm{MS}_A) = \sigma^2 + \frac{qr}{p-1}\sum \alpha_i^2$, $E(\mathrm{MS}_B) = \sigma^2 + \frac{pr}{q-1}\sum \beta_j^2$, and $E(\mathrm{MS}_{A\times B}) = \sigma^2 + \frac{r}{(p-1)(q-1)}\sum\sum \gamma_{ij}^2$. Under $H_0 : \gamma_{ij} = 0$, $F = \mathrm{MS}_{A\times B}/\mathrm{MSE} \sim F((p-1)(q-1), pq(r-1))$.

Proof. [Supplement]

Step 1. Within each cell, $Y_{ij1}, \dots, Y_{ijr}$ is a sample from $N(\mu_{ij}, \sigma^2)$ with $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$, so $E(\mathrm{SSE}) = pq(r-1)\sigma^2$ and $\mathrm{SSE}/\sigma^2 \sim \chi^2(pq(r-1))$, independent of all cell means (normal sampling theory, cells independent).

Step 2. The cell means $\bar{Y}_{ij\cdot} = \mu_{ij} + \bar{e}_{ij\cdot}$ are independent with variance σ^2/r, i.e. a two-way table without replication with error variance σ^2/r and "means" $\mu+\alpha_i+\beta_j+\gamma_{ij}$. By the computation in §9.3 (now with the extra term γ_{ij}, whose row and column sums vanish), $E(\mathrm{SS}_A) = (p-1)\sigma^2 + qr\sum \alpha_i^2$, $E(\mathrm{SS}_B) = (q-1)\sigma^2 + pr\sum \beta_j^2$ and

$$E(\mathrm{SS}_{A\times B}) = (p-1)(q-1)\sigma^2 + r\textstyle\sum_i\sum_j \gamma_{ij}^2.$$

Step 3. $\mathrm{SS}_{A\times B}$ is a function of the cell means, hence independent of SSE; under H_0 it is $\sigma^2\chi^2((p-1)(q-1))$ (Cochran's theorem applied to the cell-mean table, as in §9.3). The F law follows; the main-effect F-tests are justified the same way. ■

Definition (Test of interaction (two-way with replication))

For $H_0 : \gamma_{ij} = 0$ ($i = 1, \dots, p$, $j = 1, \dots, q$) use $F = \mathrm{MS}_{A\times B}/\mathrm{MSE}$; reject at level α if $f \geq F_\alpha((p-1)(q-1), pq(r-1))$; $P = P(F \geq f)$. The main effects are tested by $F_A = \mathrm{MS}_A/\mathrm{MSE}$ and $F_B = \mathrm{MS}_B/\mathrm{MSE}$ with denominator d.f. $pq(r-1)$.

Source	SS	d.f.	MS	F	P
Factor A	SS_A	$p-1$	MS_A	$f_A = \mathrm{MS}_A/\mathrm{MSE}$	$P(F_A \geq f_A)$
Factor B	SS_B	$q-1$	MS_B	$f_B = \mathrm{MS}_B/\mathrm{MSE}$	$P(F_B \geq f_B)$
Interaction	$\mathrm{SS}_{A\times B}$	$(p-1)(q-1)$	$\mathrm{MS}_{A\times B}$	$f_{A\times B} = \mathrm{MS}_{A\times B}/\mathrm{MSE}$	$P(F_{A\times B} \geq f_{A\times B})$
Residual	SSE	$pq(r-1)$	MSE		
Total	SST	$pqr-1$			

Example 9-4 (Machines and operators)

Three machines (A_1, A_2, A_3) and three operators (B_1, B_2, B_3) were compared by measuring productivity twice for each combination. Test the main effects and the interaction at level 5%.

	B_1	B_2	B_3	mean
A_1	9, 14	14, 16	19, 22	15.67
A_2	13, 16	18, 26	14, 18	17.5
A_3	11, 12	11, 17	15, 16	13.67
mean	12.5	17	17.33	15.61

Solution. Hypotheses: $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = 0$; $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$; $H_0 : \gamma_{ij} = 0$ $(i, j = 1, 2, 3)$.

$$\begin{aligned}
\mathrm{SS}_A &= 3 \times 2\{(15.67 - 15.61)^2 + \cdots + (13.67 - 15.61)^2\} = 44.11, \\
\mathrm{SS}_B &= 3 \times 2\{(12.5 - 15.61)^2 + \cdots + (17.33 - 15.61)^2\} = 87.44, \\
\mathrm{SS}_{A\times B} &= 2\{(11.5 - 15.67 - 12.5 + 15.61)^2 + \cdots\} = 74.22, \\
\mathrm{SST} &= (9 - 15.61)^2 + \cdots + (16 - 15.61)^2 = 288.28, \\
\mathrm{SSE} &= \mathrm{SST} - \mathrm{SS}_A - \mathrm{SS}_B - \mathrm{SS}_{A\times B} = 82.51,
\end{aligned}$$

with d.f. $2, 2, 4, 17, 9$. Hence $\mathrm{MS}_A = 22.06$, $\mathrm{MS}_B = 43.72$, $\mathrm{MS}_{A\times B} = 18.56$, $\mathrm{MSE} = 9.17$ and $f_A = 2.41$, $f_B = 4.77$, $f_{A\times B} = 2.02$, with P-values 0.146, 0.039, 0.174. Operators differ significantly in productivity at level 5%, while the machine type and the interaction are not significant.

Source	SS	d.f.	MS	F	P
Factor A	44.11	2	22.06	2.41	0.146
Factor B	87.44	2	43.72	4.77	0.039
Interaction	74.22	4	18.56	2.02	0.174
Residual	82.51	9	9.17		
Total	288.28	17			

Model checking and beyond

Check the error assumptions first by plotting studentized residuals against estimated means; for Example 9-4 no problem appears. Estimating or comparing level means after a significant result is beyond these notes.

Chapter 10. Nonparametric Inference

10.1 Introduction

Motivation. With outliers or heavy tails, t methods lose power; sign and rank methods are safer.

Most classical procedures assume the population is normal. If $X_1, \ldots, X_n$ is a random sample from $N(\mu, \sigma^2)$, then

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1),$$

and this pivot gives tests and intervals for μ, approximately valid for non-normal populations by the CLT.

With outliers or far-from-normal populations, normal-theory methods can fail badly: actual confidence below nominal, low power, needlessly long intervals.

Definition (Parametric, nonparametric and distribution-free methods)

(a) A **parametric method** assumes a specific parametric family (e.g. normal) for the population and makes inference about its parameters (μ, σ^2).
(b) A **nonparametric method** assumes no parametric form, typically only that the distribution is *continuous* (or *continuous and symmetric*).
(c) A method valid for every population distribution is **distribution-free**; e.g. a test whose statistic has a null distribution free of the population distribution.

The basic tools are the **sign** (1 if positive, 0 if negative) and the **rank** (position in increasing order) of an observation.

For example, the five observations

$$X_1 = 15, \quad X_2 = 18, \quad X_3 = 11, \quad X_4 = 36, \quad X_5 = 14$$

have ranks

$$R_1 = 3, \quad R_2 = 4, \quad R_3 = 1, \quad R_4 = 5, \quad R_5 = 2.$$

The largest value 36 gets rank 5 however large it is, so working with $R_1, \ldots, R_n$ instead of $X_1, \ldots, X_n$ limits the influence of outliers.

10.2 Wilcoxon rank-sum test

Motivation. The Wilcoxon rank-sum test compares two locations using only the ranks of the pooled sample.

Let the distribution functions be $F(x)$ and $G(x) = F(x - \Delta)$: same shape, shifted by the **shift parameter** $\Delta = \mu_Y - \mu_X$.

Definition (Data structure of the two-sample location problem)

$X_1, X_2, \ldots, X_m$: a random sample from a continuous distribution $F(x)$.
$Y_1, Y_2, \ldots, Y_n$: a random sample from the continuous distribution $F(x - \Delta)$.
The two samples are independent.

$\Delta = 0$ means that X and Y have the same distribution. $\Delta > 0$ means that the center of Y lies to the right of the center of X, and $\Delta < 0$ and $\Delta \neq 0$ are read in the same way.

Definition (Hypotheses on the shift parameter)

(i) $H_0 : \Delta = 0$ versus $H_1 : \Delta > 0$.
(ii) $H_0 : \Delta = 0$ versus $H_1 : \Delta < 0$.
(iii) $H_0 : \Delta = 0$ versus $H_1 : \Delta \neq 0$.

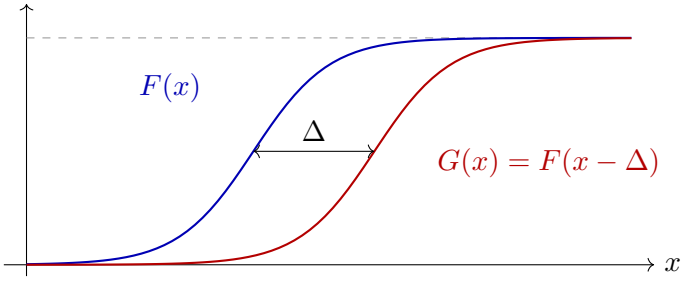


Figure 10.1: The distribution functions F and G under $H_1 : \Delta > 0$.

Pool the X_i and Y_j into the **combined sample** of size $N = m + n$. Under $H_1 : \Delta > 0$ the Y's tend to get larger ranks, so a large Y-rank sum supports H_1.

Definition 10-1 (Wilcoxon rank-sum statistic)

Let R_j be the rank of Y_j in the combined sample of the X's and Y's. The **Wilcoxon rank-sum statistic** is

$$W = \sum_{j=1}^{n} R_j.$$

10.2.1 Null distribution of W

Take $m = 3$, $n = 2$. Under $H_0 : \Delta = 0$, $\{X_1, X_2, X_3, Y_1, Y_2\}$ is a random sample of size 5 from F; with no ties, all $\binom{5}{2} = 10$ orderings of three x's and two y's are equally likely.

No.	1	2	3	4	5	6	7	8	9	10
Arrangement	*yyxxx*	*yxyxx*	*yxxyx*	*yxxxy*	*xyyxx*	*xyxyx*	*xyxxy*	*xxyyx*	*xxyxy*	*xxxyy*
W	3	4	5	6	5	6	7	7	8	9

w	3	4	5	6	7	8	9	Total
$P(W = w)$	0.1	0.1	0.2	0.2	0.2	0.1	0.1	1.0

Tables 10-1 and 10-2: the ten arrangements (each with probability 0.1) and the null distribution of W for $m = 3$, $n = 2$.

The null distribution of W does not involve F, so the rank-sum test is distribution-free. For instance, with $m = 3$, $n = 2$ and $\alpha = 0.10$, the rejection region for $H_1 : \Delta > 0$ is $W \geq 9$, whatever F is.

Tables (Appendix C of the book) or software give the null distribution; the smaller sample is the Y sample, with rank sum W_s, and the table lists $P(W_s \geq x)$ and $P(W_s \leq x^*)$. E.g. for $n = 3$, $m = 7$, $x = 25$, $x^* = 8$:

$$P(W_s \geq 25) = 0.033 = P(W_s \leq 8).$$

(The null distribution is symmetric about $n(N + 1)/2 = 16.5$, so $x^* = n(N + 1) - x$.)

Definition (Wilcoxon rank-sum test)

Take the larger sample as the X sample and the smaller one as the Y sample, and let

$$W = \text{sum of the ranks of the } Y \text{ sample in the combined sample.}$$

The rejection regions for $H_0 : \Delta = 0$ at level α are as follows.
(i) For $H_1 : \Delta > 0$: reject if $W \geq c$.
(ii) For $H_1 : \Delta < 0$: reject if $W \leq c^*$.
(iii) For $H_1 : \Delta \neq 0$: reject if $W \geq c_1$ or $W \leq c_1^*$.
The critical values satisfy

$$P(W \geq c) = \alpha, \qquad P(W \leq c^*) = \alpha, \qquad P(W \geq c_1) = \frac{\alpha}{2} = P(W \leq c_1^*).$$

Example 10-1 (Fine dust in two cities)

Cities A and B have similar population density and geography. Their monthly mean PM_{10} concentrations ($\mu g/m^3$) from March to November were recorded.

Month	3	4	5	6	7	8	9	10	11
City A	33	28	23	25	18	17	25.5	27	26
City B	22	18.5	19.5	20	16	24	22.5	21.5	21

At the 5% level, use the rank-sum test to decide whether the two cities differ.
Solution. Test $H_0 : \Delta = 0$ against $H_1 : \Delta \neq 0$, with $X = A$ and $Y = B$ ($m = n = 9$). The ranks in the combined sample of 18 values are listed below.

X_i	33	28	23	25	18	17	25.5	27	26
rank	18	17	11	13	3	2	14	16	15
Y_j	22	18.5	19.5	20	16	24	22.5	21.5	21
rank	9	4	5	6	1	12	10	8	7

Hence $W_s = 9 + 4 + 5 + 6 + 1 + 12 + 10 + 8 + 7 = 62$. From the table with $m = n = 9$,

$$P(W_s \geq 109) = 0.02 = P(W_s \leq 62).$$

The two-sided p-value is $2 \times 0.02 = 0.04 < 0.05$. We reject H_0 and conclude that the PM_{10} levels of the two cities differ.

Example 10-2 (Tire wear)

Tires of company A cost about 20% more than those of company B. Eight tires of each company were driven 10,000 km and their wear rates (%) were measured. One A measurement was lost.

Company A	15	22	27	18	13	19	25	
Company B	20	16	17	26	28	24	30	33

Can we say that tires of A are better than those of B?
Solution. The sample from A is smaller, so A is the Y sample ($n = 7$) and B is the X sample ($m = 8$). Better quality means lower wear, so we test $H_0 : \Delta = 0$ against $H_1 : \Delta < 0$.

X_i (B)	16	17	20	24	26	28	30	33
rank	3	4	7	9	11	13	14	15
Y_j (A)	13	15	18	19	22	25	27	
rank	1	2	5	6	8	10	12	

So $W_s = 44$ and $(m = 8, n = 7)$ $P(W_s \le 44) = 0.095$: not significant at 5%, but at 10% tires of A are better.

10.2.2 Ties and the normal approximation

Rounded data produce ties; tied observations get the average of their ranks (**midranks**), and the tabulated null distribution then holds only approximately.

When m and n are too large for the tables, we use a normal approximation. We standardize W by its null mean and variance and refer the result to $N(0,1)$, which is a Z test.

Theorem 10-1 (Normal approximation of W)

Under $H_0 : \Delta = 0$,

$$E(W) = \frac{n(m+n+1)}{2}, \qquad \mathrm{Var}(W) = \frac{mn(m+n+1)}{12}.$$

When m and n are large,

$$Z_W = \frac{W - E(W)}{\sqrt{\mathrm{Var}(W)}}$$

approximately follows the standard normal distribution $N(0,1)$.

Proof. [Supplement] Write $N = m + n$ and assume there are no ties.

Step 1. Under H_0 the N observations are i.i.d. from a continuous distribution. Every ordering of them is therefore equally likely, and the vector $(R_1, \dots, R_n)$ of Y-ranks is a random ordered selection of n values drawn without replacement from $\{1, \dots, N\}$. In particular each R_j is uniform on $\{1, \dots, N\}$, and each pair (R_j, R_k) with $j \neq k$ is uniform on the ordered pairs of distinct values.

Step 2. For a uniform R on $\{1, \dots, N\}$,

$$E(R) = \frac{1}{N}\sum_{r=1}^{N} r = \frac{N+1}{2},$$

$$\mathrm{Var}(R) = \frac{1}{N}\sum_{r=1}^{N} r^2 - \Big(\frac{N+1}{2}\Big)^2 = \frac{(N+1)(2N+1)}{6} - \frac{(N+1)^2}{4} = \frac{N^2-1}{12}.$$

Step 3. By exchangeability $c = \mathrm{Cov}(R_j, R_k)$ is the same for all $j \neq k$. If all N observations are ranked, the ranks sum to the constant $N(N+1)/2$, whose variance is 0:

$$0 = N\,\frac{N^2-1}{12} + N(N-1)\,c, \qquad \text{so} \qquad c = -\frac{N+1}{12}.$$

Step 4. Hence $E(W) = n(N+1)/2 = n(m+n+1)/2$ and

$$\begin{aligned}\mathrm{Var}(W) &= n\,\frac{N^2-1}{12} - n(n-1)\,\frac{N+1}{12} = \frac{n(N+1)}{12}\,[(N-1) - (n-1)] \\ &= \frac{n(N+1)(N-n)}{12} = \frac{mn(m+n+1)}{12}.\end{aligned}$$

Step 5. W is the total of a simple random sample of size n drawn without replacement from the finite population $\{1, \dots, N\}$. By the central limit theorem for sampling without replacement (Wald–Wolfowitz, Hájek), such a total is asymptotically normal when $\min(n, m) \to \infty$, because no single population value dominates the population variance. Therefore $Z_W \xrightarrow{d} N(0, 1)$. ■

Example 10-3 (Normal approximation for Example 10-2)

Use the normal approximation of W for the data of Example 10-2.
Solution. There $W = 44$ with $m = 8$ and $n = 7$, so

$$E(W) = \frac{7(8 + 7 + 1)}{2} = 56, \qquad \mathrm{Var}(W) = \frac{8 \times 7 \times (8 + 7 + 1)}{12} = 74.67,$$
$$Z_W = \frac{44 - 56}{\sqrt{74.67}} = -1.39.$$

$P(Z \le -1.39) = 0.0823$, so the one-sided p-value is 8.23%. This is very close to the exact result of Example 10-2.

Example 10-4 (New surgical method)

A new surgical method A is compared with the standard method B. Each method was applied to 10 patients, and the recovery times (days) are given below.

A	8	9	11	12	16	17	19	19	21	45
B	13	16	18	18	20	20	21	23	25	72

To decide whether A is more effective than B, carry out
(a) the t test;
(b) the Wilcoxon rank-sum test;
(c) the normal-approximation rank-sum test.
Solution. Shorter recovery means a more effective method. With B as the Y sample, we test $H_0 : \Delta = 0$ against $H_1 : \Delta > 0$.
(a) For A, $\bar{x} = 17.7$ and $\sum(x_i - \bar{x})^2 = 1010.1$. For B, $\bar{y} = 24.6$ and $\sum(y_i - \bar{y})^2 = 2600.4$. Then

$$s_p^2 = \frac{1010.1 + 2600.4}{18} = 200.6, \qquad s_p = 14.16,$$
$$t = \frac{\bar{y} - \bar{x}}{s_p\sqrt{1/m + 1/n}} = \frac{24.6 - 17.7}{14.16\sqrt{1/10 + 1/10}} = 1.09.$$

Since $t_{0.1}(18) = 1.330 > 1.09$, the p-value exceeds 10%, and the t test cannot conclude that A is more effective.
(b) With midranks for the ties, the ranks in the combined sample are as follows.

X_i (A)	8	9	11	12	16	17	19	19	21	45
rank	1	2	3	4	6.5	8	11.5	11.5	15.5	19
Y_j (B)	13	16	18	18	20	20	21	23	25	72
rank	5	6.5	9.5	9.5	13.5	13.5	15.5	17	18	20

So $W_s = 128$. With $m = n = 10$ the table gives $P(W_s \ge 128) = 0.045$. At the 5% level, A is more effective than B.
(c) We have

$$E(W) = \frac{10 \times 21}{2} = 105, \qquad \mathrm{Var}(W) = \frac{10 \times 10 \times 21}{12} = 175,$$

$$Z_W = \frac{128 - 105}{\sqrt{175}} = 1.74,$$

and $P(Z \geq 1.74) = 0.0409$. The p-value is 4.09%, so again A is more effective than B.

Why the t test failed here

In Example 10-4 the rank-sum test gives $p \approx 4$–4.5% but the t test $p > 10\%$, because the outliers 45 and 72 inflate s_p. If they were errors or exceptional patients and removed, then

$$\bar{x} = 14.67, \quad \sum(x_i - \bar{x})^2 = 182.00, \qquad \bar{y} = 19.33, \quad \sum(y_i - \bar{y})^2 = 104.00,$$
$$s_p^2 = \frac{182.00 + 104.00}{16} = 17.88, \qquad s_p = 4.23, \qquad t = \frac{19.33 - 14.67}{4.23\sqrt{1/9 + 1/9}} = 4.957.$$

Since $t_{0.0005}(16) = 4.015$, the p-value is below 0.05%, and A is clearly better. With outliers or markedly non-normal data, the rank-sum test is often more efficient than the t test.

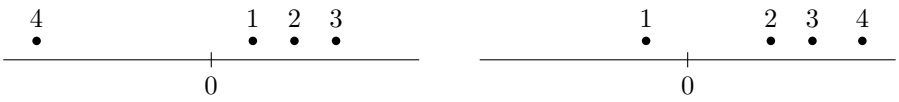
10.3 Wilcoxon signed-rank test

Motivation. The signed-rank test targets the centre of a symmetric distribution, also for paired comparisons.

Definition (Data structure of the one-sample location problem)

$X_1, \ldots, X_n$ is a random sample from $F(x - \theta)$, continuous and symmetric about the **location parameter** θ (mean, if it exists, and median).

Consider two samples of $n = 4$ observations (Figure 10.2). Both have three positive observations. In (a) the negative observation lies far from 0, while in (b) it lies close to 0.



(a) negative value relatively far from 0 (b) negative value relatively close to 0

Figure 10.2: Two samples with $n = 4$; the numbers are the absolute ranks.

The **absolute rank** of an observation is its rank by distance from 0. To test whether the location is 0, use the sum of absolute ranks of the positive observations:

$$\text{(a) } T = 1 + 2 + 3 = 6, \qquad \text{(b) } T = 2 + 3 + 4 = 9.$$

The farther the data are shifted to the right of 0, the larger the ranks or the number of the positive observations, so this sum grows. In general, to test $H_0 : \theta = \theta_0$ we set

$$Z_i = X_i - \theta_0 \qquad (i = 1, \ldots, n).$$

Under H_0, $Z_1, \ldots, Z_n$ follow a distribution symmetric about 0. Let R_i^+ be the absolute rank of $|Z_i|$ among $|Z_1|, \ldots, |Z_n|$. The **Wilcoxon signed-rank statistic** is

$$W^+ = \text{sum of the absolute ranks } R_i^+ \text{ of the positive } Z_i.$$

Under H_0 the absolute ranks of positive and negative Z_i are alike and each sign is $\pm$ with probability $\frac{1}{2}$, so the null distribution of W^+ is free of F: the test is distribution-free.

Appendix C tabulates $P(W^+ \geq x)$ and $P(W^+ \leq x^*)$; for $n = 4$, $P(W^+ \geq 9) = 0.125 = P(W^+ \leq 1)$, since of $2^4 = 16$ sign patterns only $\{2, 3, 4\}$ and $\{1, 2, 3, 4\}$ have rank sum ≥ 9.

Definition (Wilcoxon signed-rank test)

(i) Compute $Z_i = X_i - \theta_0$ for $i = 1, \ldots, n$.
(ii) Find the absolute rank R_i^+ of $|Z_i|$ among $|Z_1|, \ldots, |Z_n|$.
(iii) Compute $W^+ =$ the sum of R_i^+ over the positive Z_i.
The rejection regions for $H_0 : \theta = \theta_0$ are as follows.
For $H_1 : \theta > \theta_0$: reject if $W^+ \geq c$.
For $H_1 : \theta < \theta_0$: reject if $W^+ \leq c^*$.
For $H_1 : \theta \neq \theta_0$: reject if $W^+ \geq c_1$ or $W^+ \leq c_1^*$.
The critical values satisfy

$$P(W^+ \geq c) = \alpha, \quad P(W^+ \leq c^*) = \alpha, \quad P(W^+ \geq c_1) = \frac{\alpha}{2} = P(W^+ \leq c_1^*).$$

Example 10-5 (Smartwatch heart-rate error)

A smartwatch maker claims its heart-rate error (absolute difference from the true rate) averages 5 bpm; a consumer group suspects more. Ten watches on 10 similar people, against a medical monitor, gave these errors (bpm).

3.5	2.6	7.8	6.0	6.8	3.1	8.8	8.5	7.7	4.5

Is the consumer group's claim supported?
Solution. Test $H_0 : \theta = 5$ against $H_1 : \theta > 5$. With $\theta_0 = 5$, the values $Z_i = X_i - 5$ are

$$-1.5,\ -2.4,\ 2.8,\ 1.0,\ 1.8,\ -1.9,\ 3.8,\ 3.5,\ 2.7,\ -0.5.$$

negative Z_i	abs. rank	positive Z_i	abs. rank
-0.5	1	1.0	2
-1.5	3	1.8	4
-1.9	5	2.7	7
-2.4	6	2.8	8
		3.5	9
		3.8	10

So $W^+ = 2+4+7+8+9+10 = 40$. With $n = 10$ the table gives $P(W^+ \geq 40) = 0.116$. The p-value 11.6% is above 5%, so the evidence is insufficient to conclude that the error exceeds 5 bpm.

Ties get average ranks, and the null distribution of W^+ then holds approximately; for large n use the normal approximation.

Theorem 10-2 (Normal approximation of W^+)

Under $H_0 : \theta = \theta_0$,

$$E(W^+) = \frac{n(n+1)}{4}, \qquad \mathrm{Var}(W^+) = \frac{n(n+1)(2n+1)}{24}.$$

When n is large,

$$Z_{W^+} = \frac{W^+ - E(W^+)}{\sqrt{\mathrm{Var}(W^+)}}$$

approximately follows the standard normal distribution $N(0,1)$.

Proof. [Supplement] Assume there are no ties, and let $Z = X - \theta_0$ have a continuous distribution symmetric about 0 under H_0.

Step 1. The sign of Z is independent of $|Z|$. For $a > 0$, symmetry gives

$$P(Z > 0,\ |Z| \le a) = P(0 < Z \le a) = \tfrac{1}{2}P(|Z| \le a) = P(Z > 0)\,P(|Z| \le a).$$

Step 2. For $k = 1, \dots, n$ let $I_k = 1$ if the observation with absolute rank k is positive, and $I_k = 0$ otherwise. The absolute ranks depend only on $|Z_1|, \dots, |Z_n|$, and by Step 1 the signs are i.i.d. fair coin flips independent of these. Hence $I_1, \dots, I_n$ are i.i.d. Bernoulli$(\tfrac{1}{2})$, and

$$W^+ = \sum_{k=1}^{n} k\, I_k.$$

Step 3. With $E(I_k) = \tfrac{1}{2}$ and $\mathrm{Var}(I_k) = \tfrac{1}{4}$,

$$E(W^+) = \frac{1}{2}\sum_{k=1}^{n} k = \frac{n(n+1)}{4},$$
$$\mathrm{Var}(W^+) = \frac{1}{4}\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{24}.$$

Step 4. The summands $Y_k = k(I_k - \tfrac{1}{2})$ are independent with mean 0 and $|Y_k| = k/2$. Write $s_n^2 = \mathrm{Var}(W^+)$. Then

$$\frac{\sum_{k=1}^{n} E|Y_k|^3}{s_n^3} = \frac{\frac{1}{8}\sum k^3}{s_n^3} = \frac{n^2(n+1)^2/32}{\big(n(n+1)(2n+1)/24\big)^{3/2}} = O(n^{-1/2}) \to 0.$$

This is Lyapunov's condition, so the Lyapunov CLT gives $Z_{W^+} \xrightarrow{d} N(0,1)$. ∎

10.3.1 Paired comparison

For heterogeneous units, homogeneous pairs beat random splitting (Section 6.2). With n pairs $(X_1, Y_1), \dots, (X_n, Y_n)$ form the differences

$$D_i = Y_i - X_i \qquad (i = 1, \dots, n).$$

$H_0 : \Delta = 0$ vs. $H_1 : \Delta > 0$ becomes $H_0 : \theta = 0$ vs. $H_1 : \theta > 0$ for the D_i; the signed-rank test on the D_i is a nonparametric paired comparison.

Example 10-6 (Twins and a new curriculum)

For a new curriculum A, 16 pairs of identical twins with equal scores were recruited; one twin of each pair took the old curriculum B, the other A, then both sat the same test.

A	70	85	73	75	65	80	71	80
B	65	43	45	80	84	71	52	42
$Z = A - B$	5	42	28	−5	−19	9	19	38
A	51	72	76	79	59	84	56	57
B	78	62	38	80	54	87	38	43
$Z = A - B$	−27	10	38	−1	5	−3	18	14

At the 5% level, is the new curriculum more effective?
Solution. Let θ be the location of Z, and test $H_0 : \theta = 0$ against $H_1 : \theta > 0$. Absolute ranks (with midranks for ties) are as follows.

negative Z_i	-1		-3		-5		-19		-27		
abs. rank	1		2		4		10.5		12		
positive Z_i	5	5	9	10	14	18	19	28	38	38	42
abs. rank	4	4	6	7	8	9	10.5	13	14.5	14.5	16

So $W^+ = 106.5$. With $n = 16$, Theorem 10-2 gives

$$E(W^+) = \frac{16 \times 17}{4} = 68, \qquad \mathrm{Var}(W^+) = \frac{16 \times 17 \times 33}{24} = 374,$$
$$Z_{W^+} = \frac{106.5 - 68}{\sqrt{374}} = 1.99.$$

$P(Z \geq 1.99) = 0.0233$, so the one-sided p-value 2.33% is below 5%. We reject H_0 and conclude that the new curriculum A is more effective than B.

Chapter 11. Time Series Analysis

11.1 Types of time series data

Motivation. Time-ordered data are usually dependent; first identify the components (trend, seasonality, randomness).

A **time series** is data observed in time order, used to find patterns, forecast, or study dynamics. A **continuous time series** is observed at every instant, a **discrete time series** at particular times; most series are generated continuously but recorded at fixed intervals.

An important element of the analysis is the **time lag**, the interval between successive observation times. A time series is written

$$\{Z_t : t = 1, 2, 3, \dots\} \qquad \text{or} \qquad Z_1, Z_2, Z_3, \dots$$

The aims of time series analysis fall into two groups.

(i) **Forecasting**: assuming the past pattern persists, analyse the data so far and forecast the future.

(ii) **Understanding and control**: model the system (stochastic process) that generates the series in order to understand or control it.

The first step is always the **time series plot**, with time t on the horizontal axis and Z_t on the vertical axis. It shows the features of the series at a glance and guides the choice of model.

11.1.1 Components of a time series

A series has a **systematic component**, split into a **trend component** (long-term change) and a **seasonal** (cyclical) **component** (repeated short-term change), and a **random** (**irregular**) **component**.

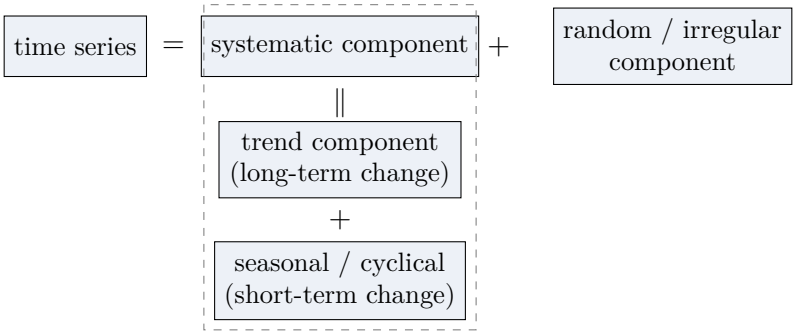


Figure 11.1: Components of a time series.

Definition (Components)

(a) The **random (irregular) component** is variation caused by random factors, unrelated to any regular movement over time. A series made only of a random component around a level is modelled by

$$Z_t = \beta_0 + \epsilon_t. \tag{11.1}$$

(b) The **trend component** is a persistent increase or decrease over time. It need not be linear and may be quadratic or S-shaped. A linear trend gives the **linear trend model**

$$Z_t = \beta_0 + \beta_1 t + \epsilon_t, \tag{11.2}$$

where $\beta_0 + \beta_1 t$ is the trend component and ϵ_t is the irregular component. Curved trends use

$$Z_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \epsilon_t \quad \text{or} \quad Z_t = \beta_0 + \beta_1 t + \cdots + \beta_k t^k + \epsilon_t.$$

(c) The **seasonal component** is variation that repeats with the week, month or season. Examples are sales of summer goods, rainfall and temperature, which rise in summer and fall in winter.
(d) The **cyclical component** repeats over a long period not tied to the seasons: **business cycles** (Kondratiev 50–60 years, Juglar 9–10, Kitchin about 40 months), or the 11-year sunspot cycle.

A series whose systematic part is purely seasonal (no trend) can be described with trigonometric or indicator functions. For monthly data with a 12-month seasonal period,

$$Z_t = \beta_0 + \beta_1 \sin\left(\frac{2\pi t}{12}\right) + \beta_2 \cos\left(\frac{2\pi t}{12}\right) + \epsilon_t. \tag{11.3}$$

The systematic part takes the same value at $t = 1$ and $t = 13$, so it has period 12 months. Many economic series (airline passengers, sales of ice cream or roasted sweet potatoes) have both a trend and seasonality:

$$Z_t = \beta_0 + \beta_1 t + \beta_2 \sin\left(\frac{2\pi t}{12}\right) + \beta_3 \cos\left(\frac{2\pi t}{12}\right) + \epsilon_t. \tag{11.4}$$

Here $\beta_0 + \beta_1 t$ is the trend, and the sine and cosine terms form the 12-month seasonal component.

In some series the size of the fluctuations grows over time together with the trend, as in many economic indicators. Then one first takes logarithms and fits a model of type (11.4):

$$\log Z_t = \beta_0 + \beta_1 t + \beta_2 \sin\left(\frac{2\pi t}{12}\right) + \beta_3 \cos\left(\frac{2\pi t}{12}\right) + \epsilon_t.$$

The log transform stabilises the variability and simplifies the model. Forecasts are then on the log scale, so they must be converted back by the **exponential transformation** $\exp(x) = e^x$.

Components are often not clearly separable, and two trend lines in different periods indicate a **change point**; the time series plot reveals such features.

11.2 Methods of time series analysis

Motivation. A survey of trend models and smoothing methods developed in this chapter.

A traditional approach is to regress the variable on time and analyse the trend. Expressing Z_t as a polynomial trend in t plus an irregular component gives

$$Z_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \cdots + \beta_k t^k + \epsilon_t, \tag{11.5}$$

the **linear polynomial trend model** (or simply **polynomial trend model**). It is a special case of multiple regression,

$$Z_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \cdots + \beta_k X_{kt} + \epsilon_t,$$

in which the explanatory variables are determined by time: $X_{1t} = t$, $X_{2t} = t^2, \ldots$. If the trend is a nonlinear function of t, we may use a **nonlinear trend model** such as

$$Z_t = \exp(\beta_0 + \beta_1 X_t)\, \epsilon_t.$$

Since the generating system may change, recent observations deserve more weight in forecasting.

Smoothing methods remove seasonal and irregular components to reveal the trend: the **moving average method** and the **exponential smoothing method**.

(a) The moving average gives equal weight to the most recent observations only, not to all observations.

(b) Exponential smoothing weights recent observations more, with exponentially decaying weights; it adapts flexibly and is cheap to compute.

Exponential smoothing serves forecasting; moving averages serve **seasonal adjustment**. Other approaches: the **decomposition method**, Box–Jenkins ARMA/ARIMA models (identification, estimation, diagnosis), transfer-function models, and regression with autocorrelated errors.

This chapter covers trend analysis, moving averages and exponential smoothing: empirical, convenient methods that detect a trend without explaining it.

11.3 Trend analysis

Motivation. For a level or a clear trend, least squares on time gives forecasts and prediction intervals.

The choice of model depends on the shape of the series.

(a) A series that moves randomly around a fixed level calls for the constant mean model.

(b) A series with a steady upward or downward trend calls for a linear or a **quadratic trend model**.

(c) A series repeating the same pattern with a fixed period calls for a **seasonal trend model**. If it also has a steady linear trend, we use a **linear seasonal trend model**.

(d) Nonlinear trend models are used mainly for series growing geometrically or moving nonlinearly.

Definition (Three stages of model building)

(i) **Identification**: plot the series, study its features, and choose a tentative model.
(ii) **Estimation**: estimate the parameters by least squares or maximum likelihood.
(iii) **Diagnosis**: analyse the residuals to check the fit. If the fit is inadequate, modify the model or choose another.

Trend models are fitted as in regression; if an assumption fails (e.g. equal variance), transform (e.g. logs), replot, and choose a tentative model.

11.3.1 Constant mean model

A series moving randomly around a fixed level is described by the **constant mean model**

$$Z_t = \beta_0 + \epsilon_t,$$

where the errors ϵ_t are independent with mean 0 and variance σ^2, and β_0 is the unknown constant mean. From the past data $Z_1, \dots, Z_n$ the least squares estimator of β_0 is

$$\hat{\beta}_0 = \frac{1}{n}\sum_{t=1}^{n} Z_t = \overline{Z}.$$

Let $Z_n(l) = E(Z_{n+l} \mid Z_n, Z_{n-1}, \dots)$ be the **l-step-ahead forecast** made at origin n. Under the constant mean model

$$Z_n(l) = \hat{\beta}_0 = \overline{Z} \qquad (l = 1, 2, \dots).$$

Definition (Forecast error, variance estimate and prediction interval)

The mean squared forecast error is

$$E[Z_{n+l} - Z_n(l)]^2 = \Big(1 + \frac{1}{n}\Big)\sigma^2,$$

and σ^2 is estimated by

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{t=1}^{n}(Z_t - \overline{Z})^2.$$

If the ϵ_t are normal, a $100(1-\alpha)\%$ **prediction interval** for Z_{n+l} is

$$\overline{Z} \pm t_{\alpha/2}(n-1)\,\hat{\sigma}\Big(1 + \frac{1}{n}\Big)^{1/2},$$

where $t_{\alpha/2}(n-1)$ is the upper $\alpha/2$ point of the t distribution with $n-1$ degrees of freedom.

[Supplement] Why these hold: $Z_{n+l} - \overline{Z} = \epsilon_{n+l} - \bar{\epsilon}$, and ϵ_{n+l} is independent of $\bar{\epsilon} = \frac{1}{n}\sum_{t\le n}\epsilon_t$. So the error has mean 0 and variance $\sigma^2 + \sigma^2/n$. Under normality it is $N\big(0, (1+\frac{1}{n})\sigma^2\big)$ and independent of $\hat{\sigma}^2$, with $(n-1)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-1)$. Hence $(Z_{n+l} - \overline{Z})/(\hat{\sigma}\sqrt{1+1/n}) \sim t(n-1)$, which gives the interval.

A prediction interval (L, U) for a future *random* value Z_{n+l} satisfies $P(L < Z_{n+l} < U) \ge 95\%$ when $\alpha = 0.05$. It is not a confidence interval for a parameter.

When a new observation Z_{n+1} arrives, the forecasts from origin $n+1$ are

$$\begin{aligned} Z_{n+1}(l) &= \frac{1}{n+1}(Z_1 + \cdots + Z_n + Z_{n+1}) = \frac{n}{n+1}Z_n(l) + \frac{1}{n+1}Z_{n+1} \\ &= Z_n(l) + \frac{1}{n+1}[Z_{n+1} - Z_n(l)]. \end{aligned}$$

The update combines the old forecast $Z_n(l)$ and the error $e_n(l) = Z_{n+1} - Z_n(l)$, needing no past data.

Example 11-1 (Statistics course enrolment)

The (hypothetical) numbers of students taking a statistics course each semester are given below.

Sem.	Students	Sem.	Students	Sem.	Students	Sem.	Students
2012/S1	1,372	2015/S1	1,423	2018/S1	1,420	2021/S1	1,435
2012/S2	1,388	2015/S2	1,337	2018/S2	1,406	2021/S2	1,376
2013/S1	1,478	2016/S1	1,366	2019/S1	1,372	2022/S1	1,347
2013/S2	1,404	2016/S2	1,378	2019/S2	1,489	2022/S2	1,389
2014/S1	1,406	2017/S1	1,461	2020/S1	1,425	2023/S1	1,349
2014/S2	1,486	2017/S2	1,418	2020/S2	1,302	2023/S2	1,364

Using the constant mean model, answer the following.
(a) Find the forecast and the 95% prediction interval for enrolment in the first semester of 2026.
(b) If 1,500 students enrol in the first semester of 2024, find the updated forecast.
Solution. Here $n = 24$, $\overline{Z} = 1399.625$ and $\hat{\sigma} = 47.83$. The first semester of 2026 is $l = 5$ steps ahead, and $Z_n(5) = \overline{Z} = 1399.625$. The forecast for 2024/S1 is also 1399.625.
(a) With $t_{0.025}(23) = 2.069$,

$$\overline{Z} \pm t_{\alpha/2}(n-1)\,\hat{\sigma}\Big(1 + \frac{1}{n}\Big)^{1/2} = 1399.625 \pm 2.069 \times 47.83 \times \Big(1 + \frac{1}{24}\Big)^{1/2} = 1399.625 \pm 100.001.$$

The 95% prediction interval is (1298.624, 1500.626), so roughly 1,300 to 1,500 students are expected.
(b) The forecast error is $Z_{n+1} - Z_n(l) = 1500 - 1399.625 = 100.375$. Hence

$$Z_{n+1}(l) = Z_n(l) + \frac{1}{n+1}\big[Z_{n+1} - Z_n(l)\big] = 1399.625 + \frac{100.375}{25} = 1403.64.$$

11.3.2 Linear trend model

A series following $Z_t = \beta_0 + \beta_1 t + \epsilon_t$ is analysed by the regression methods of Chapter 8.

Illustration 11-1: women in senior professional and technical jobs

A hypothetical monthly series (January 2018 to January 2024) of women in senior professional and technical occupations shows a clear linear trend. The linear trend model gives $R^2 = 0.8373$ and these tables.

Source	SS	df	MS	F	p-value
Model	2,170,954	1	2,170,954	365.48	$< 2.2 \times 10^{-16}$
Error	421,736	71	5,940		
Total	2,592,690	72			

Variable	Coefficient	SE	T	p-value
Intercept	−4,578	262.1	−17.46	$< 2 \times 10^{-16}$
Date	0.2689	0.01407	19.12	$< 2 \times 10^{-16}$

The coefficients are significant and R^2 exceeds 80%, so the trend model fits fairly well. The fitted model is

$$\widehat{\text{Female}} = -4578 + 0.269 \times \text{Date},$$

where Date is the number of days since 1 January 1970 (e.g. 1 January 2018 is day 17,532). The residual time plot shows runs of equal sign: positively correlated errors, violating independence; a regression with autocorrelated errors would be better.

11.4 Moving averages

Motivation. Averaging the last m observations tracks a drifting level and, with m the period, removes seasonality.

In $Z_t = \beta_0 + \epsilon_t$ let β_0 drift slowly. Instead of $\hat{\beta}_0 = \overline{Z}$, use the mean of the last m observations:

$$M_n = \frac{1}{m}\sum_{t=n-m+1}^{n} Z_t = \frac{1}{m}(Z_{n-m+1} + Z_{n-m+2} + \cdots + Z_n). \tag{11.6}$$

M_n is the **simple moving average**, and it serves as the forecast of Z_{n+l} for every lead l:

$$Z_n(l) = M_n.$$

Large m smooths strongly but reacts slowly; small m follows local change; m equal to the seasonal period removes seasonality. Drawbacks: no forecast for the first m observations, and equal weights regardless of age.

For seasonal adjustment one uses instead of (11.6) the **centered moving average** with $m = 2k+1$:

$$M_n = \frac{1}{2k+1}(Z_{n-k} + \cdots + Z_{n+k}). \tag{11.7}$$

Illustration 11-2: quarterly births in Korea

The national quarterly numbers of births from 2000/Q1 to 2024/Q1 (97 quarters, Statistics Korea) are, row by row:

178245	152324	151913	157607	161437	136920	131907	129670	140157
123005	119727	114022	133304	121051	117707	122974	132259	115630
115593	113476	118216	106313	108074	106104	119423	110753	111082
110501	126767	117315	128434	124306	127560	111416	115866	111050
120664	107525	110135	106525	120550	111369	117792	120460	128997
116100	116491	109677	128802	117652	123317	114779	119432	105518
109651	101854	116004	107103	110808	101520	117952	110126	108263
102079	112366	102337	102192	89348	98529	89532	89638	80072
89760	82040	80480	74542	83030	75395	73683	70568	73595
68263	68900	61579	70125	66129	66551	57757	68339	60149
64175	56523	64468	56142	56743	52618	60474		

Centered moving averages with $m = 3$ ($k = 1$) and $m = 11$ ($k = 5$) give the following (excerpt of Table 11-3; NA means not computable).

t	Year/Q	Actual	$m = 3$	$m = 11$
1	2000/1	178245	NA	NA
2	2000/2	152324	160827.3	NA
3	2000/3	151913	153948	NA
4	2000/4	157607	156985.7	NA
5	2001/1	161437	151988	NA
6	2001/2	136920	143421.3	143901.1
7	2001/3	131907	132832.3	138062.6
8	2001/4	129670	133911.3	136333.5
9	2002/1	140157	130944	133527.9
10	2002/2	123005	127629.7	129900.6
⋮	⋮	⋮	⋮	⋮
89	2022/1	68339	62081.67	62903.36
90	2022/2	60149	64221	62463.73
91	2022/3	64175	60282.33	60872.18
92	2022/4	56523	61722	60358.09
93	2023/1	64468	59044.33	NA
94	2023/2	56142	59117.67	NA
95	2023/3	56743	55167.67	NA
96	2023/4	52618	56611.67	NA
97	2024/1	60474	NA	NA

Larger m smooths more: with $m = 11$ the trend stands out, $m = 3$ still follows short-term movements; the cost is 1 ($m = 3$) or 5 ($m = 11$) lost values at each end.

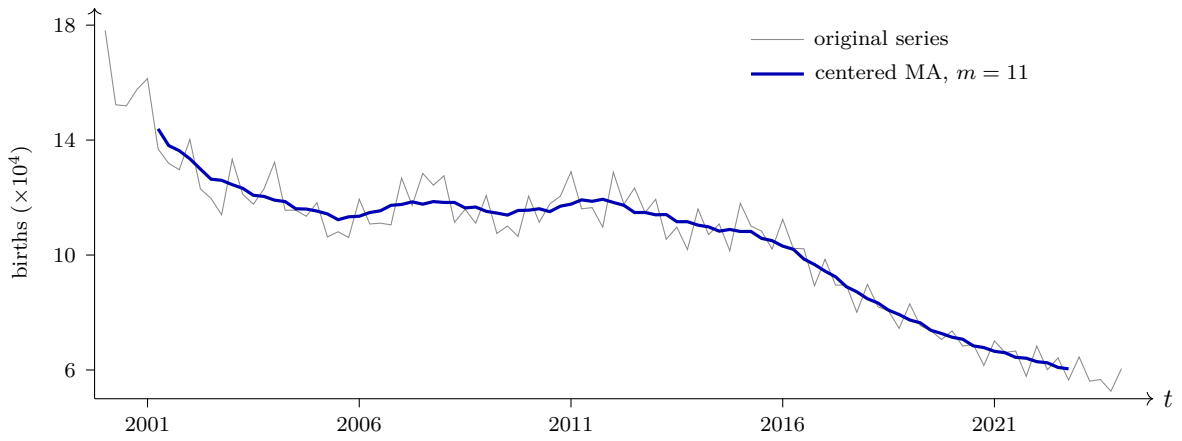


Figure 11.2: Quarterly births and their centered moving average ($m = 11$).

If the series has a linear trend as in (11.2), or a quadratic trend, double or triple moving averages are used instead.

11.5 Exponential smoothing

Motivation. Exponential smoothing uses all observations, weights recent ones more, and updates with two numbers.

Consider again the model (11.1), whose level may change over time. Given $Z_1, \dots, Z_n$, exponential smoothing forecasts Z_{n+1} by weights that decay geometrically into the past:

$$Z_n(1) = c \sum_{t=0}^{n-1} (1-w)^t Z_{n-t} = c\big[Z_n + (1-w)Z_{n-1} + \cdots + (1-w)^{n-1} Z_1\big]. \tag{11.8}$$

The **normalizing constant** c makes the weights sum to 1, i.e. $c \sum_{t=0}^{n-1} (1-w)^t = 1$. As n grows, $c \to w$. So for large n the one-step forecast is

$$Z_n(1) = \sum_{j \geq 0} w(1-w)^j Z_{n-j} = wZ_n + w(1-w)Z_{n-1} + w(1-w)^2 Z_{n-2} + \cdots \tag{11.9}$$

This is a sensible forecast for model (11.1), because it is unbiased:

$$E\big(Z_n(1)\big) = w \sum_{j \geq 0} (1-w)^j E(Z_{n-j}) = w \sum_{j \geq 0} (1-w)^j \beta_0 = \frac{w}{1-(1-w)}\, \beta_0 = \beta_0.$$

Definition (Simple exponential smoothing)

The weight w ($0 < w < 1$) in (11.8)–(11.9) is the **smoothing constant**, usually between 0.05 and 0.3. The smoothed forecast is the **simple exponentially smoothed value**

$$S_n = Z_n(1),$$

and from origin n every lead has the same forecast: $Z_n(l) = S_n$ for $l = 1, 2, \dots$. With S_{n-1} the smoothed value at time $n-1$,

$$S_n = wZ_n + (1-w)S_{n-1} = S_{n-1} + w(Z_n - S_{n-1}). \tag{11.10}$$

By (11.10) S_n combines S_{n-1} and Z_n only; computing $Z_n(l) = S_n$ recursively is **simple exponential smoothing**.

11.5.1 Choosing S_0 and w

To start (11.10) we need $S_1 = wZ_1 + (1-w)S_0$, hence an initial value S_0 and a constant w.

(a) *Initial value.* Use prior data for S_0 if available, else choose it subjectively; for large n its weight $(1-w)^n$ is tiny.

(b) *Smoothing constant.* Small w reacts slowly, large w follows the latest data; usually $w \in [0.05, 0.3]$, fixed over time.

A data-based choice uses the **1-step-ahead prediction error** at time $t-1$,

$$e_{t-1}(1) = Z_t - S_{t-1},$$

and picks the w that minimises

$$\mathrm{SSE}(w) = \sum_{t=1}^{n} e_{t-1}^2(1).$$

For rapidly changing levels, w near 1 and $S_0 = Z_1$; for slowly changing ones, w near 0 and $S_0 = \overline{Z}$.

Example 11-2 (KOSPI index)

Monthly closing values of the KOSPI index from January 2018 to December 2024 are given below.

Yr	1	2	3	4	5	6	7	8	9	10	11	12
18	2566.46	2427.36	2445.85	2515.38	2423.01	2326.13	2295.26	2322.88	2343.07	2029.69	2096.86	2041.04
19	2204.85	2195.44	2140.67	2203.59	2041.74	2130.62	2024.55	1967.79	2063.05	2083.48	2087.96	2197.67
20	2119.01	1987.01	1754.64	1947.56	2029.6	2108.33	2249.37	2326.17	2327.89	2267.15	2591.34	2873.47
21	2976.21	3012.95	3061.42	3147.86	3203.92	3296.68	3202.32	3199.27	3068.82	2970.68	2839.01	2977.65
22	2663.34	2699.18	2757.65	2695.05	2685.9	2332.64	2451.5	2472.05	2155.49	2293.61	2472.53	2236.4
23	2425.08	2412.85	2476.86	2501.53	2577.12	2564.28	2632.58	2556.27	2465.07	2277.99	2535.29	2655.28
24	2497.09	2642.36	2746.63	2692.06	2636.52	2797.82	2770.69	2674.31	2593.27	2556.15	2455.91	2399.49

(a) With $S_0 = \overline{Z}$, compute the simple exponentially smoothed values $S_1, \ldots, S_{83}$ for $w = 0.3$ and for $w = 0.1$.
(b) Plot the series with the smoothed values $S_0, \ldots, S_{83}$ for $w = 0.1$ and $w = 0.3$, and describe the effect of w.
Solution.
(a) The series has length $T = 84$, so

$$S_0 = \overline{Z} = \frac{1}{T}\sum_{t=1}^{T} Z_t = 2478.221.$$

For $w = 0.3$, the one-step forecast at $n = 1$, using only Z_1, is

$$S_1 = S_0 + w(Z_1 - S_0) = 2478.221 + 0.3 \times (2566.46 - 2478.221) = 2504.693.$$

Likewise S_2 uses S_1 and Z_2:

$$S_2 = S_1 + w(Z_2 - S_1) = 2504.693 + 0.3 \times (2427.36 - 2504.693) = 2481.493.$$

S_2 uses the data up to time 2 and forecasts Z_3. Continuing gives all of $S_0, \ldots, S_{83}$. Selected values (Table 11-5 lists S_{t-1}, the forecast of Z_t, beside Z_t):

	S_0	S_1	S_2	S_3	$\cdots$	S_{83}
$w = 0.3$	2478.221	2504.693	2481.493	2470.8	$\cdots$	2580.833
$w = 0.1$	2478.221	2487.045	2481.076	2477.554	$\cdots$	2596.527

(b) The smoothed series is much gentler (Figure 11.3): $w = 0.3$ stays close to the data, $w = 0.1$ is smoother and lags at the 2020–2022 turning points. [Supplement] The one-step SSE is about 2.75×10^6 ($w = 0.3$) against 6.41×10^6 ($w = 0.1$), so $w = 0.3$ is preferred.

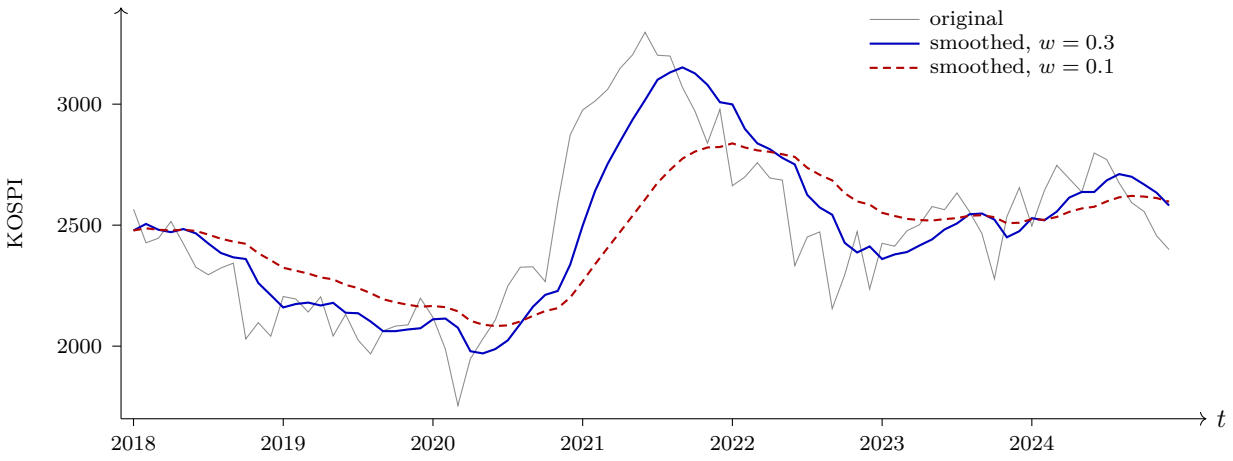


Figure 11.3: KOSPI and its forecasts S_{t-1} by simple exponential smoothing ($w = 0.3$, $w = 0.1$).

Chapter 12. Sample Surveys

12.1 Basic concepts of sample surveys

Motivation. A well-designed sample beats a slow, costly census; it needs a precise population and awareness of errors.

12.1.1 What is a sample survey

To learn about a population we can survey all of it or only part of it and estimate the whole from that part.

Definition (Census and sample survey)

(a) Surveying the entire population is a **census** (complete enumeration).
(b) Drawing a sample from part of the population and inferring population characteristics from it is a **sample survey**.

A census is exact in theory but often impossible or more error-prone in practice: surveying every Korean teenager's attitudes would take years and tens of thousands of interviewers, and the results would be outdated. Opinion surveys use samples.

Definition (Populations, frames and units)

(a) The **target population** is the one we want to study, defined precisely (are students on leave, or graduate students, "university students"?).
(b) The **sampling frame** (**frame**) is the list or rule identifying the units available for sampling.
(c) The **sampled population** is the one actually covered by the frame; a telephone-directory frame for "all adults in Korea" excludes unlisted adults.
(d) A **sampling unit** is an element of the population that can be selected. In a survey of students' attitudes it is each student; in a survey of farm household incomes it is each household.
(e) The **sampling design** specifies which units are selected and how. It is chosen to suit the parameters to be estimated and the structure of the population, so that estimation is as efficient as possible.

12.1.2 Errors and bias

Definition (Sampling and nonsampling error)

(a) **Sampling error** is the statistical error that arises because we infer about the whole population from part of it.
(b) **Nonsampling error** is every other error: the gap between target and sampled populations, interviewer mistakes, poorly worded questionnaires, and so on.

Sampling error shrinks like $1/\sqrt{n}$; nonsampling error does not depend on n (and can be larger in a census), so total error stops decreasing beyond some n.

Sampling error is computable and controlled through n; nonsampling error must be reduced by care in planning, collection and analysis. Its main sources:

(i) **Faulty questions**: unclear or leading questions distort answers.

(ii) **Wrong answers and interviewer inexperience**: incorrect or misunderstood answers (**response error**), or careless interviewers.

(iii) **Nonresponse error** (nonresponse bias): selected people do not respond. A low response rate makes the sample unrepresentative, especially when nonrespondents have particular tendencies.

(iv) **Coverage error**: some units are missing from (or wrongly included in) the frame, or the frame differs greatly from the target population.

(v) **Processing error**: errors in recording or in computer entry.

Besides $E(\hat{\theta})-\theta$, survey **bias** covers systematic distortions from nonsampling errors; its most serious source is a flawed design.

Case study: the 2015 UK general election polls

Before the 2015 UK election, five weeks of polls predicted no Commons majority, with Labour and the Conservatives level (the Conservatives had not won a majority since 1992). In fact the Conservatives won 331 seats, 91 more than Labour. The British Polling Council's inquiry found two main causes.
(i) *Wrongly chosen sampled population.* Frames over-represented Labour supporters (urban workers who answer surveys easily) and missed older Conservative voters.
(ii) *Neglect of nonresponse.* Respondents differed from nonrespondents. Conservative supporters likely had a lower response rate, so their real support was underestimated.

Similar errors: national attitudes surveyed through parents at particular schools, or landline-only telephone polls.

12.2 Sampling methods

Motivation. Simple random, systematic, stratified, cluster and stratified cluster sampling trade off cost and precision.

12.2.1 Simple random sampling

Definition (Simple random sampling)

Drawing n of N units so that all $\binom{N}{n}$ samples are equally likely is **simple random sampling** (SRS). In practice:
(i) number the units 1 to N;
(ii) select n of the numbers at random.

A random number generator is normally used. For a small population we can also draw lots from shuffled slips of paper.

Theorem (Sample mean under SRS without replacement)

Let a finite population have values $x_1,\dots,x_N$ with mean $\mu=\frac{1}{N}\sum_k x_k$ and variance $\sigma^2=\frac{1}{N}\sum_k(x_k-\mu)^2$. For an SRS of size n without replacement with sample mean $\overline{X}$ and sample variance S^2,

$$E(\overline{X})=\mu,\qquad \operatorname{Var}(\overline{X})=\frac{N-n}{N-1}\cdot\frac{\sigma^2}{n},\qquad E(S^2)=\frac{N}{N-1}\sigma^2.$$

Consequently $\left(1-\frac{n}{N}\right)\frac{S^2}{n}$ is an unbiased estimator of $\mathrm{Var}(\overline{X})$.

Proof. [Supplement]

Step 1. Let $\delta_k = 1$ if unit k is in the sample and $\delta_k = 0$ otherwise, so that $\overline{X} = \frac{1}{n}\sum_{k=1}^{N} x_k\delta_k$. By symmetry of SRS, for $k \neq l$,

$$P(\delta_k = 1) = \frac{n}{N}, \qquad P(\delta_k = \delta_l = 1) = \frac{n(n-1)}{N(N-1)}.$$

Hence $E(\overline{X}) = \frac{1}{n}\sum_k x_k \frac{n}{N} = \mu$.

Step 2. From Step 1,

$$\mathrm{Var}(\delta_k) = \frac{n}{N}\left(1-\frac{n}{N}\right) = \frac{n(N-n)}{N^2}, \qquad \mathrm{Cov}(\delta_k, \delta_l) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}.$$

Step 3. Since $\sum_k \delta_k = n$ is constant, $\sum_k x_k\delta_k$ and $\sum_k d_k\delta_k$ (with $d_k = x_k - \mu$) have the same variance. Using $\sum_k d_k = 0$, so that $\sum_{k\neq l} d_k d_l = -\sum_k d_k^2 = -N\sigma^2$,

$$\begin{aligned}\mathrm{Var}\left(\sum_k d_k\delta_k\right) &= \frac{n(N-n)}{N^2}\left[\sum_k d_k^2 - \frac{1}{N-1}\sum_{k\neq l} d_k d_l\right] = \frac{n(N-n)}{N^2}\, N\sigma^2\left(1+\frac{1}{N-1}\right) \\ &= \frac{n(N-n)}{N-1}\,\sigma^2.\end{aligned}$$

Dividing by n^2 gives $\mathrm{Var}(\overline{X}) = \frac{N-n}{N-1}\cdot\frac{\sigma^2}{n}$.

Step 4. Let $X_1, \ldots, X_n$ be the sample values. Each X_i has mean μ and variance σ^2, and

$$E\sum_{i=1}^{n}(X_i - \overline{X})^2 = E\sum_{i=1}^{n}(X_i-\mu)^2 - n\,E(\overline{X}-\mu)^2 = n\sigma^2 - \frac{N-n}{N-1}\sigma^2 = \frac{N(n-1)}{N-1}\sigma^2.$$

So $E(S^2) = \frac{N}{N-1}\sigma^2$. Finally,

$$E\left[\left(1-\frac{n}{N}\right)\frac{S^2}{n}\right] = \frac{N-n}{N}\cdot\frac{N}{N-1}\cdot\frac{\sigma^2}{n} = \mathrm{Var}(\overline{X}).$$

■

12.2.2 Systematic sampling

Definition (Systematic sampling)

Systematic sampling takes a random unit among the first $k = N/n$ (the **sampling interval**) and every k-th unit thereafter.

E.g. $n = 200$ from $N = 10000$: $k = 50$, and start 23 gives units $23, 73, 123, \ldots, 9973$. It is simpler than SRS and spreads the sample evenly.

Exit polls use systematic sampling with k fixed in advance, since N and n are unknown; the joint exit polls of the three Korean broadcasters (2022 presidential, 2024 National Assembly elections) used $k = 5$.

Systematic sampling is convenient and is common in large surveys. If the list has a periodic pattern whose period matches k, however, it can produce serious bias.

12.2.3 Stratified random sampling

Definition (Stratified random sampling)

Drawing an SRS within each of several groups (**strata**) and pooling them is **stratified (random) sampling**.

Strata homogeneous within and heterogeneous across ensure every kind of unit appears, raising precision; stratification is the theoretically most important design.

E.g. if students' attitudes differ by college and year but are alike within them, small samples per stratum describe each accurately and precision improves greatly.

12.2.4 Cluster sampling

Definition (Cluster sampling)

Cluster sampling chooses some **clusters** (subpopulations) by SRS and surveys all their units (**single-stage**) or an SRS within each (**two-stage cluster sampling**).

It works well when clusters resemble each other and the population, and is cheaper: if attitudes depend on year but not major, survey all or some students of a few random departments.

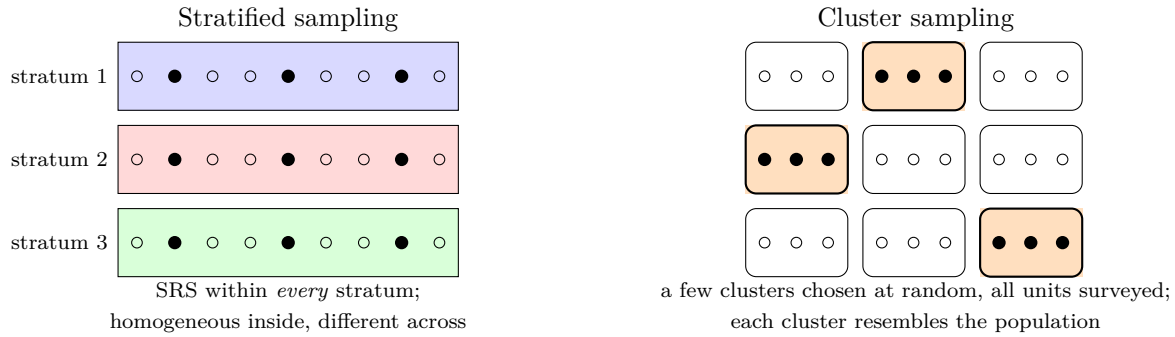


Figure 12.1: Stratified sampling versus (single-stage) cluster sampling; filled dots are sampled units.

12.2.5 Stratified cluster sampling

Definition (Stratified cluster sampling)

Stratified cluster sampling draws clusters (primary units) within each stratum and surveys them completely (**stratified single-stage cluster sampling**) or by SRS of secondary units (**stratified two-stage cluster sampling**).

E.g. with attitudes differing between colleges but little between departments: colleges as strata, departments as clusters, students drawn by SRS from chosen departments.

12.3 Estimation under stratified random sampling

Motivation. Estimators, standard errors and intervals for means and proportions under stratified sampling.

A population of N units has h strata of sizes N_i; **proportional allocation** splits n in proportion to stratum size:

$$n_i = n\Big(\frac{N_i}{N}\Big) \qquad (i = 1, 2, \ldots, h).$$

For means, let μ_i and σ_i^2 be the mean and variance of stratum i, with $\sigma_i^2 = \frac{1}{N_i}\sum(x-\mu_i)^2$ over the stratum. For proportions, let P_i be the proportion of units in stratum i with the attribute.

Definition 12-1 (Estimation of the population mean)

(i) Estimator:

$$\overline{X}_{\text{st}} = \frac{N_1}{N}\overline{X}_1 + \frac{N_2}{N}\overline{X}_2 + \cdots + \frac{N_h}{N}\overline{X}_h = \frac{1}{N}\sum_{i=1}^{h} N_i \overline{X}_i.$$

(ii) Variance of the estimator:

$$\operatorname{Var}(\overline{X}_{\text{st}}) = \frac{1}{N^2}\sum_{i=1}^{h} N_i^2 \Big(\frac{N_i - n_i}{N_i - 1}\Big)\frac{\sigma_i^2}{n_i}.$$

(iii) Standard error of the estimator:

$$\widehat{\operatorname{SE}}(\overline{X}_{\text{st}}) = \sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2 \Big(1 - \frac{n_i}{N_i}\Big)\frac{S_i^2}{n_i}}\,.$$

Here $\overline{X}_i$ and S_i^2 are the sample mean and sample variance of stratum i, and the subscript st stands for "stratified".

Justification. [Supplement] The samples from different strata are independent SRSs without replacement.

Step 1. By the theorem of Section 12.2.1, $E(\overline{X}_i) = \mu_i$. Hence $E(\overline{X}_{\text{st}}) = \frac{1}{N}\sum_i N_i\mu_i = \mu$, so $\overline{X}_{\text{st}}$ is unbiased.

Step 2. By independence across strata and the same theorem,

$$\operatorname{Var}(\overline{X}_{\text{st}}) = \frac{1}{N^2}\sum_i N_i^2 \operatorname{Var}(\overline{X}_i) = \frac{1}{N^2}\sum_i N_i^2 \frac{N_i - n_i}{N_i - 1}\cdot\frac{\sigma_i^2}{n_i},$$

which is (ii).

Step 3. The same theorem shows that $(1 - n_i/N_i)S_i^2/n_i$ is unbiased for $\operatorname{Var}(\overline{X}_i)$. Summing these estimates with weights N_i^2/N^2 gives an unbiased estimator of $\operatorname{Var}(\overline{X}_{\text{st}})$, and its square root is (iii). ■

Definition 12-2 (Estimation of the population proportion)

(i) Estimator:

$$\hat{P}_{\text{st}} = \frac{N_1}{N}\hat{P}_1 + \frac{N_2}{N}\hat{P}_2 + \cdots + \frac{N_h}{N}\hat{P}_h = \frac{1}{N}\sum_{i=1}^{h} N_i \hat{P}_i.$$

(ii) Variance of the estimator:

$$\operatorname{Var}(\hat{P}_{\text{st}}) = \frac{1}{N^2}\sum_{i=1}^{h} N_i^2 \Big(\frac{N_i - n_i}{N_i - 1}\Big)\frac{P_i(1-P_i)}{n_i}.$$

(iii) Standard error of the estimator:

$$\widehat{\operatorname{SE}}(\hat{P}_{\text{st}}) = \sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2 \Big(1 - \frac{n_i}{N_i}\Big)\frac{\hat{P}_i(1-\hat{P}_i)}{n_i}}\,.$$

Here $\hat{P}_i = X_i/n_i$ is the sample proportion of stratum i, where X_i is the number of sampled units in stratum i with the attribute.

Justification. [Supplement] Code each unit as 1 (has the attribute) or 0. Then $\mu_i = P_i$, $\sigma_i^2 = P_i(1-P_i)$ and $\overline{X}_i = \hat{P}_i$, so (i) and (ii) are the special cases of Definition 12-1.

For the sample variance of 0–1 data, $S_i^2 = \frac{n_i}{n_i-1}\hat{P}_i(1-\hat{P}_i)$. The exactly unbiased variance estimate is therefore

$$\Big(1-\frac{n_i}{N_i}\Big)\frac{\hat{P}_i(1-\hat{P}_i)}{n_i-1}.$$

Formula (iii) with n_i in the denominator differs from it by the factor $(n_i-1)/n_i \approx 1$. Example 12-2 uses the exact version with $n_i - 1$. ■

Finite population correction and confidence intervals

$\frac{N_i-n_i}{N_i-1}$ in part (ii) of Definitions 12-1, 12-2 is the **finite population correction factor**; the estimated standard error uses $\frac{N_i-n_i}{N_i} = 1-\frac{n_i}{N_i}$ for unbiasedness (Section 12.2.1), and part (iii) replaces σ_i^2, P_i by S_i^2, $\hat{P}_i$.

The 95% confidence intervals for the mean and the proportion are

$$\bar{x}_{\mathrm{st}} \pm z_{0.025}\widehat{\mathrm{SE}}(\overline{X}_{\mathrm{st}}) = \bar{x}_{\mathrm{st}} \pm 1.96\sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(1-\frac{n_i}{N_i}\Big)\frac{s_i^2}{n_i}},$$

$$\hat{p}_{\mathrm{st}} \pm z_{0.025}\widehat{\mathrm{SE}}(\hat{P}_{\mathrm{st}}) = \hat{p}_{\mathrm{st}} \pm 1.96\sqrt{\frac{1}{N^2}\sum_{i=1}^{h} N_i^2\Big(1-\frac{n_i}{N_i}\Big)\frac{\hat{p}_i(1-\hat{p}_i)}{n_i}}.$$

The half-widths $z_{0.025}\widehat{\mathrm{SE}}(\overline{X}_{\mathrm{st}})$ and $z_{0.025}\widehat{\mathrm{SE}}(\hat{P}_{\mathrm{st}})$ are called the **margin of error**, also known as the error limit or allowable error.

Example 12-1 (Household living expenses)

To survey monthly living expenses per household in a small city, the city was divided by standard of living into 3 regions (strata), and a stratified sample gave the following data (unit: 100,000 won).

Stratum	N_i	n_i	$\bar{x}_i$	s_i^2
1	2,800	30	18.5	12.2
2	5,600	60	15.2	11.3
3	3,400	30	10.7	10.5
Total	$N = 11{,}800$	$n = 120$		

Estimate the mean monthly living expenses per household and give a 95% confidence interval.

Solution. The estimate is

$$\bar{x}_{\mathrm{st}} = \frac{1}{11800}(2800\times 18.5 + 5600\times 15.2 + 3400\times 10.7) = 14.69.$$

The estimated variance is

$$\widehat{\mathrm{Var}}(\overline{X}_{\mathrm{st}}) = \frac{1}{11800^2}\Big\{2800^2\Big(1-\frac{30}{2800}\Big)\Big(\frac{12.2}{30}\Big) + 5600^2\Big(1-\frac{60}{5600}\Big)\Big(\frac{11.3}{60}\Big) + 3400^2\Big(1-\frac{30}{3400}\Big)\Big(\frac{10.5}{30}\Big)\Big\} = 0.09342.$$

So $\widehat{\mathrm{SE}}(\overline{X}_{\mathrm{st}}) = \sqrt{0.09342} = 0.3056$, and

$$\bar{x}_{\mathrm{st}} \pm z_{0.025}\widehat{\mathrm{SE}}(\overline{X}_{\mathrm{st}}) = 14.69 \pm 1.96\times 0.3056 = 14.69 \pm 0.60.$$

The 95% confidence interval is (14.09, 15.29), i.e. about 1.41 to 1.53 million won.

Example 12-2 (Support for a university policy)

A university grouped similar colleges into 3 strata A, B, C and surveyed students on leave about a university policy by stratified sampling.

College (stratum)	N_i	n_i	$\hat{p}_i$
A	3,500	35	0.56
B	4,500	45	0.60
C	2,000	20	0.40

Estimate the support rate and give a 95% confidence interval.
Solution. The estimate is

$$\hat{p}_{\text{st}} = \frac{1}{10000}(3500 \times 0.56 + 4500 \times 0.60 + 2000 \times 0.40) = 0.546.$$

Using $n_i - 1$ in the denominators,

$$\widehat{\text{Var}}(\hat{P}_{\text{st}}) = \frac{1}{10000^2}\Big\{3500^2\Big(1 - \frac{35}{3500}\Big)\Big(\frac{0.56 \times 0.44}{35-1}\Big) + 4500^2\Big(1 - \frac{45}{4500}\Big)\Big(\frac{0.60 \times 0.40}{45-1}\Big) + 2000^2\Big(1 - \frac{20}{2000}\Big)\Big(\frac{0.40 \times 0.60}{20-1}\Big)\Big\} = 0.00247.$$

So $\widehat{\text{SE}}(\hat{P}_{\text{st}}) = \sqrt{0.00247} = 0.0497$, and

$$\hat{p}_{\text{st}} \pm z_{0.025}\widehat{\text{SE}}(\hat{P}_{\text{st}}) = 0.546 \pm 1.96 \times 0.0497 = 0.546 \pm 0.097.$$

The 95% confidence interval is (0.449, 0.643), i.e. 44.9% to 64.3%. (The book centres the interval at 0.564 by a transposition slip and reports (0.467, 0.661); the margin of error 0.097 is the same.)

12.4 Problems in practical surveys

Motivation. A published margin of error covers only sampling error, not errors from misused randomness or poor questions.

A typical poll report: *automated wireless calls to virtual mobile numbers, sample 1,003, response rate* 2.3%, *sampling error* ±3.1 *points at* 95% *confidence.* The 3.1% is the margin of error of Section 12.3, covering sampling error only. Professional firms keep nonsampling error small, so it is then a good guide; in non-expert surveys nonsampling error can grow sharply, typically through misunderstood randomness and poor wording.

12.4.1 The concept of randomness

Random sampling selects units with probabilities fixed in advance; samples not drawn from the whole population by statistical theory incur severe bias (Section 12.1.2).

E.g. a student studying "generational differences in attitudes to household chores and gender equality" hands questionnaires to high-school students in one apartment complex for their parents. Childless families and the elderly are excluded, and one complex's households share socio-economic traits, so the sampled population differs greatly from the target.

The 2015 UK polls (Section 12.1) failed the same way: "random" read as "haphazard", or convenience chosen over design.

12.4.2 Writing the questionnaire

Questions must neither lead respondents to a particular answer nor confuse them with vague wording.

(a) *Leading questions.* "Should ancestral rites be kept in the traditional way, to honour the ancestors and teach tradition to our children?" and "Should they be simplified to fit changing values?" both invite "yes"; asked separately, the two proportions may sum to over 100%.

(b) *Ambiguous questions.* "Are you satisfied with your current life?" leaves "current life" undefined; ask about several aspects (work, family, leisure, friends) and combine. In "Do you have many friends?" "many" may mean 3 to one person and not 10 to another.

(c) *Campaigning disguised as a poll.* "Do you know that candidate X devoted his life to the democracy movement?" promotes a candidate and abuses surveys.

(d) *Pretesting.* A **pilot survey** finds such problems: asking students "monthly leisure budget? (1) under 100,000 won (2) 100–500 thousand (3) 500 thousand–1 million (4) over 1 million" yields mostly (1), revealing nothing about the distribution; revise the categories.

(e) *Length.* With too many questions, respondents tire and answer carelessly. Limit the number of items, and offering fair compensation for the respondent's time also helps.

Every item must be reviewed carefully. However sophisticated the sampling design, a badly written questionnaire yields distorted information.